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镍基高温合金CM247LC定向凝固的偏析行为研究

STUDY ON SEGREGATION BEHAVIOR OF DIRECTIONAL SOLIDIFIED NICKEL-BASED SUPERALLOY CM247LC

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STUDY ON SEGREGATION BEHAVIOR OF DIRECTIONAL SOLIDIFIED NICKEL-BASED SUPERALLOY CM247LC

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Abstract

Single-crystal or columnar-crystal Ni-based superalloys based on directional-solidification technology are widely used in the forming of aerospace and heavy-duty gas-turbine blades. To ensure the high-temperature performance of Ni-based superalloy products, the temperature gradient and solidification velocity of the directional-solidification process must be deeply optimized to obtain the required microstructural features (single crystal or columnar crystals with small deviation angle) and compositional features, so as to fully exploit the γ/γ′ strengthening of Ni-based superalloys.

This thesis takes the widely used directionally solidified Ni-based superalloy CM247LC as the object, and, using experiments and computer-simulation techniques, investigates the stability of the directionally solidified microstructure and the segregation behaviour of the various elements in the alloy, and further analyses the effect of element segregation on the precipitation of the harmful TCP (topologically close-packed) phase.

During Bridgman directional solidification, the temperature gradient at the solid–liquid interface decreases continuously with increasing distance from the water-cooling plate, and the compositional distribution, interface stability and morphological evolution ahead of the interface change continuously as solidification proceeds. The compositional distribution at the interface obeys the solidification-redistribution rules; the fully solidified solid, the mushy zone where solid and liquid coexist, and the solute diffusion layer ahead of the interface can all be calculated from the phase diagram and the effective partition coefficient. CM247LC has many alloying elements and an obvious liquid-density difference, so the solute distribution ahead of the interface is further affected by complex flow-field factors.

In this study, CM247LC specimens were prepared in an ALD directional-solidification furnace at a pulling speed of 41 μm/s, and the axial composition and as-

cast microstructure were characterized; then, with computer-simulation tools at the same pulling speed, the temperature field, flow field and solute diffusion field were processed and analysed. The main work and conclusions are as follows:

(1) Using the CAFE microstructure-morphology module of ProCAST, simulations under different cellular-automata and solidification parameters were carried out, and appropriate temperature-field boundary conditions and shell thermophysical parameters were determined by comparison with the experimental microstructure.

(2) The composition of each element on the central axis was measured by EDS at equal intervals; through Pearson-coefficient analysis the elements were divided into four classes by their linear relationships, and a Spearman trend test showed that W decreases and Ta and Ti increase along the axis; the partition coefficients k_0 and liquidus slopes m were obtained from JMatPro phase-diagram data.

(3) Based on the Scheil model, the measured values of each element were log-linearly fitted by gradient descent to obtain the effective partition coefficient k_E at 41 μm/s; global phase-stability calculations based on the measured compositions show a high tendency of TCP-phase precipitation at the composition peaks of Ta and Ti.

(4) The compositions of the dendrite core and the interdendritic region were measured to compute the microsegregation coefficient k'_x, and the phase fractions of each element in γ, γ′ and MC carbide were analysed by JMatPro Scheil-Gulliver. The measured and computed micro- and macro-segregation behaviours are essentially consistent; in particular, liquid Al first rises (to 6.33 wt% at 1290 °C) and then falls; k_Al increases from 0.91 at the liquidus and crosses 1 at about 1290 °C, so Al is weakly enriched in the solid macroscopically.

(5) The temperature-field–flow-field coupling module of ProCAST was used to reconstruct the thermal history of each test point of the CM247LC specimen at

a pulling speed of 41 μm/s, from which the local temperature gradient G and the interface advancing velocity R at the liquidus were derived. JMatPro shows a density inversion of about 0.14 % of the interdendritic liquid near 1350 ℃, a possible driver of thermosolutal convection; in the coupled simulation (linear thermal density model) G oscillates on a 15–20 mm scale, whose origin — radial thermal asymmetry or solutal convection — cannot be decided from the present data; through the cooling-rate identity $\dot{T} = G \cdot R$, R oscillates in anti-phase with G in the steady stage. After normalization, the concentration fluctuation of elements with $k_E<1$ is in phase with G, whereas that of elements with $k_E>1$ is in anti-phase; the solute boundary layer responds quasi-statically ($\omega\tau\approx0.016\ll1$), but the response via the growth rate predicts the opposite sign, the sign of the alignment can be explained by the export/retention phase of the convection, a mechanism that still awaits verification with flow-field data. Point-by-point cross-correlation confirms the alignment quantitatively: the correlation coefficients of Ta, Ti and Hf with G are +0.66, +0.53 and +0.43, those of Ni and Co are −0.62 and −0.55, W is near zero (autocorrelation-corrected p=0.02–0.03 for Ta and Ni), and carbide/eutectic pools tend to lie where G is locally high.

(6) Statistically, the fluctuation amplitude correlates with the partition-coefficient deviation $|k_0-1|$ with a Spearman coefficient of 0.93, far exceeding its correlation with atomic number (0.26) or liquid diffusivity; the geometric asymmetry between $k>1$ and $k<1$ accounts for the small amplitude of the strong absorber W, and the mass-balance closure relation accounts for the dilution component of the “anti-phase” matrix elements.

(7) The measured k_E being closer to 1 than the phase-diagram values is a static boundary-layer effect of the effective partition coefficient (inverted effective Péclet number about 0.9–1.5); the layer-thickness-weighted correction due to

the velocity oscillation is only of order 10^{-3}. Overall, the composition fluctuation in directionally solidified CM247LC is the superposition of a fine-scale sampling of interdendritic carbide/eutectic pools (amplitude ordered by $|k_0-1|$) and a mesoscale channel/plume modulation driven by density inversion.

Keywords: Nickel-base superalloy; CM247LC; Directional solidification; Effective partition coefficient; Thermosolutal convection; Composition fluctuation

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Chapter 1 Introduction

1.1 Nickel-Based Superalloys

Nickel-based superalloys are high-performance materials capable of withstanding extreme temperatures. Owing to their outstanding high-temperature mechanical properties and microstructural stability, they are widely employed in high-temperature hot-end components such as aircraft engine turbine blades and heavy-duty gas turbine blades [1,2].

Directionally solidified nickel-based superalloys evolved from wrought nickel-based superalloys. In 1950, the primary manufacturing process transitioned from forging to casting; the subsequent introduction of directional solidification technology eliminated grain boundaries in both the transverse and longitudinal directions [2]. However, large-scale nickel-based alloy blades for gas turbines are susceptible to various casting defects arising from complex convective flows during solidification. In particular, macrosegregation during solidification necessitates costly homogenization heat treatment for its remediation [4].

CM247LC is a low-carbon nickel-based superalloy developed for casting applications. It was derived by systematically fine-tuning the composition of the Mar-M247 alloy to enable the production of directionally solidified turbine blades [5]. Compared with the Mar-M247 baseline alloy, directionally solidified CM247LC castings demonstrate superior resistance to grain boundary cracking, along with a more favorable carbide microstructure and enhanced toughness [6].

1.1.1 Microstructure

(a) (b)

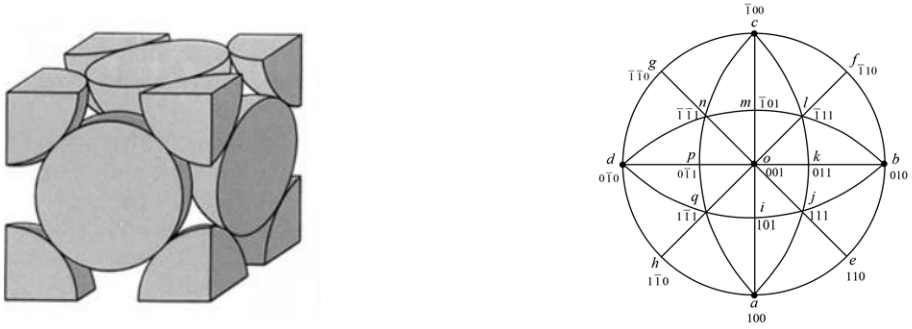


Figure. 1.1 Lattice structure of the γ phase (a) Face-centered cubic crystal structure ; (b) Standard projection of the (001) plane[2]

The exceptional high-temperature properties of nickel-based superalloys originate from a coherent two-phase microstructure comprising the γ matrix phase and the γ′ precipitate phase. As shown in Figure 1.1(a), the γ phase adopts a face-centered cubic (FCC) crystal structure belonging to the cubic crystal system, characterized by three mutually perpendicular axes of twofold symmetry. Figure 1.1(b) presents the standard stereographic projections of all crystallographic planes in the FCC structure, with the (001) plane serving as the projection plane. The high-temperature strength of nickel-based superalloys is primarily governed by the γ′ phase, which exhibits an anomalous strengthening behavior whereby its yield strength increases with rising temperature. In recent years, the volume fraction of the γ′ phase in certain nickel-based alloys has exceeded 60% and approached as high as 75% [1]. The γ′ phase is composed predominantly of Al, Ti, Nb, and Ta combined with Ni, and precipitates from the γ matrix in an ordered $L1_2$ superlattice structure [7].

Metal carbides generally exhibit high thermal stability and constitute an important microstructural constituent of nickel-based superalloys. Carbides in these alloys are classified into two types: Type I and Type II. Type I carbides, designated as MC carbides, form through the reaction of carbon with reactive or refractory elements. They appear as discrete particles with an FCC crystal structure, with forming elements including Cr, Mo, W, Nb, Ti, Ta, and Hf. The formation of MC carbides within the γ matrix consumes solid-solution strengthening elements and refractory elements that promote γ′ precipitation, thereby disrupting the phase stability of the alloy during service [8]. Type II carbides, designated as $M_{23}C_6$, are derived from the decomposition of MC carbides in W-, Mo-, and Cr-containing nickel-based superalloys during low-temperature heat treatment or prolonged service at temperatures between 760 and 980 °C [9].

Complex multi-component nickel-based alloy systems are also prone to the formation of topologically close-packed (TCP) phases, which are highly detrimental to mechanical performance. These phases arise from structural evolution or chemical instability during alloy heat treatment or component service. Since refractory elements are the primary constituents of TCP phases, their formation depletes solid-solution strengthening elements—such as W, Mo, Cr, Co, and Re—from the γ matrix. The loss of these elements significantly reduces the alloy's

high-temperature fracture life [10]. Furthermore, the intrinsic brittleness of TCP phases degrades the overall toughness of the alloy, and platelet-shaped TCP phases can act as preferential crack initiation sites, ultimately leading to material failure [2].

1.1.2 Alloying Elements

The high strength of nickel-based alloys derives primarily from solid solution strengthening and precipitation hardening. Solid solution strengthening arises from interactions between solute atoms and dislocations within the matrix: alloying additions lower the stacking fault energy, thereby increasing the resistance to cross-slip and dislocation motion and raising the material's strength. Precipitation hardening, by contrast, relies on the interaction between dislocations and the uniformly distributed precipitate particles within the matrix [4]. Figure 1.2 lists the alloying elements most commonly used in nickel-based alloys.

Molybdenum, chromium, cobalt, and tungsten are common γ-matrix solid-solution strengtheners in nickel-based alloys. Cobalt content typically ranges from 2 to 15 wt%; cobalt additions improve microstructural stability, lower the γ′ solvus temperature, and reduce the stacking fault energy. Chromium is another widely used solid-solution strengthener, with a typical content of 10–20 wt%. The formation of a protective Cr_2O_3 scale enhances resistance to hot corrosion and oxidation; chromium additions also lower the γ′ solvus temperature and reduce the γ/γ′ interfacial energy.

Tungsten content is typically 5–8 wt%. Tungsten strengthens the matrix through solid solution and exhibits a synergistic strengthening effect with rhenium. Tungsten additions raise the incipient melting temperature but reduce microstructural stability and increase susceptibility to hot corrosion. Tungsten also impairs castability; the Ta/W ratio serves as an indicator of alloy castability, and increasing this ratio can reduce casting defects caused by convective instability during solidification. Nickel-based alloys may contain trace amounts of carbon, which suppresses the tendency of refractory elements (W, Re) to segregate to the dendrite cores, thereby improving microstructural stability.

Molybdenum content is typically 0–3 wt%. Molybdenum strengthens the γ matrix through solid solution and can substitute for tungsten on an atom-for-atom basis; molybdenum additions are generally believed to reduce the degree of tungsten segregation.

Aluminum, titanium, and tantalum are the principal γ′-strengthening elements and occupy equivalent sublattice sites in the $L1_2$ superlattice structure. Aluminum content is generally 3–6 wt%. Aluminum additions reduce alloy density and improve castability; aluminum also forms a protective Al_2O_3 scale, enhancing high-temperature oxidation resistance.

Titanium content is generally 0–5 wt%. Titanium is a γ′-forming element and raises the γ/γ′ interfacial energy; it typically reduces oxidation resistance but improves hot corrosion resistance. Lowering the Al/Ti ratio reduces oxidation resistance while enhancing hot corrosion resistance [11].

Tantalum content is approximately 4–12 wt%. Tantalum acts as both a γ′-forming element and a solid-solution strengthener. Tantalum additions raise the γ/γ′ interfacial energy and thereby improve castability.

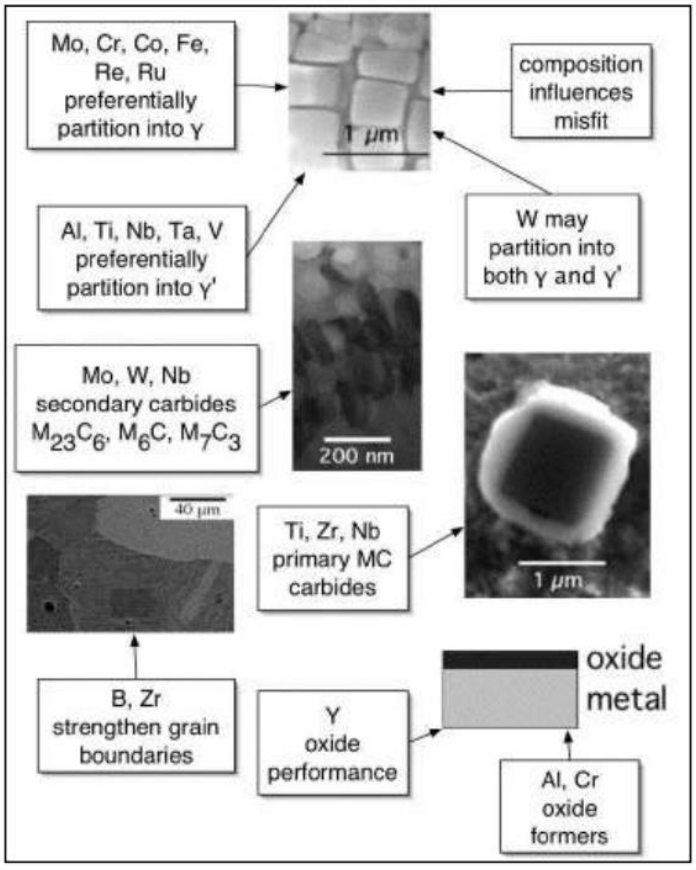


Figure 1.2 Effect of various alloying elements in nickel-based superalloys[2]

1.1.3 Micro-Segregation

During solidification, alloying elements inevitably exhibit various forms of segregation. Segregation not only governs microstructural evolution but also affects the consistency of mechanical properties. Because nickel-based superalloys are commonly produced by directional solidification—particularly for single-crystal components—microscale segregation between and within dendrites has become the primary focus of research on alloying element partitioning in these alloys.

Qin Jiarun et al. [12], in their study of the DD3 single-crystal superalloy, found that W and Co exhibit negative segregation, partitioning preferentially to the dendrite cores, with W showing the stronger tendency; Al, Ti, Mo, and Cr exhibit positive segregation, partitioning to the interdendritic regions, with Ti showing the strongest tendency.

Qi Yongshun et al. [16] reported that W, Re, and Co segregate to the dendrite cores, with segregation intensity following the order Re > W > Co; Al, Cr, and Ta segregate to the interdendritic regions, with intensity following Ta > Al > Cr. Among all solutes, Cr and Co were identified as weakly segregating elements.

Xu Zhiqiang et al. [17], in their study of the solidification behavior of the GH4151 alloy, found that W is a negatively segregating element that concentrates in the dendrite cores; Mo, Nb, and Ti are positively segregating elements that concentrate in the interdendritic regions, with segregation intensity following Nb > Ti > Mo; Co, Cr, and Al show essentially no segregation.

Zhou Xuefeng et al. [12] investigated the effect of withdrawal rate on solute microsegregation. Over the withdrawal-rate range of 10–80 μm/s, W and Re exhibited negative segregation, with the degree of segregation increasing as the withdrawal rate increased; Ta and Al exhibited positive segregation, with the degree of segregation likewise increasing with withdrawal rate; Mo exhibited positive segregation, but its segregation decreased with increasing withdrawal rate; Cr and Co had partition coefficients close to unity, and their segregation was insensitive to withdrawal rate.Mei Zihan et al. [14] examined microsegregation behavior over a wider withdrawal-rate range of 0–600 μm/s, finding that Al, Mo, Cr, Ti, and Ta

showed a tendency toward positive segregation, whereas Co and W showed a tendency toward negative segregation. W, Ti, and Ta exhibited the most severe segregation, and the overall degree of segregation for these elements first increased and then decreased with increasing withdrawal rate.

Gao Shuang et al. [15] investigated solute segregation in nickel-based alloys at varying Ta additions. Al, Ti, Mo, and Ta were found to partition to the interdendritic regions. With increasing Ta content, the partition coefficients (k) of Al, Ti, and Mo increased toward unity, indicating reduced segregation; the k value of Ta itself, however, decreased from 0.83 to 0.54, indicating intensified segregation. The segregation behavior of Co, which partitions to the dendrite cores, was largely unaffected by Ta content.

1.2 Directional Solidification Process

In directionally solidified nickel-based superalloys, grains preferentially grow along the <100> direction, and blades oriented along this axis achieve optimal mechanical properties, requiring heat flow to align accordingly during casting [1]. The two principal production methods are High-Rate Solidification (HRS) and Liquid Metal Cooling (LMC).

HRS (Bridgman method) generates a one-dimensional axial temperature field via a thermal barrier separating hot and cold zones, with the solid–liquid interface controlled by pulling rate. While suitable for small castings, the temperature gradient decreases significantly for larger components as solidification progresses [18]. LMC immerses the casting in a liquid metal coolant, enhancing convective heat dissipation and enabling higher pulling rates and larger temperature gradients at the solid–liquid interface compared to HRS [19].

Controlling the temperature gradient and crystal growth rate is critical for microstructure optimization. A high temperature gradient improves interface stability, but excessive solidification rates can cause columnar grains to deviate from the preferred <100> orientation. Prior studies confirm that impurity defects in single-crystal superalloys correlate strongly with furnace position, and that pulling speed significantly influences microstructure [20]; for DZ22 alloys, increased cooling rate refines both primary and secondary dendrite arm spacing [21]. Under complex, component-specific conditions, computer-aided solidification simulation has become the most effective tool for determining optimal directional solidification parameters.

1.3 Solidification Simulation

Grain size and crystallographic orientation are critical factors affecting turbine blade high-temperature performance. Computational simulation enables efficient prediction of crystal morphology by coupling temperature, flow, and solute fields across multiple physical domains.

Microscopic dendrite models primarily include the phase-field method and cellular automata (CA) models [20]. The phase-field method, based on free energy minimization, can simulate dendritic growth in multi-component alloys when coupled with thermodynamic databases [20,23], but its integration with flow fields remains limited [24].

For CA models, Rappaz (1993) proposed a 2D grain prediction model [25], which Gandin and Rappaz extended to 3D in 1997 [26] and further integrated with finite element temperature fields in 1999 to form the CAFE model [27]. Wang et al. (2003) developed the CAFD model for simulating temperature evolution, grain growth, and microsegregation in directionally solidified blades [28–30]. These models have since been broadly applied to single-crystal blades [31,32], steel and aluminum alloys [33–36], and laser cladding [37], with commercial implementations in ProCAST [33,35,38] and Calcosoft [36].Volume-averaged methods [39,40] capture coupling among temperature, flow, and composition fields but cannot resolve individual grain shapes or growth processes [41].

Temperature field accuracy is central to solidification simulation, relying on precise thermophysical data derived from thermodynamic calculations [42]. Software such as JMatPro and Thermo-Calc integrates thermodynamic databases for phase diagram calculation, solidification prediction, and alloy composition optimization [46,47], with extensive applications in nickel-based superalloys including segregation analysis and phase precipitation in low-cobalt alloys [48], Mar-M247 [49], Nimonic 105 [50], 718Plus [51], and γ/γ′ microstructure interpretation [52].

1.4 Research Objectives and Scope

Based on the fundamental principles of metal solidification, this study integrates computer-aided numerical simulation with directional solidification experiments to establish a

numerical model for the directional solidification of Ni-based superalloys. Numerical and experimental investigations are conducted to examine the evolution of the temperature field, solute segregation behavior, and resulting grain morphology during directional solidification. The main research activities are summarized as follows:

(1) Based on the fundamental principles of crystal nucleation and growth, heat transfer theory, and the capabilities of numerical simulation software, a cellular automaton–finite element (CAFE) model is developed to simulate the directional solidification of Ni-based superalloys.

(2) By combining directional solidification experiments with numerical process simulations, the effects of nucleation parameters, grain growth parameters, and thermal boundary conditions on the simulated grain morphology are systematically investigated and compared. Furthermore, the evolution of the temperature field within the alloy specimens during directional solidification is obtained through numerical simulations.

(3) The elemental compositions of the specimens obtained from directional solidification experiments are measured to investigate the segregation behavior of individual alloying elements in Ni-based superalloys. Particular attention is paid to the relationship between the evolving temperature field and elemental macrosegregation during directional solidification.

Chapter 2: Fundamentals

2.1 Solute Redistribution

Materials researchers have developed comprehensive theoretical frameworks to describe solute diffusion during solidification under one-dimensional steady-state temperature fields and the resulting segregation patterns [59]. The equilibrium solidification model assumes complete diffusion between solid and liquid phases, yielding a homogeneous single-phase solid solution structure. Under this assumption, solute distribution follows the Lever Rule, with interfacial concentrations described by Equations (2.1) and (2.2).

$$C_S^* = \frac{C_0 k_0}{1-f_s(1-k_0)} \tag{2.1}$$

$$C_L^* = \frac{C_0}{1-f_s(1-k_0)} \tag{2.2}$$

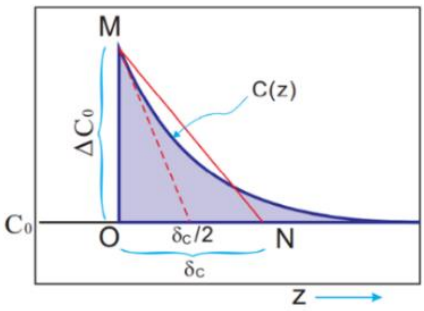


Figure. 2.1 Solute boundary layer[59]

Solute segregation behavior during solidification can be classified into three models according to the extent of liquid-phase mixing: Complete liquid-phase mixing: Strong convection or agitation ensures compositional homogeneity throughout the liquid phase. Diffusion-only liquid phase: No convective mixing occurs, representing minimal liquid-phase homogenization. Combined diffusion and partial mixing: Both diffusion and limited convective mixing contribute to solute redistribution. Assuming negligible solid-state diffusion but complete liquid-phase mixing (due to vigorous convection), the solute distribution follows the Scheil equation (Equation 2.3).

$$C_S^* = k_0 C_0 (1-f_s)^{k_0-1} \tag{2.3}$$

When only diffusion governs liquid-phase transport, solute redistribution reduces to a diffusion-controlled problem. Under steady-state advancement of the solid-liquid interface, the Tiller equation (Equation 2.4) describes the non-uniform solute distribution in the liquid phase.

$$C_L(x)=C_0(1+\frac{1-k_0}{k_0}\exp(-\frac{R}{D_L}x)) \tag{2.4}$$

As illustrated in Figure 2.1, at $x = \infty$, the concentration [C] approaches C_0. Defining $\delta(\delta = D_L/R)$ as the characteristic thickness of the enrichment layer, larger δ values (or smaller diffusion coefficients k_0) intensify solute enrichment, steepen the concentration gradient at the interface, and reduce the boundary layer thickness. When partial liquid-phase mixing occurs, an effective distribution coefficient k_E governs the solute distribution pattern.

$$k_E=\frac{C_S^*}{C_0}=\frac{k_0}{k_0+(1-k_0)\exp(-\frac{R}{D_L}\delta)} \tag{2.5}$$

$$C_S^*=k_E C_0(1-f_s)^{k_E-1} \tag{2.6}$$

Under one-dimensional directional solidification conditions with negligible solid-state diffusion, the compositional evolution can be divided into three distinct stages:

1. Initial transient stage: For $k < 1$ under local equilibrium, the initial solid composition (kC_0) is lower than the nominal composition C_0. Solute rejection from the advancing solid phase creates a diffusion-enriched layer ahead of the solid-liquid interface. This enrichment depresses the local liquidus temperature, progressively increasing the equilibrium concentration and decreasing the interface temperature until solute rejection and diffusion reach steady state.

2. Steady-state stage: The solid-phase composition equals C_0, with the solute flux rejected by solidification balanced by diffusive transport away from the interface.

3. Final eutectic stage: As solidification approaches completion, accumulated solute concentrates in the residual liquid with limited diffusive escape. The interfacial equilibrium concentration rises progressively until reaching the eutectic composition C_E.

2.2 Phase Diagram Calculations

2.2.1 Phase Diagram computational models

Phase diagram calculations integrate thermodynamic theory with numerical simulation methods. Several computational models are employed depending on the phase characteristics [43,44]:

1. Stoichiometric phases: Line compound models that progressively increase system

complexity.

2. Disordered solid solutions: Conventional solution models for phases without fixed stoichiometry.

3. Ordered phases with solubility: Sublattice models accounting for site occupancy preferences.

4. Order-disorder transitions: Hybrid energy models combining conventional solution and sublattice approaches.

The Gibbs free energy of a phase φ in a multi-component system is expressed as:

$$G^{\varphi}=G^{0}+G^{ideal}+G^{xs} \tag{2.7}$$

where G^{0} denotes the reference state free energy of a single phase, G^{ideal} represents the ideal mixing entropy contribution, and G^{xs} accounts for excess free energy arising from non-ideal interactions.Multi-component phase diagram calculations rely on thermodynamic descriptions of constituent binary systems. Two principal modeling approaches exist—the equivalent model and the sublattice model—both reducible to Equation (2.7). Under the assumption that interactions in multi-component alloys are significantly weaker than in binary subsystems, these models predict multi-component phase equilibria by extrapolating from binary thermodynamic data.

$$G=\sum_{i=1}^{p} n_i G_i^{\varphi}=\text{minimum} \tag{2.8}$$

In the equation, `$n_i$ ` denotes the number of moles of phase i, and `$G_i^{\varphi}$ ` denotes the Gibbs free energy of phase i. When the minimum Gibbs free energy for a given state is reached, the chemical potential `$\mu_n$ ` of each component n is the same across all phases and is related to the Gibbs free energy by the following equation:

$$G=\sum_{i=1}^{n} \mu_i x_i \tag{2.9}$$

This thermodynamic equilibrium condition forms the foundation for computational phase diagram (CALPHAD) methods, enabling prediction of phase stability, composition, and fraction under specified temperature and composition conditions.

2.2.2 Micro-segregation Model

The Lever rule and the Scheil model represent the extreme cases of good diffusion and

complete lack of diffusion of solute atoms in the solid phase during solidification, respectively. The general equation for mass conservation during the diffusion of a single solute in a binary alloy is:

$$\left(C_l\text{-}C_s^*\right)df_s=f_s dC_l+\frac{\delta_s}{2L}dC_s^* \tag{2.10}$$

where L is the system length, specifically referring to the distance over which diffusion behavior needs to be modeled in the target model:

$$\frac{C_l}{C_0}=(1\text{-}uf_s)^{-p/u}C_s^* \tag{2.11}$$

$u=1-2\alpha' k$,$p=1-k$ and $C_s^* = kC_l$, and α' is obtained from the following calculation:

$$\alpha'=\alpha\left[1-\exp\left(-\frac{1}{\alpha}\right)\right]-0.5\exp\left(-\frac{1}{2\alpha}\right) \tag{2.12}$$

In the equation,α is the Fourier criterion ,$\alpha=\frac{D_s t_f}{L^2}$, and t_f in the Fourier criterion is the local solidification time. For a dendritic system of length L, the secondary dendrite armd_2 is generally taken, and$\Delta T'=m(C_l^*-C_l^m)$, so we have:

$$\alpha=\frac{4D_s m(C_l^*-C_l^m)}{d_2{}^2\dot{T}} \tag{2.13}$$

The Fourier criterion introduces the concepts of local cooling rate ($\dot{T}$) and local solidification time (t_f). For directional solidification, the definition of the local cooling rate ($\dot{T}$) is shown in Figure 2.2. Assuming an average temperature gradient exists for the slurry zone model of directional solidification, this temperature gradient can be defined as the distribution of the temperature difference between the high temperature at the top of the model (T_h) and the low temperature at the bottom (T_c) along the height (H) of the model.

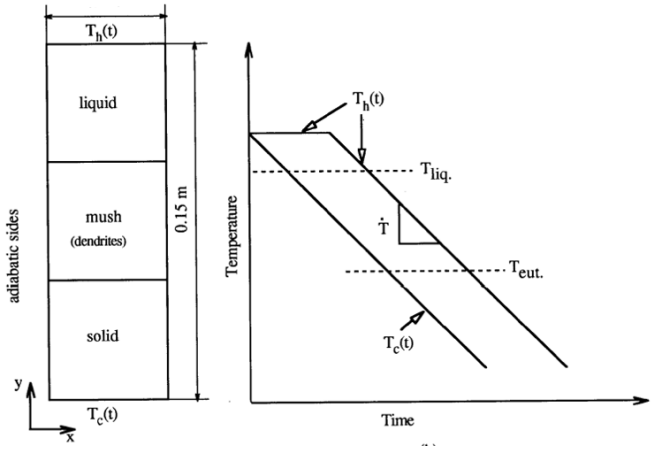


Figure. 2.2 Schematic diagram of the local cooling rate [66]

$$G=(T_h-T_c)/H \tag{2.14}$$

$$\dot{T}=GR=\frac{dT}{dy}\cdot\frac{dy}{dt}=\frac{dT}{dt} \tag{2.15}$$

$$t_f=-\frac{\Delta T'}{\dot{T}} \tag{2.16}$$

From the aforementioned theoretical analysis, the relationship between α and α' corresponds to distinct diffusion models: when the mass Biot number (α) is large, diffusion follows the Lever rule; conversely, when α is small, diffusion approximates the Scheil model. Specifically, when $\alpha \leq 0.1$, $\alpha' \approx \alpha$, indicating minimal solid-state diffusion consistent with Scheil assumptions. For intermediate values where α represents a moderately large number, α' = 0.5. In the micro-diffusion model, when no diffusion occurs within the solid phase (α = α' = 0), the simplified micro-diffusion model reduces to the Scheil equation. Conversely, when α is sufficiently large (typically α > 100), complete solid-state diffusion can be assumed, corresponding to the Lever micro-diffusion model.

The Lever rule applies to the majority of high-quality castings but is unsuitable for continuous casting processes. It is primarily applicable to small-scale solidification systems, such as elemental microsegregation in ductile iron containing trace carbon. The phase diagram calculations based on the Lever rule assume complete diffusion in both solid and liquid phases at every time step during cooling. Consequently, all phases achieve perfect equilibrium without compositional gradients existing within them.

The Scheil micro-diffusion model assumes negligible diffusion within the solid phase, with thermal equilibrium established only at the solid-liquid interface. The theoretical foundation of the Scheil model lies in local equilibrium at the dendritic scale. This model presumes that the coexisting dendritic regions within the solid-liquid interface maintain uniform temperature fields, with complete homogenization of liquid-phase composition and uniform solid-phase mixing. After solidification, the composition remains unchanged, neglecting back-diffusion governed by Gibbs-Thomson effects or solute redistribution driven by concentration gradients. This simplification eliminates the need to consider cooling rates and dendritic morphological parameters [55-58].

Phase diagram calculations based on the Scheil model yield the lowest final solidification

(freezing range) temperature [54]. The Scheil model neglects solid-state diffusion and disregards dendritic microscale structural evolution. For solid-phase diffusion (back-diffusion) calculations, reverse diffusion is typically assumed to follow dendritic morphologies—cylindrical, plate-like, or spherical geometries—with solutions obtained for the system of equations at a given time increment dt [58].

This theoretical framework enables selection of appropriate solidification models based on process conditions—cooling rates, system dimensions, and desired prediction accuracy—thereby improving the fidelity of phase diagram calculations for practical alloy systems.

2.3 Macro-Segregation Calculations

Key parameters for calculations in the ProCAST software’s Macro-Segregation and modules include: diffusion coefficients D_l^m of each solute in the liquid phase; element-solid-liquid partition coefficients and liquidus line slopes calculated from phase diagrams; and diffusion coefficientsD_s^m of each element in the solid phase. Solute field calculations adhere to the following conservation relationships:

Mass conservation between the solid and liquid phases:

$$\frac{\partial(f_l\rho_l)}{\partial t}+\nabla\cdot\left(f_l\rho_l v_l\right)=-\frac{\partial(\rho_s f_s)}{\partial t} \tag{2.17}$$

Mass conservation of solutes in the liquid phase:

$$f_l\rho_l\frac{\partial c_l^m}{\partial t}+f_l\rho_l v_l\cdot\nabla c_l^m=\nabla\cdot\left(f_l\rho_l D_l^m\nabla c_l^m\right)+(c_l^m-c_{sl}^m)\frac{\partial\left(\rho_s f_s\right)}{\partial t}+\frac{S\rho_s D_s^m}{l}(c_s^m-c_{sl}^m) \tag{2.18}$$

Mass conservation of solutes in the solid phase:

$$f_s\rho_s\frac{\partial c_s^m}{\partial t}=(c_{sl}^m-c_s^m)[\frac{\partial(\rho_s f_s)}{\partial t}+\frac{S\rho_s D_s^m}{l}] \tag{2.19}$$

Where: m is the solute element; $l=f_s d_2/6-$ is the diffusion length; $S=2/d_2-$ is the composition in the inter-phase region; d_2- is the distance between secondary dendrite arms. At the microscopic scale, the transfer of solute m across the solid-liquid interface caused by the concentration gradient in the solid phase can be regarded as the product of the concentration S in the interface region and the average solute flux of element m at the interface.The flux of solute across the solid-liquid interface is proportional to the difference between the average composition at the interface（c_{sl}^m） and the average composition in the solid phase（c_s^m）.

This proportionality constant is calculated using the solid-phase diffusion length (l), the solid-phase density（ρ_s） , and the solid-phase diffusion coefficient of solute m (D_s^m). This function ensures that the interfacial concentration and the diffusion length are related to the solid fraction, without requiring any assumptions about the diffusion rate of solute element m in the solid.

2.4 Growth Kinetics

During liquid solidification, both the temperature field and the solute concentration field undergo continuous and mutually coupled changes. The dendritic growth mechanism is therefore complex, requiring simultaneous consideration of heat transfer, solute diffusion, capillary effects, and interfacial dynamics.

Ivantsov [60] proposed in 1947 a steady-state paraboloid-of-revolution diffusion solution for needle crystals, applicable to the calculation of both the thermal and solutal fields. The specific analytical form of the Ivantsov function is given as Eq. (2.20):

$$\Omega=\mathrm{Iv}(P_e)=P_e\exp(P_e)E_1(P_e) \tag{2.20}$$

$\mathrm{Iv}(P_e)$ for the Ivantsov function; here, $E_1(P_e)$ denotes the first exponential integral; P_e denotes the Peclet number; and Ω denotes the dimensionless supersaturation.

Langer and Müller-Krumbhaar [62] subsequently proposed the marginal stability theory (also referred to as LM-K theory). Through experimental analysis they obtained the stability constant δ, which determines the dendrite tip radius r and growth velocity v under a given undercooling ΔT.

$$r=\lambda_i=2\pi(\delta_c s)^{\frac{1}{2}} \tag{2.21}$$

In the equation, r is the radius of the dendrite tip; the thickness of the diffusion layer at the interface front $\delta_c=D/V$; and the capillary length $s=\Gamma/\Delta T_0$.

Lipton et al. [63] established the LGK dendrite growth model on the basis of Ivantsov steady-state diffusion theory, applicable to the free-dendrite growth regime at small undercooling and small Péclet numbers.

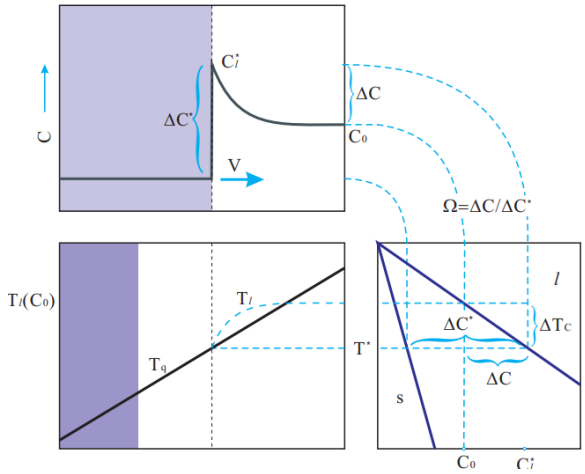


Figure. 2.3 Compositional Undercooling at the Tip of an Isolated Dendrite[59]

As illustrated in Fig. 2.3, a positive temperature gradient always exists ahead of the solid–liquid interface during directional solidification. Enrichment of low-melting-point solutes in the liquid phase depresses the liquidus temperature, giving rise to a finite compositional undercooling under the positive thermal gradient (mG_c-G=Φ). The total undercooling at the growing dendrite tip comprises three contributions: thermal diffusion undercooling (ΔT_t), solutal enrichment undercooling (ΔT_c), and curvature undercooling arising from the Gibbs–Thomson effect (ΔT_r). Invoking the Ivantsov function:

$$\Delta T_t=\theta_t I(P_t) \tag{2.22}$$

$$\Delta T_r=2\Gamma/R \tag{2.23}$$

$$\Delta T_c=mC_0[1-[1-pI(P_c)]^{-1}] \tag{2.24}$$

$$\Delta T=\Delta T_c+\Delta T_t+\Delta T_r=\theta_t I(P_t)+mC_0\left[1-[1-pI(P_c)]^{-1}\right]+2\Gamma/R \tag{2.25}$$

According to the theory of critical stability, dendrites grow to the tip size determined by the stability limit,$R_s=\lambda_i=2\pi(\delta_c s)^{\frac{1}{2}}$. Based on the dendrite tip growth kinetics derived from the classical LGK model, we obtain:

$$v(\Delta T)=\frac{D_l m_l c_l^*(k-1)}{\pi^2\Gamma}[Iv^{-1}(\Omega)]^2 \tag{2.26}$$

where Ω is the supersaturation，$\Omega=\frac{\Delta C}{\Delta C^*}$; Iv^{-1} is the inverse of the Ivantsov function , ,$Iv^{-1}(\Omega)=0.4567(\frac{\Omega}{1-\Omega})^{1.195}$.

2.5 Cellular Automation

2.5.1 Cellular Automation algorithm

The Cellular Automation algorithm follows the following rules: the model space is divided into uniformly arranged cells of equal size; the neighboring cells of each cell must be defined (e.g., nearest neighbors or second-nearest neighbors); each cell has different variable values (temperature, crystallographic direction) and states (liquid, solid);At each time step, the process by which a cell transitions from the liquid to the solid state depends on the variable values and states of its neighboring cells[25] . In a three-dimensional model, the cell shape is an ellipsoidal octahedron, and the cell size is defined as the distance from its center to any (111) face, which numerically corresponds to the time integral of the grain tip growth rate:

$$L_v^t=L_v^{t_c}+\frac{1}{\sqrt{3}}\int_{t_c}^{t} v(\Delta T_v(\tau))\cdot d\tau \qquad (2.27)$$

In this equation, `$L_v^{t_c}$ ` corresponds to the initial dimensions of the cell when the eccentric octahedron captures it; at the moment of nucleation, `$L_v^{t_n}=0$ ` and `$C_v\equiv v$ `. The `<100>` direction corresponds to the diagonal of the octahedron, which forms an Euler angle (`$\phi_1$, $\phi_2$, $\phi_3$ `) with respect to the (X, Y, Z) axes of the cubic cell lattice.Once the outer contour of the octahedron of cell$v$ encloses the center of its neighboring cellμ , the center of the cell μ and the initial dimensions of its corresponding eccentric octahedron are calculated in a manner similar to that of a two-dimensional cellular automaton. This cellular automaton algorithm enables the realization of the feature where dendrites exhibit a specific preferred growth direction.

2.5.2 Nucleation and Growth

In this study, the nucleation density used in the cellular automaton simulation follows a Gaussian distribution, mathematically expressed as `$N(\mu,\sigma^2)$ `, with its probability density function shown in Equation (2.28).

$$\frac{dn}{d(\Delta T)}=\frac{n_{max}}{\Delta T_\sigma\sqrt{2\pi}}e^{-[(\Delta T-\overline{\Delta T})/\sqrt{2}\Delta T_\sigma]^2} \qquad (2.28)$$

The increment in grain density induced by the supercooling increment $d(\Delta T)$ is$\frac{dn}{d(\Delta T)}$, and the total nucleation density $n(\Delta T)$ is the integral of the nucleation density distribution

over the range at a given supercooling:

$$n(\Delta T)=\int_0^{\Delta T}\frac{dn}{d(\Delta T')}[1-f_s(T')]d(\Delta T') \quad (2.29)$$

wheref_s denotes the solid-phase volume fraction.

The number of grains formed during the microtime step δt ,δn_v , is the increment in grain density multiplied by the total volume.

$$\delta n_v=n_v[\Delta T+\delta(\Delta T)]-n_v(\Delta T)=\int_0^{\Delta T+\delta(\Delta T)}\frac{dn}{d(\Delta T')}d(\Delta T') \quad (2.30)$$

The calculation of the local undercooling ΔT requires consideration of the radius of curvature and compositional undercooling. The Gibbs-Thompson equation for a melt-growth system is[59] :

$$T_\phi=T_m+mC_{L\phi}-\Gamma_m T_m(\frac{1}{r_1}+\frac{1}{r_2}) \quad (2.31)$$

$$\Delta T=T^*-T_\phi \quad (2.32)$$

where$(\frac{1}{r_1}+\frac{1}{r_2})$ is the radius of curvature;Γ is Gibbs −

Thompsoncoefficients，$\Gamma=\Gamma_m\cdot T_m$ (Γ_m is the average capillary length,T_m is the freezing point at zero curvature, and Γ_m depends on the interfacial free energy at the solid-liquid interface).

The crystal growth rate is calculated using the LGK growth kinetics model:

$$v(\Delta T)=\frac{D_l m_l c_l^*(k-1)}{\pi^2\Gamma}[Iv^{-1}(\Omega)]^2 \quad (2.33)$$

The liquidus slope and solid-liquid phase distribution ratio for multi-component alloys are obtained as the weighted average of the corresponding coefficients for each individual solute element:

$$C_0=\sum C_i \quad (2.34)$$

$$m_l=\sum(m_i C_i)/C_0 \quad (2.35)$$

$$k_l=\sum(m_i C_i k_i)/(mC_0) \quad (2.36)$$

In the equation, C_i represents the mass fraction of element i in the alloy; m_i represents the liquidus slope of element i in the alloy; and C_i represents the solid-liquid partition coefficient of element i. ProCAST assumes that the undercooling at the dendrite tip is proportional to the growth rate, as given by $v(\Delta T)=a_2\Delta T^2+a_3\Delta T^3$. By entering the parameters

in Table 2.1, the value of a_2、a_3 can be automatically calculated.

Table 2.1 Parameters Required for Growth Kinetic Calculation

Parameter Symbol	Definition and Units
C_i	Mass fraction of element i (wt%)
m_i	Liquidus slope of element i (K/(wt%))
k_i	Solid-liquid partition coefficient of element i
D_{li}	Liquid-phase diffusion rate of element i (m^2/s）
Γ	Gibbs-Thompson coefficient（m·K）

2.5.3 Competitive Growth

The growth rate of a grain is closely related to its own crystallographic orientation and the temperature field in which it is located. In the cellular automaton model, all new grains are generated at cell nodes and possess three principal crystallographic directions relative to the system coordinate system. Figure 2.4 shows the outer contours of a single grain after growth, with crystallographic directions (20° , 20° , and 20°). In (a1) and (a2), the temperature field is uniform, while in (b1) and (b2), temperature gradients are present. The three-dimensional cellular automaton model assumes that the grain expands only along the [001], [010], and [001] crystal directions, with the expansion rate depending on the magnitude of the temperature gradient in each growth direction and the angle relative to the coordinate system.

Figure 2.5 shows a cross-sectional view of the crystal outlines along the <100> crystal direction, illustrating the competitive growth of secondary dendrite arms extending from two mutually perpendicular primary dendrite stems, resulting in the grain outline shown in Figure 2.4(a2).

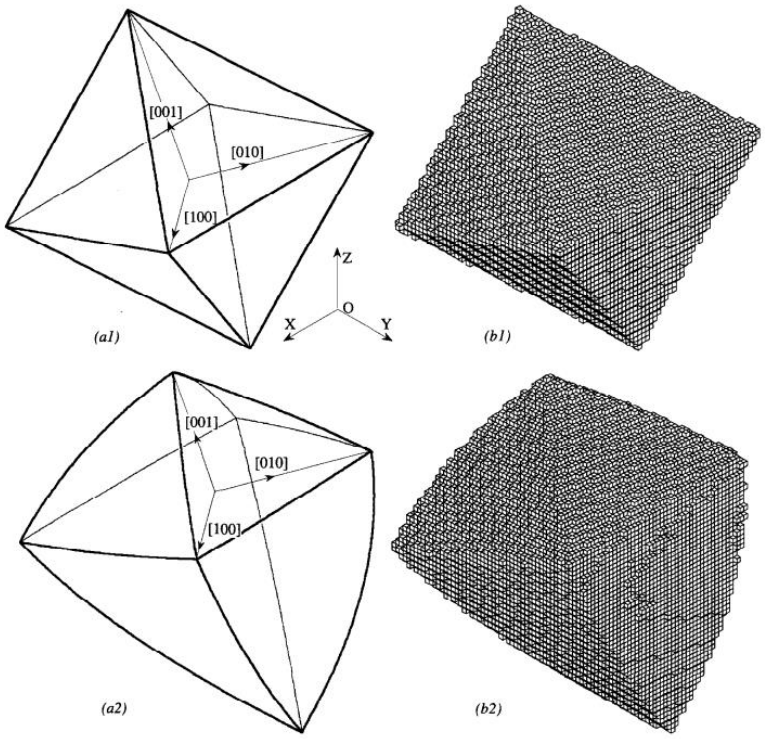


Figure. 2.4 Envelopes of a Single Dendritic Grain[26]

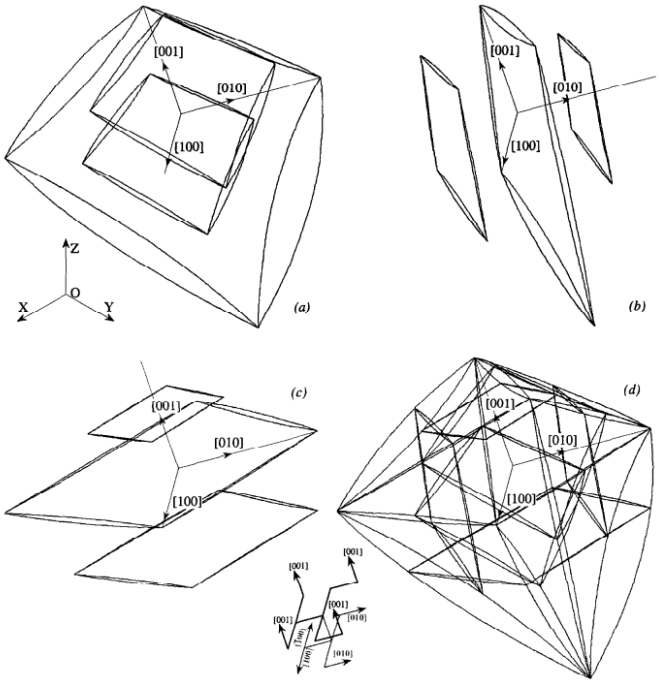


Figure. 2.5 Cross-sections of the Envelopes of a Single Dendritic Grain[26]

Figure 2.6 shows a schematic diagram of the competitive growth of columnar grains . In the figure, the crystallographic orientation of the columnar grains in the central region forms a large angle of deviation with the direction of heat flow in the temperature field system. If the liquidus isotherm T_L moves at a velocity V_L under a temperature gradient G, the length of the

undercooled zone at the leading edge of the dendritic crystal tips on both sides—where the growth direction is parallel to the heat flux direction—is $\Delta z_0 = \Delta T(V_L)$ /G, which is relatively short. Columnar crystals with a larger deviation in crystallographic orientation require a higher dendrite tip growth rate (v_θ), which implies a greater supercooling ($\Delta T_\theta = \Delta T(v_\theta)$) and a longer supercooled zone; consequently, columnar crystals with crystallographic orientations closer to the heat flow direction are more likely to be retained during competitive growth.

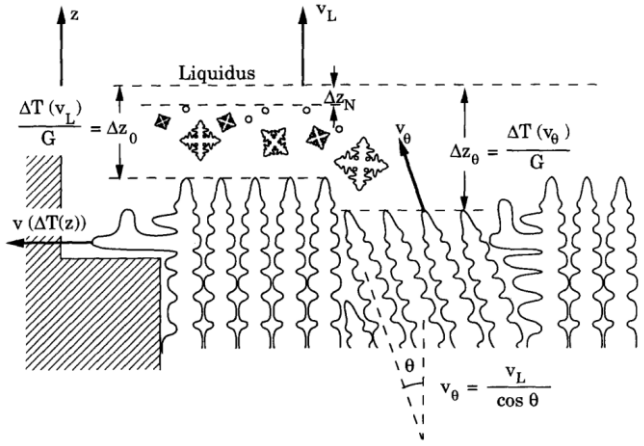


Figure 2. 6 Schematic diagram of competitive growth[64]

2.5.4 Coupled Scheme

Finite Element (FE) cells are used to calculate the macroscopic heat flow in castings, while the cellular automaton (CA) grid serves as the smallest computational unit in the cellular automaton model, having a relatively smaller size than finite elements, and is used to compute microstructural morphology. CAFE coupled computation requires combining FE calculations with the CA grid within a single model, predicting the evolution of microstructure through the macroscopic temperature field and latent heat release[64]. The main coupling modes between the CA grid and FE are full coupling and weak coupling.

The calculation of temperature in the weakly coupled mode is based on a macroscopic finite element mesh. Weak coupling can be simply defined as the interpolation of the temperatureT_v of the cell v from the temperature T_n at finite element node n. Since the macro-time step Δt of the finite element calculation does not meet the requirements of the cellular automaton calculation, Δt it is divided into micro-time steps δt during cellular automaton operations. When performing cellular automaton calculations, for cells located within a finite element element,$t_m \in [t, t+\Delta t]$.

$$T_v^{t_m}=\sum_{nE_v}\phi_n(X_v)T_n^{t_m} \quad (2.37)$$

$$T_n^{t_m}=T_n^t\frac{t+\Delta t-t_m}{\Delta t}+T_n^{t+\Delta t}\frac{t_m-t}{\Delta t} \quad (2.38)$$

$$t_m\in[t,t+\Delta t]$$

The $\phi_n(X_v)$ in the equation is the shape function measured at the center of the cell X_v. The nucleation and growth calculations of the cellular automaton are performed in micro-time steps until $t_m=t+\sum\delta t=t+\Delta t$.

The advantage of the weakly coupled mode is that it can be performed after the finite element calculation is complete; crystal structure calculations can be carried out as a post-processing step using the finite element mesh and temperature history files.

The fully coupled mode is a more accurate coupling method, where the macro-time step Δt from the finite element simulation is similarly divided into micro-time steps δt, and the temperature and solid-state ratio of the finite element nodes are recalculated after each micro-time step. The enthalpy change for each node within a macro-time step Δt is linear:

$$\delta H_n=\frac{\delta t}{\Delta t}\Delta H_n \quad (2.39)$$

The fully coupled mode assumes that the heat release rate at each node remains constant within a macro-time step Δt. For finite element nodes where the temperature is above the liquidus or below the solidus, the new temperature $\{T_n^{t+\Delta t}\}$ can be simply derived from the enthalpy change $\{H_n^{t+\Delta t}=H_n^t+\Delta H_n\}$. When the finite element is in the solidification state, the following energy conservation must be satisfied within a micro-time step δt:

$$\Delta H_n=\rho c_p\delta T_n-L\delta f_{s,n} \quad (2.40)$$

In this equation, ρc_p and L represent the volumetric specific heat and latent heat of fusion, respectively. The change in temperature at the finite element node δT_n is automatically determined once the solid-phase transformation ratio $\delta f_{s,n}$ is calculated, and $\delta f_{s,n}$ is obtained by summing the relevant terms:

$$\delta f_{s,n}=\frac{\sum_v\phi_n(X_v)\delta f_{s,n}}{\sum_v\phi_n(X_v)} \quad (2.41)$$

New temperature values are calculated at each micro-time stepδt, $\{T_n^{t_m+\delta t}=T_n^{t_m}+\delta T_n\}$; at the end of each macro-time step ΔT, new temperatures $\{T_n^{t+\Delta t}\}$ and new solid-phase fractions $\{f_{s,n}^{t+\Delta t}\}$ are obtained at the finite element node positions, and the next macro-time step ΔT is

performed. Within a micro-time step δt, there are three possible scenarios for changes in the solid-phase fraction ($\delta f_{s,v}$) within a cell v:

(1) No change in the solid-phase volume fraction

The cell is already fully solidified at the start of the micro-time step or remains in the liquid phase throughout the entire micro-time step; in this case, $\delta f_{s,v}=0$.

(2) Transition from the liquid state to the paste state

Due to nucleation occurring within the microtime step δt or the cell being enveloped by the growth of neighboring cells, the cell's state indicator changes from the liquid state (0) to a non-zero indicator. The increment in the solid-phase fraction is calculated using the Scheil micro-diagonalization model based on the cell's actual temperature $T_v^{t_m}$:

$$\delta f_{s,v}=1-\left(\frac{T_v^{t_m}-T_m}{T_L-T_m}\right)^{1/(k-1)} \tag{2.42}$$

In this equation, T_L denotes the liquidus of the alloy, T_m is the melting point of the pure metal, and k is the partition coefficient. This equation is a truncated form of the Scheil relationship and does not fully adhere to solute conservation; however, it provides a good approximation for most solidifications with small undercooling.

(3) Solidification of the paste-state cell

A cell v with a non-zero state indicator continues to solidify according to the Scheil microsegregation model:

$$\delta f_{s,v}=\frac{-\delta H_v}{\rho c_p(T_L-T_m)(k-1)[1-f_{s,v}^{t_m}]^{(k-2)}+L} \tag{2.43}$$

The enthalpy change δH_v of cell v is derived from a linear interpolation of the enthalpy change at the finite element node to which the cell belongs. When the cell temperature drops to the eutectic temperature T_E, an isothermal transformation is assumed to occur, and the cell temperature is maintained at T_E. This process continues until the solid fraction reaches 1, during which $\delta f_{s,v}$ is calculated as: $\delta f_{s,v}=-\delta H_v/L$.

The simulation in this paper employs a fully coupled mode.

2.6 Chapter Summary

This chapter presents the fundamental principles of solute redistribution, phase diagram calculation, the governing equations and constraints for macrosegregation modeling in

numerical simulations, and the theoretical framework of grain morphology simulation using the cellular automaton–finite element (CAFE) method. The main conclusions are summarized as follows:

(1) Based on the principles of solute redistribution, solute elements with partition coefficients $k < 1$ are continuously rejected into the liquid phase as the solid–liquid interface advances during the directional solidification of Ni-based superalloys. Under conditions of sufficient mixing in the liquid phase, these solute elements exhibit an approximately exponential macrosegregation profile along the solidification direction of the specimen. In contrast, when liquid-phase mixing is limited, a solute-enriched boundary layer develops ahead of the solid–liquid interface for elements with $k < 1$. Variations in the local fluid flow within this boundary layer can significantly influence the extent of macrosegregation. Therefore, elemental macrosegregation during the directional solidification of Ni-based superalloys is governed by the combined effects of solute redistribution and fluid flow.

(2) Phase diagram calculation integrates thermodynamic principles with computational modeling techniques. Its theoretical foundation is closely related to the diffusion behavior of solute atoms in the solid phase. Depending on the assumptions regarding solid-state diffusion, solidification calculations can be performed using the Scheil model or the lever-rule model. The results of these calculations provide an essential basis for investigating elemental segregation behavior and for conducting numerical simulations of grain morphology during alloy solidification.

(3) In the subsequent chapters, the cellular automaton–finite element (CAFE) method will be employed to simulate grain characteristics, including grain number and crystallographic orientation deviation, during the directional solidification of Ni-based superalloys. This chapter briefly introduces the fundamental principles of grain nucleation and growth within the CAFE framework, providing the theoretical basis for modeling preferred crystallographic orientation and competitive grain growth.

Chapter 3 Microstructural Analysis and Simulation of Directional Solidification

3.1 Directional Solidification Experiments

The experimental alloy CM247LC had the following composition (wt%): Ni-8.1Cr-5.6Al-1.0Ti-9.2Co-0.007Zr-0.015B-0.07C-3.2Ta-9.5W-0.5Mo. The alloy was melted in a vacuum furnace and heated to 1580°C, then poured into a mold shell preheated to 1560°C. After a 120-second holding period, a Ø20 mm × 180 mm circular cross-section bar was produced using the vertical pulling method in a vacuum directional solidification furnace (ALD directional solidification furnace) at a pulling rate of 41 μm/s. The hot and cold zones of the apparatus were maintained at 1560°C and 120°C, respectively. After solidification, the mold shell was cooled to room temperature (20°C) and the test bar was extracted.The outer surface of the bar was ground, polished, and subjected to macroscopic etching.

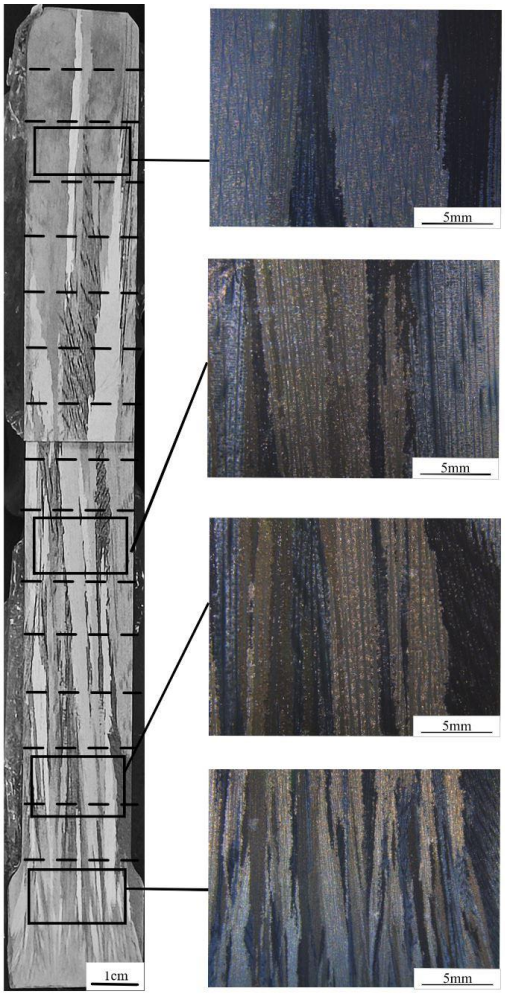


Figure. 3.1 Grain morphology of directional solidification test specimens

As shown in Figure 3.1, the grain diameter of the test bar increases along the length from the quenching end (water-cooled plate): at 1 cm it is primarily 0.5–1 mm; at 5 cm, 1–2 mm; at 9 cm, ~2 mm; at 13 cm, ~2 mm; and at 17 cm, 2–4 mm. In the Bridgman directional solidification process, the high specific heat capacity of the alloy results in a pronounced temperature gradient only near the initial quenching end, while subsequent cooling relies mainly on radial radiative heat loss, producing a reduced temperature gradient. The combined effects of the gradually decreasing temperature gradient, intrinsic columnar grain orientation deviations, and competitive growth of neighboring crystals lead to the gradual increase in grain size observed on the outer surface of the test bar.During the experiment, 8 test bars were solidified simultaneously, partially obstructing circumferential radiative heat dissipation for each individual bar. This produced asymmetrical grain distributions, with one side exhibiting fewer grains and the opposite side more grains, as shown in Figure 3.1. Figure 3.2 presents backscattered electron (BSE) images of a longitudinal cross-section at a specific height of the test bar. The calculated secondary dendrite arm spacings for Figure 3.2(a), (b), (c), and (d) are 43 μm, 89.5 μm, 91.6 μm, and 91.6 μm, respectively.Based on the numerical relationship $\lambda=\alpha(\Delta T')^{n}(GR)^{-n}$[21].It can be inferred that the temperature conditions in regions other than the initial end are relatively similar.

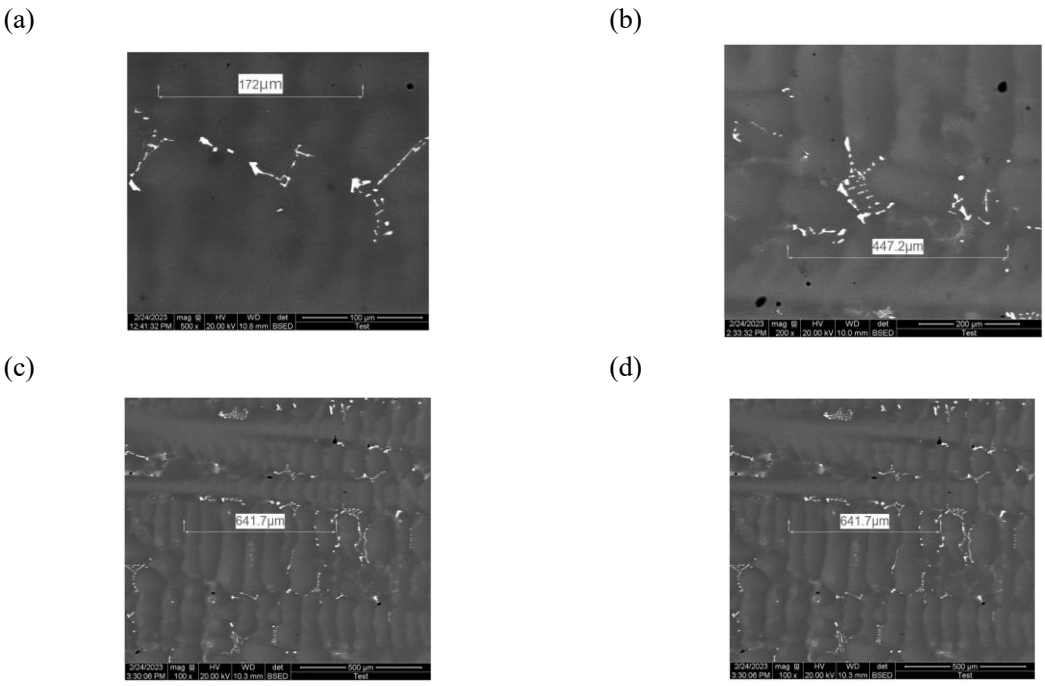


Figure 3.2 Microstructural morphology of the directional solidification test bar cross-section: (a) at 2 cm; (b) at 5 cm; (c) at 9 cm; (d) at 17 cm

3.2 Directional Solidification Simulation Model and Parameters

For the directional solidification experiments described in the previous section, the following simplifications and assumptions were applied in the simulation study:

① The melt temperature is assumed to be uniform prior to solidification.

② Filling is assumed to occur instantaneously; forced convection during pouring is neglected, and the molten metal is considered stationary at the onset of solidification.

③ Ingot shrinkage during solidification and associated liquid-phase flow are neglected.

④ The interfacial heat transfer coefficient and boundary temperatures are treated as constants.

⑤ Variations in circumferential radiative heat loss are ignored.

⑥ The gating system is thermally isolated from the external environment by a vacuum and maintains a constant temperature.

The primary components involved in directional solidification for investment casting are illustrated in Figure 3.3, comprising four parts: the molten alloy, water-cooled plate, the mold shell, and the cover box.

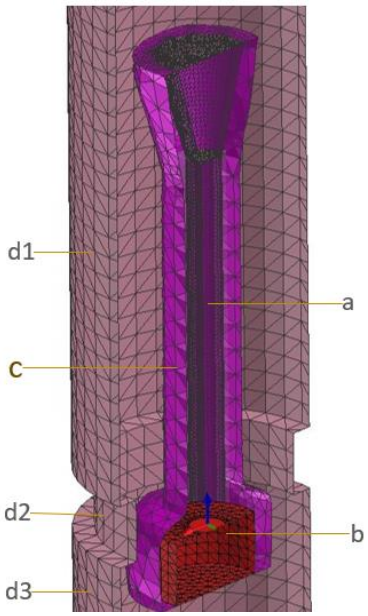


Figure 3.3 Schematic diagram of directionally solidified investment casting
a- Alloy melt; b- water-cooled plate; c-Mold shell; d-Chamber
(d1-Hot zone; d2-Insulated zone; d3-Cold zone)

3.2.1 Temperature Field Parameters

Based on the internal–external relationship, the physical parameters of the temperature

field are divided into three categories:

1. Thermal properties of the molten alloy, including thermal conductivity, enthalpy change, and solidification path.

2. Parameters related to direct heat exchange between the molten alloy and the external environment, including the thermal conductivities between the molten alloy and the water-cooled plate, the thermal conductivity between the molten alloy and the mold shell (Figure 3.3a–c), and the material thermal conductivities and specific heats of the water-cooled plate and the mold shell (Figure 3.3b–c).

3. Parameters related to indirect heat transfer between the molten alloy and the external environment, including the radiation coefficient between the mold shell and the clamping box (Figure 3.3c–d) and the thermal conductivity between the water-cooled plate and the bottom water-cooled section.

Based on directional relationships, heat transfer is further categorized as **axial** or **radial**. Axial heat transfer, parallel to the direction of columnar crystal growth, refers to heat transfer between the molten alloy and the mold shell, as well as radiative heat transfer between the mold shell and the clamping box. Radial heat transfer, perpendicular to the columnar growth direction, refers to heat transfer between the molten alloy and the water-cooled plate, as well as heat transfer between the water-cooled plate and the bottom water-cooled system.

Other boundary conditions for the temperature field include the initial superheat of the molten alloy, the initial temperature of the water-cooled plate, the temperatures of the hot and cold zones of the clamping box, and the initial temperature of the mold shell. The thermophysical parameters of the mold shell and water-cooled plate are shown in Figure 3.4.

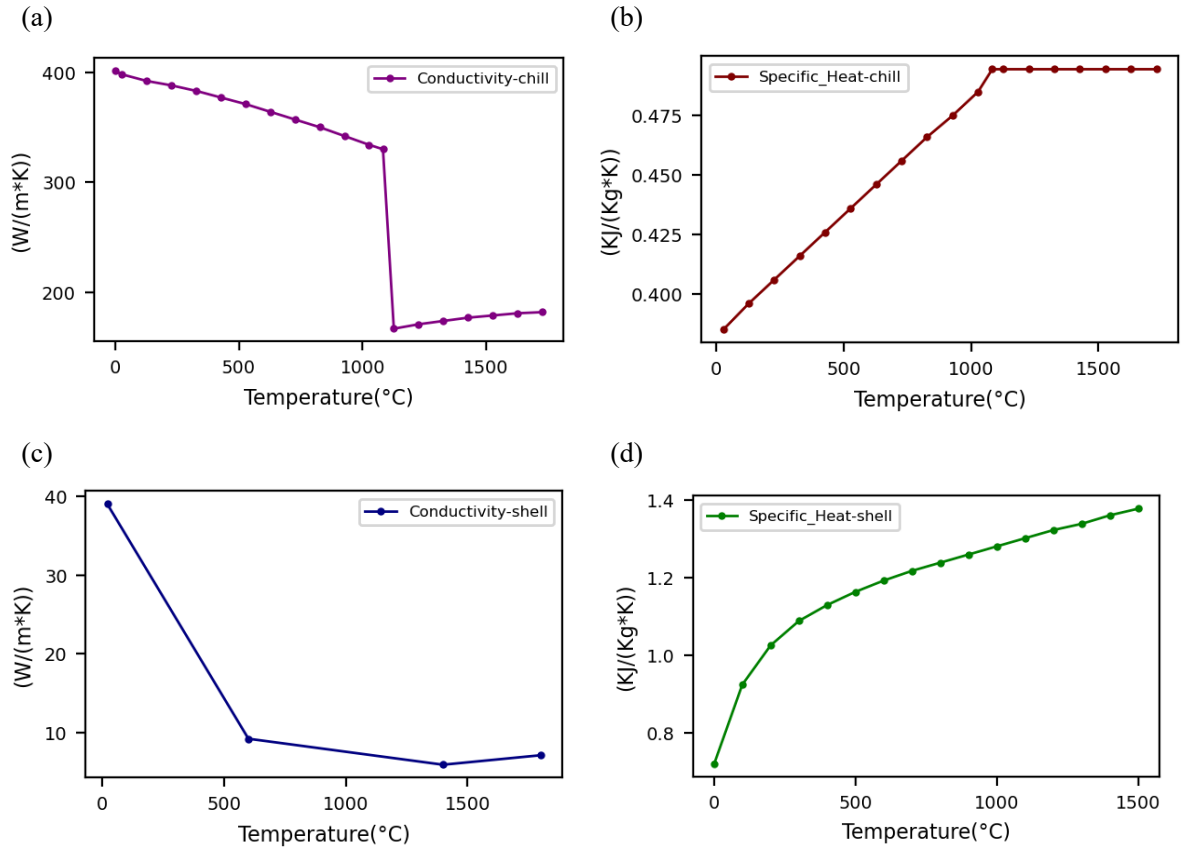


Figure. 3.4 Thermal Physical Parameters of Shell and water-cooled plate:
(a) Thermal conductivity of the water-cooled plate ; (b) Specific heat capacity of the water-cooled plate; (c) Thermal conductivity of the mold shell and (d) Specific heat capacity of Shell

As shown in Figures 3.4(a) and (b), the thermal conductivity of the water-cooled disk remains relatively stable at operating temperatures ranging from 25 to 600℃ , maintaining a value between 400 and 360 W/m·K , while its specific heat capacity remains stable at 0.4KJ/Kg·K； . As shown in Figures 3.4(c) and (d), the thermal conductivity of the mold shell remains at6 W/m·K within the alloy’ s solidification temperature range, with a specific heat capacity of approximately 1.3 KJ/Kg·K .The temperature field parameters used in the simulation are shown in Table 3.1.

Table 3.1 Temperature Field Boundary Conditions Used in Simulations

Parameter	Unit	Value
Shell-Cast *k*	$W/m^2 \cdot K$	200
Chill-Cooling Water *k*	$W/m^2 \cdot K$	2500/9000
Chill-Cast *k*	$W/m^2 \cdot K$	1000/2000/3000/4000
ε of Hot Zone	-	0.8
ε of Insulation Zone	-	0
ε of the cold zone	-	0.7
T_0 of the hot zone	℃	1560
T_0 of the cold zone	℃	120
T_0 of insulation panel	℃	120
T_0 of Shell	℃	1560/1370
T_0 of Chill	℃	150

3.2.2 CM247LC Alloy Parameters

The alloy melt is the subject of this simulation study. The primary thermal properties of the alloy melt are enthalpy change and thermal conductivity, which were automatically calculated by the ProCAST software based on the composition of the CM247LC alloy. The specific data are shown in Figure 3.5.

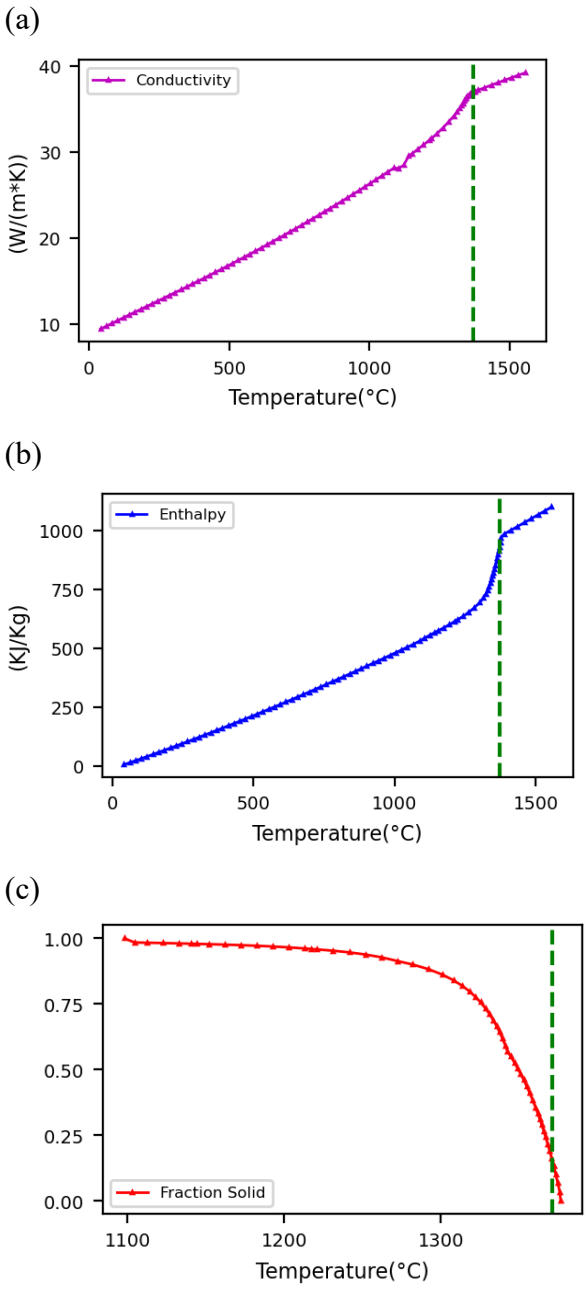


Figure. 3.5 Thermophysical parameters of CM247LC alloy:
(a) Thermal conductivity; (b) Enthalpy;
(c) Solidification path

Table 3.2 lists the relevant parameters for the of the CM247LC alloy required for CAFE calculations , where k^m_P and m_P are the partition coefficients and liquidus slopes

calculated by the ProCAST software using its own thermodynamic database;k^m_J andm_J are the partition coefficients and liquidus slopes obtained after phase diagram calculations using the JMatPro software.

Table 3.2 CM247LC Solidification Parameters[70-76]

	C_{in} (wt%)	D_l $10^{-9}(m^2/s)$	k^m_J	m_J (K/wt%)	k^m_P	m_P (K/wt%)
Al	5.6	2.1	1.059	80.9	0.935	-13.98
C	0.07	4.32	0.112	-300	0.23	-54
Co	9.2	1.83	1.097	81.5	1.14	0.54
Cr	8.1	1.59	1.012	-106	0.81	-7.46
Hf	1.4	1.37	0.107	-16	0.095	-7.72
Mo	0.5	1.6	0.764	-200	0.639	-6.11
Ta	3.2	1.68	0.57	-27	0.826	-4.74
Ti	0.7	2.10	0.756	-274	0.808	-13.1
W	9.5	2.51	1.232	37.8	1.03	-2.45
Zr	0.015	2.1	0.032	-300	0.0409	-14.99

3. 3 CAFE Parametric Simulation

3.3.1 Cell Size

When performing CAFE simulations, the cell size must be set l_{cell} . The cell size l_{cell} should be of the same order of magnitude as the secondary dendrite arm spacing; numerically, it can be several times the secondary dendrite arm spacing[27] .In this subsection, simulations were performed using cell sizes of 100 μm and 50μm , with cell densities of $1\times10^{12}/m^3$ and $8\times10^{12}/m^3$, respectively. Since $\delta t\leq\delta t_{max}=l_{cell}/v(\Delta T_{max})$, using a smaller cell size implies a smaller micro-time step δt in CAFE calculations, resulting in a significant increase in simulation time and memory usage.

Table 3.3 Cell Size

	NC100	NC50
l_{cell}	100 μm	50 μm

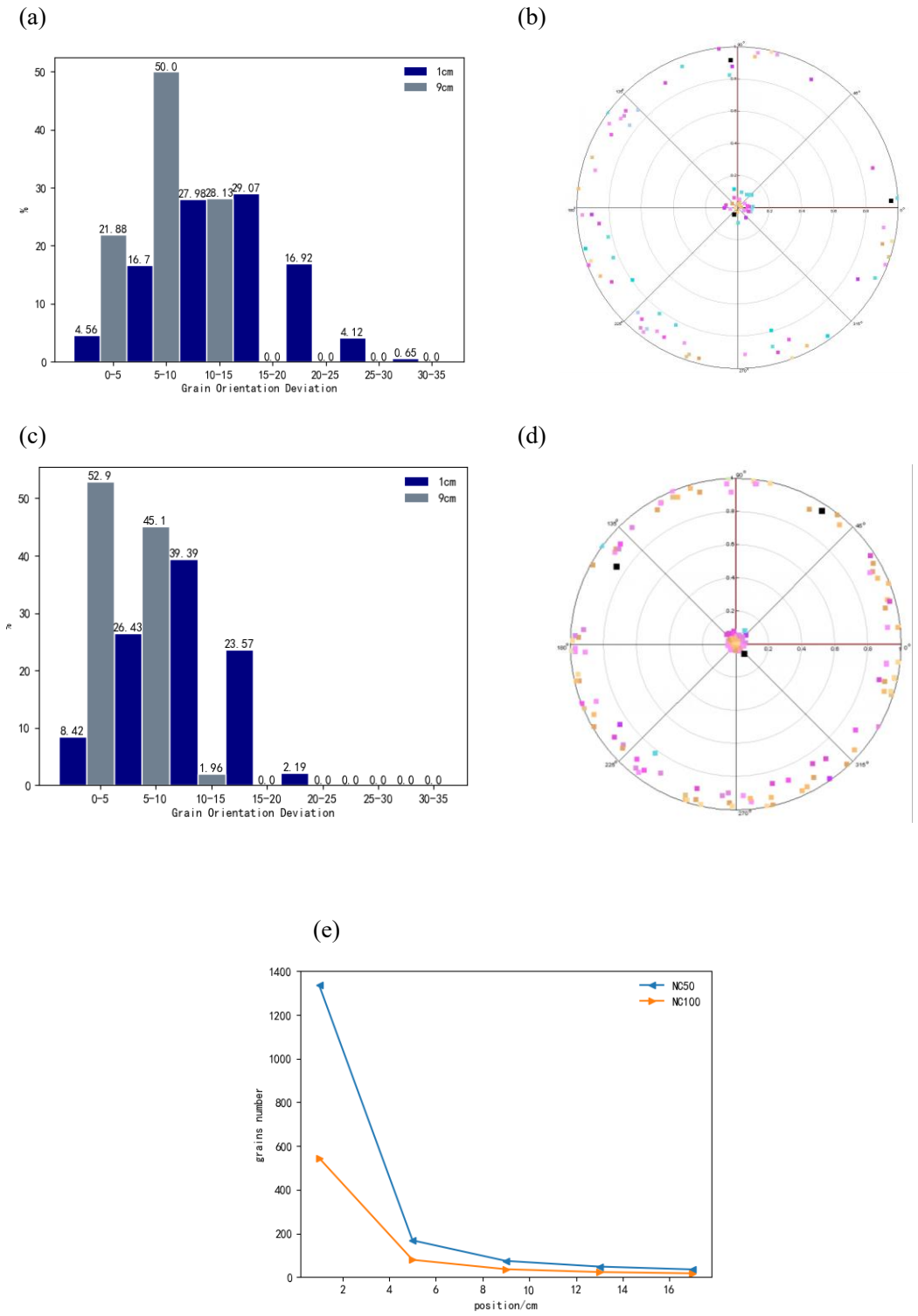


Figure 3.6 Comparison of Grain Data for Different Cell Sizes (a) NC50 grain orientation deviation; (b) NC50 pole figure; (c) NC100 grain orientation deviation; (d) NC100 pole diagram; (e) Trend chart of grain number

The cell size (l_{cell}) has a significant impact on the calculated grain morphology and grain count, as shown in Figure 3.6(e). When the cell size is halved and the cell volume density is eight times that of the original, the number of grains generated on the cross-section of the NC50 test bar (small cell size case) is twice that of the NC100 case (large cell size case) .Reducing the cell size significantly decreases the grain offset. As shown in the cross-sections of the specimens at a distance of 9 cm from the cooling plate in Figures 3.6(a), (b), (c), and (d), the offset angles between the columnar grains and the <100> grain direction in the NC50 case are all less than 10° ;while in the NC100 case, the deviation angles between the columnar grains and the <100> grain direction were partially within the 10 – 15° range ;On the cross-sections of the initial solidification zone 1 cm from the cooling plate, the deviation angles between the columnar grains and the <100> crystal orientation in the NC50 case are all less than 20° , while in the NC100 case, some of the deviation angles fall within the 20 – 30° range.

3.3.2 Nucleation Parameters

The nucleation density during solidification of an alloy melt depends on the alloy's intrinsic properties and the characteristics of the nucleation surface. In this subsection, four maximum nucleation numbers—differing by one order of magnitude on a one-dimensional scale—are selected to simulate and compare the effects of varying nucleation density on microstructure morphology. The specific parameters are listed in Table 3.4.

Table 3.4 Maximum Number of Nucleation

	NS6	NS8	NS10	NS12
$n_{max}(m^{-2})$	10^6	10^8	10^{10}	10^{12}

(a) (b)

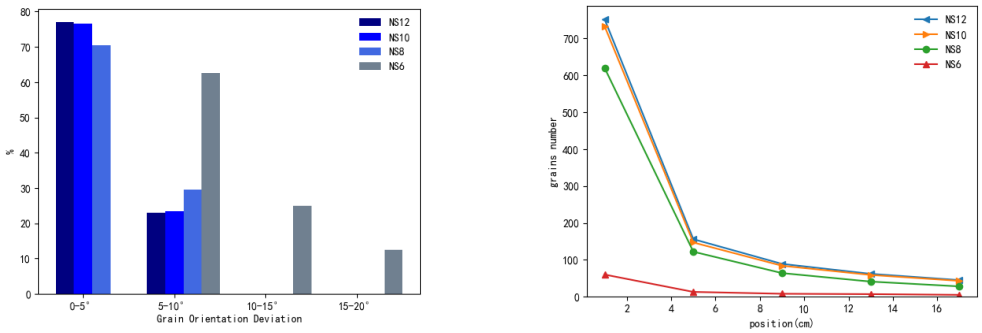


Figure. 3.7 Comparison of grain data under variable nucleation parameters:
(a) Grain orientation deviation, (b) Trend in grain number

Figure 3.7(a) shows the histogram of grain orientation deviation distribution for a cross-section of the test rod located 9 cm from the cooling disk. For the N12, N10, and N8 cases, the deviation angles of the columnar grains relative to the <100> grain orientation are partially within the 5–10° range, whereas for the N6 case, they fall within the 10–20° range. Figure 3.7(b) presents a quantitative analysis of grain size in the test rod cross-section at distances of 1, 5, 9, 13, and 17 cm from the nucleation end, measured along the normal direction of the water-cooled disk, and displayed as a line graph. It can be observed that the number of grains decreases from N12, N10, and N8 to N6; the number of two-dimensional grains in the N6 case is one-tenth that of the other cases. Figure 3.8 shows images of the external and cross-sectional grain morphologies obtained from simulations using different nucleation parameters. Upon observation and comparison, it can be seen that the morphological simulation results for N12, N10, and N8 are relatively similar, while the grain structure in case N6 is noticeably coarse with a sparse number of grains.A comparison of multiple cases reveals that changes in the maximum nucleation density cause overall changes in grain size; a decrease in the maximum nucleation density reduces the total number of grains and increases the average grain size.

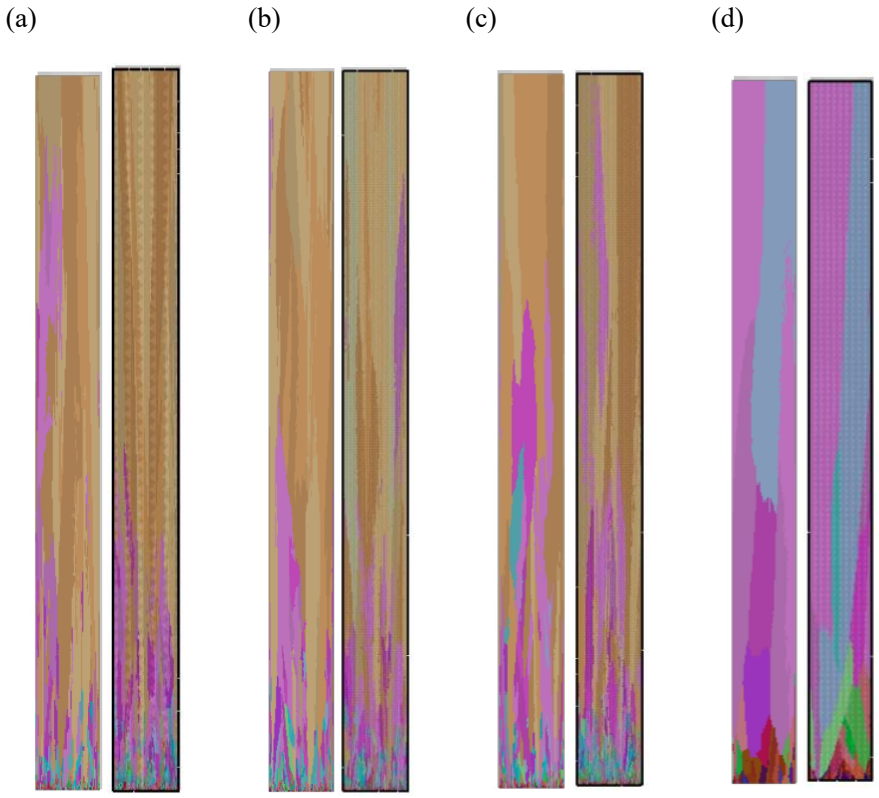


Figure. 3.8 Comparison of grain morphology under varying nucleation parameters:
(a) NS12; (b) NS10; (c) NS8; (d) NS6

3.3.3 Growth kinetics parameters

The CAFE module in ProCAST assumes that the growth rate at the dendrite tip follows a power-law relationship with the undercooling at the interface front, where $v(\Delta T)=a_2\Delta T^2+a_3\Delta T^3$; see Equations (3.1) and (3.2) for the specific calculation formulas. The parameters required for growth kinetics calculations include: liquid-phase diffusion coefficients D_l , Gibbs-Thompson coefficients Γ , and phase diagram data m_l、k . It holds that $v(\Delta T)$ is inversely proportional to Γ and directly proportional to the liquid-phase diffusion coefficient D_l . The phase diagram data for the CM247LC alloy selected in this section are shown in Table 3.2 of the ProCAST dataset. In existing microstructural simulations of nickel-based alloys, the phase diagram calculation parameters and surface tension parameters adopted by researchers are not consistent[29~35] . Table 3.5 lists the different parameters selected by other researchers for reference, as well as the growth rates calculated using these parameters.

$$a_2=\frac{2k\Gamma(1-k)-\rho D^2}{2kmC_0\pi^2\Gamma(1-k)^2} \tag{3.1}$$

$$a_3=\frac{D}{\pi\Gamma(mC_0)^2(1-k)} \tag{3.2}$$

Table 3.5 Growth kinetic parameters[29~35]

	Unit	NG1	NG2	NG3	NG4
Γ	K·m	1×10^{-7}	2×10^{-7}	3×10^{-7}	3.65×10^{-7}
D_l	m^2/s	3×10^{-9}	3×10^{-9}	2×10^{-9}	1×10^{-9}
a_2	$ms^{-1}K^{-2}$	10×10^{-9}	5×10^{-9}	2×10^{-9}	1.4×10^{-9}
a_3	$ms^{-1}K^{-3}$	8×10^{-7}	4×10^{-7}	2×10^{-7}	1.1×10^{-7}

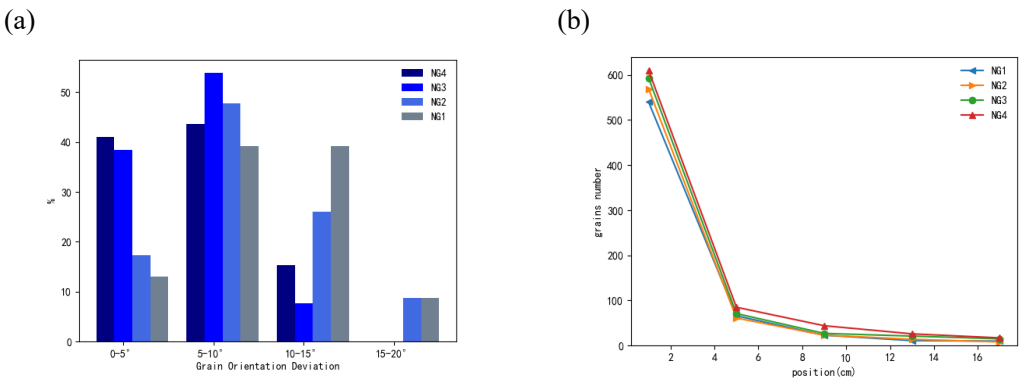
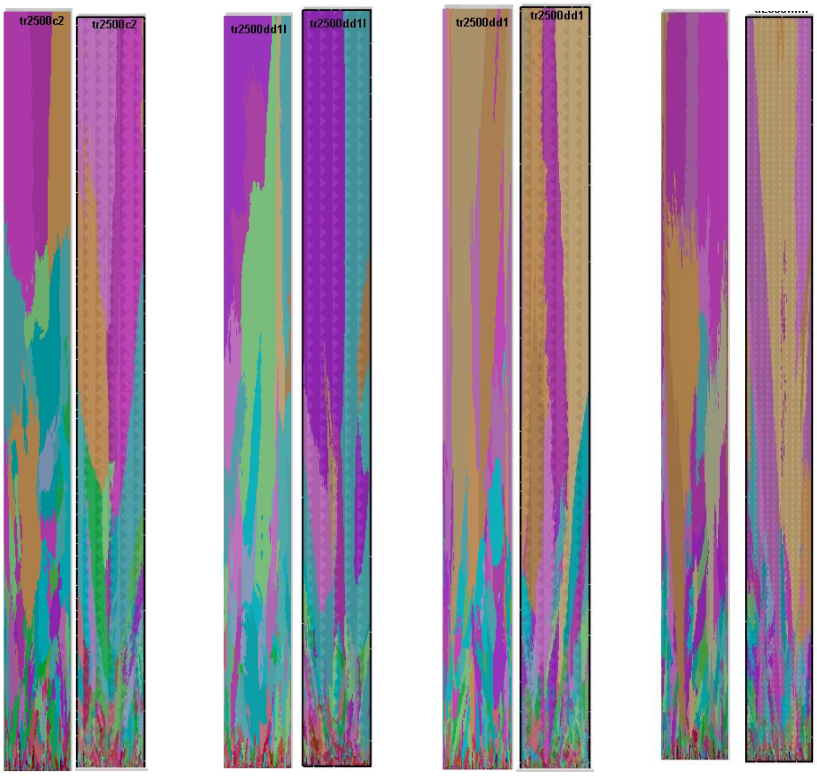


Figure. 3.9 Comparison of grain data under variable growth kinetics



(a) NG1 (b) NG2 (c) NG3 (d) NG4

Figure 3.10 Comparison of grain morphology in s with varying growth rates (a) NG1; (b) NG2; (c) NG3; (d) NG4

As shown in Table 3.5, the crystal growth rates for cases NG1 through NG4 decrease sequentially while remaining within the same order of magnitude. Figure 3.10 illustrates that as the crystal growth rate decreases, the grain morphology becomes progressively more regular and columnar, and the number of grains increases slightly.

Figure 3.9(a) presents the histogram of grain offset distribution in the cross-section of the test rod at a distance of 9 cm from the water-cooled plate. Figure 3.9(b) provides a quantitative analysis of the number of grains in cross-sections located 1, 5, 9, 13, and 17 cm from the nucleation end, measured along the normal direction of the water-cooled plate. It can be observed that as the growth rate decreases from NG4 to NG1, the number of grains increases slightly while the grain displacement gradually decreases; the variation in grain count remains within 10%.Overall, a lower dendrite tip growth rate prolongs the time required for a dendritic cell to capture neighboring cells, resulting in an increased total number of grains and a reduction in the crystallographic misorientation of the columnar grains.

3.4 Temperature Field Simulation

The growth kinetics parameters used in this section are:a_2 = 3.874×10^{-8} $ms^{-1}K^{-2}$,a_3 = 6.197×10^{-9} $ms^{-1}K^{-3}$, and the maximum face nucleation density is $10^{12}/m^3$. The relevant phase diagram data for the CM247LC alloy used in the growth kinetics calculations are shown in Table 3.2.

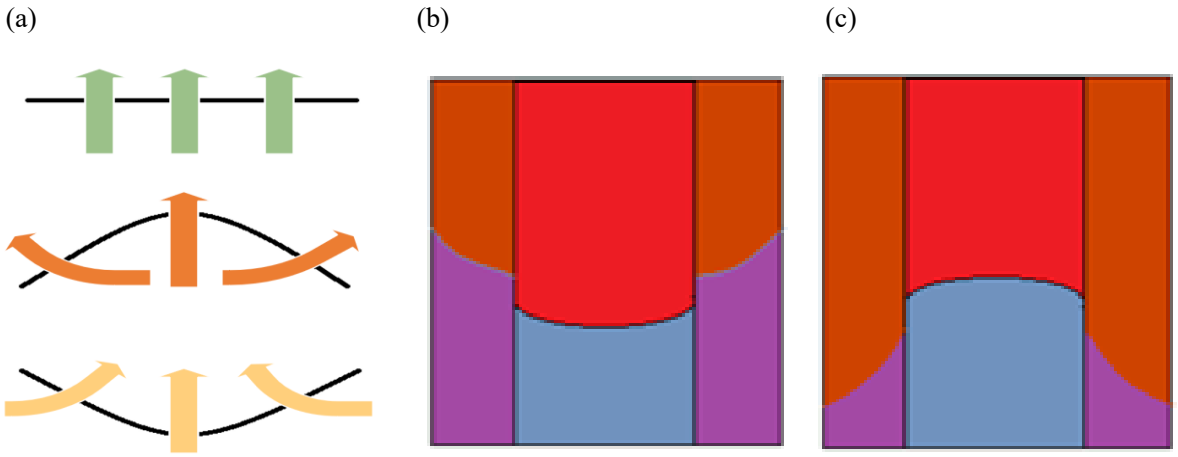


Figure 3.11 Interface shape and heat flux: (a) Schematic diagram of heat flux distribution; (b) concave interface; (c) convex interface

As shown in Figure 3.11, the temperature field during directional solidification strongly affects the morphology of the solid–liquid interface. Under unidirectional heat flux, the interface remains straight. According to competitive growth theory, grains with crystallographic orientations aligned with the heat flux grow faster and are less likely to be eliminated. In this subsection, the influence of temperature field conditions on grain morphology is examined by varying the heat transfer coefficient between the water-cooled plate and the alloy. The coefficients used in each case are summarized in Table 3.6.

Table 3.6 Heat Transfer Coefficient between Cooling Disk and Alloy

	TA1	TA2	TA3	TA4
h($W/m^2\cdot K$)	1000	2000	3000	4000

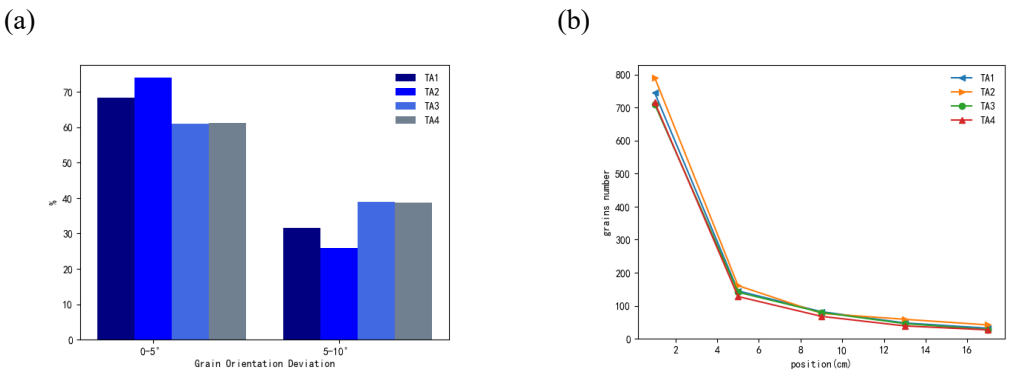


Figure. 3.12 Comparison of grain data under variable heat transfer coefficients: (a) Grain orientation deviation; (b) Trend in grain number

Figure 3.12(a) presents the histogram of grain displacement in the cross-section 9 cm from the water-cooled plate, while Figure 3.12(b) quantifies the number of grains at 1, 5, 9, 13, and 17 cm from the nucleation end, measured perpendicular to the plate. As shown in Figure 3.13, grain count decreases from TA2, TA1, and TA3 to TA4, with a corresponding increase in grain offset. In the TA2 and TA1 cases, radial heat flux is minimal, resulting in the most linear solid–liquid interface.

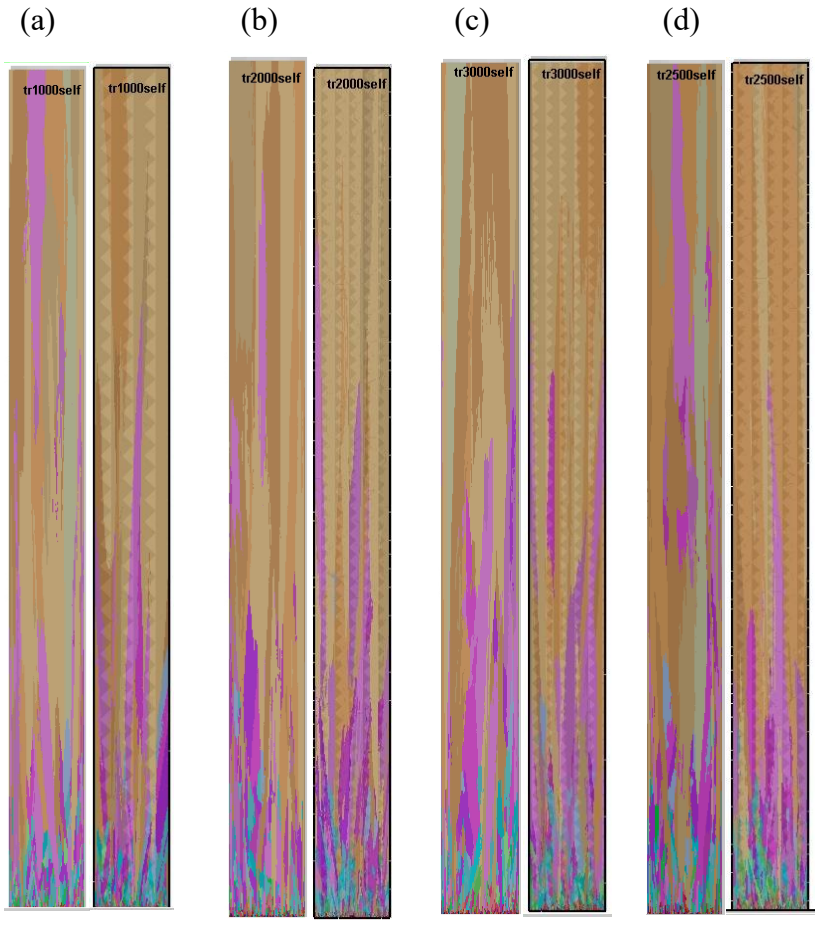


Figure 3.13 Comparison of grain morphology for varying heat transfer coefficients: (a) TA1; (b) TA2; (c) TA3; (d) TA4

3.5 Validation of Microstructural Morphology

Figure 3.14 shows the grain morphology images obtained from the experiments in the first subsection of this chapter, along with the simulated grain structure images obtained after adjusting the simulation parameters as described above. It can be seen that there is a good correlation between grain size and competitive growth behavior.



Figure. 3.14 Comparison of experimental results with simulated images

3.6 Macro-segregation Simulation

The segregation results obtained using the ProCAST macrosegregation module are shown in Figure 3.15, where Figure 3.15(a – j) illustrates the macroscopic distribution of the 10 solute elements in CM247LC during the solidification process. For solute elements Co and W with k > 1, the ProCAST macrosegregation module is ineffective in predicting solute elements that tend

to be distributed into the solid phase; the other solute elements with k < 1 are divided into two categories: the first category consists of elements where k < 1 but close to 1, such as Al, Cr, Mo, Ta, and Ti; the second category consists of elements where k < 1 and close to 0, such as B, C, and Hf.For the first category of elements with k < 1, most of the solidified solid phase region retains the nominal composition, while the paste region exhibits a trend of decreasing composition; for the second category of elements with k < 1, the solute composition in the solidified solid phase region is abnormally high, several times higher than the nominal composition, whereas the paste region shows no significant variation. Overall, the macroscopic segregation simulation results are insufficient for quantitative analysis and comparison with actual conditions.

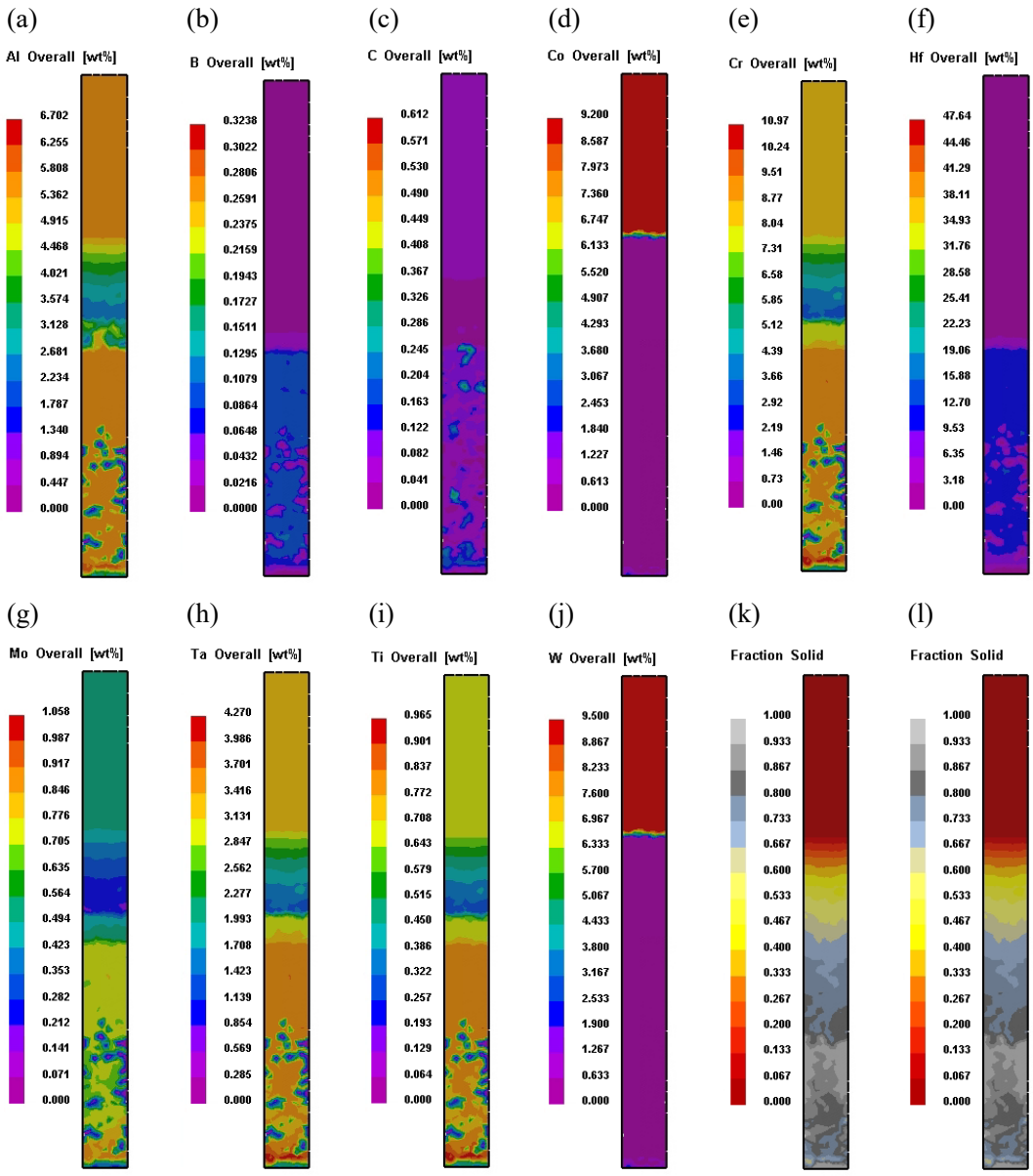


Figure 3.15 Macro-segregation simulation results

3.7 Chapter Summary

In this chapter, directional solidification experiments and numerical simulations were combined to investigate the grain morphology and temperature field characteristics of the Ni-based superalloy CM247LC. Particular attention was paid to the effects of simulation parameters on the predicted grain morphology. The main findings are summarized as follows:

(1) Directional solidification experiments were conducted at a withdrawal rate of 41 μm/s, and the crystallographic characteristics of the directionally solidified Ni-based superalloy specimens were characterized.

(2) A simplified numerical model of the directional solidification process was established. The computational mesh was generated, and the material properties of the casting and mold were defined separately. The parameters involved in the cellular automaton–finite element (CAFE) simulation and the criteria for their selection were also introduced. The grain morphology simulation results indicated that, when the maximum nucleation rate was sufficiently high, a smaller cellular size resulted in a finer grain structure and a lower degree of grain orientation deviation. Furthermore, a lower dendrite tip growth velocity led to smaller grain sizes and consequently reduced grain orientation deviation.

(3) The evolution of thermal conditions during directional solidification was simulated by prescribing the movement velocity of a multi-zone thermal enclosure. The axial heat flow during solidification was adjusted by varying the interfacial heat transfer coefficient. The results demonstrated that the highest grain number and the lowest grain orientation deviation were achieved when the radial heat flow was minimized relative to the axial heat flow.

(4) The dendrite growth kinetic coefficients and selected nucleation parameters for the solidification simulation of the Ni-based superalloy CM247LC were determined. Validation results demonstrated that the selected processing conditions and material parameters provided good agreement between the numerical predictions and experimental observations.

Chapter 4: Compositional Segregation Analysis

4.1 Composition Measurement Experiments

The directionally solidified test rods from the experiment described in the previous chapter were cut along the longitudinal centerline using a wire-cutting machine, and then cross-sectioned into seven small segments to facilitate sample grinding, polishing, and electron microscope analysis. EDS point 0 lies about 1 mm from the chill face; the seven cut segments were not corrected for kerf; the model is 180 mm long while the real bar is shorter (shell shrinkage, cutting loss); positions x_j = 1 + 5j mm are nominal. After grinding and polishing the central cross-section, no etching or heat treatment was performed. An energy dispersive spectrometer (EDS) attached to a scanning electron microscope (SEM) was used to measure the alloy element content at 34 test points along the central axis of the test bar cross-section at 5-mm intervals. The scanning area for each test point was100μm×100μm . The specific test point locations and testing sequence are shown in Figures 4.1 and 4.2.

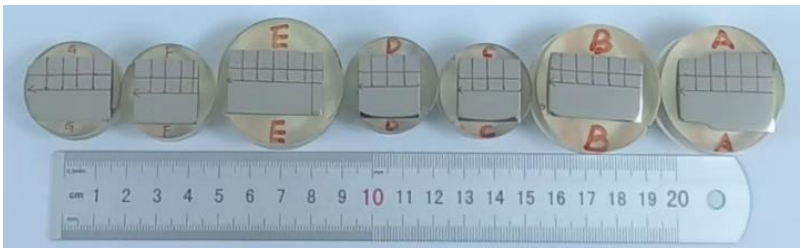


Figure. 4.1 Schematic diagram of concentration test

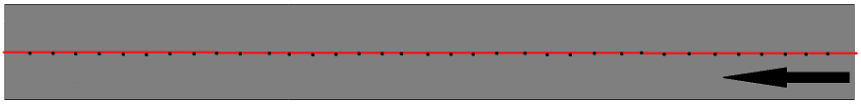


Fig 4.2 Concentration test axis and direction

4.2 Composition Measurement Results and Analysis

4.2.1 Analysis of Inter-Elemental Relationships

SEM- EDS analysis has low accuracy for low-atomic-number elements; therefore, boron, carbon, and low-concentration zirconium were excluded. To assess solute distribution correlations during directional solidification, this study used Pearson correlation coefficients to analyse the linear relationships among the nine solute elements and a Spearman trend test to analyse the rising or falling trend of each element along the bar.

The Pearson coefficient measures linear correlation as the quotient of covariance cov(X,Y) and standard deviation σ_X, as given in Equation (4.1):

$$\rho = [n\sum XY - \sum X\sum Y] / \sqrt{[n\sum X^2 - (\sum X)^2]}\sqrt{[n\sum Y^2 - (\sum Y)^2]} \quad (4.1)$$

Based on Pearson coefficients, solute elements fall into four categories: Category 1 (Co, Cr, Al) — strong positive correlation (>0.5), trending with Ni mass fraction; Category 2 (Ta, Ti, Hf) — strong positive correlation (>0.5), trending opposite to Ni; Category 3 (W) — weak positive correlation with Category 1 and weak negative correlation with Category 2; Category 4 (Mo) — weak correlation with all others. Categories 1 and 2 exhibit strong negative correlation with each other: as solidification progresses and Ni mass fraction decreases, Category 1 elements decrease while Category 2 elements increase.

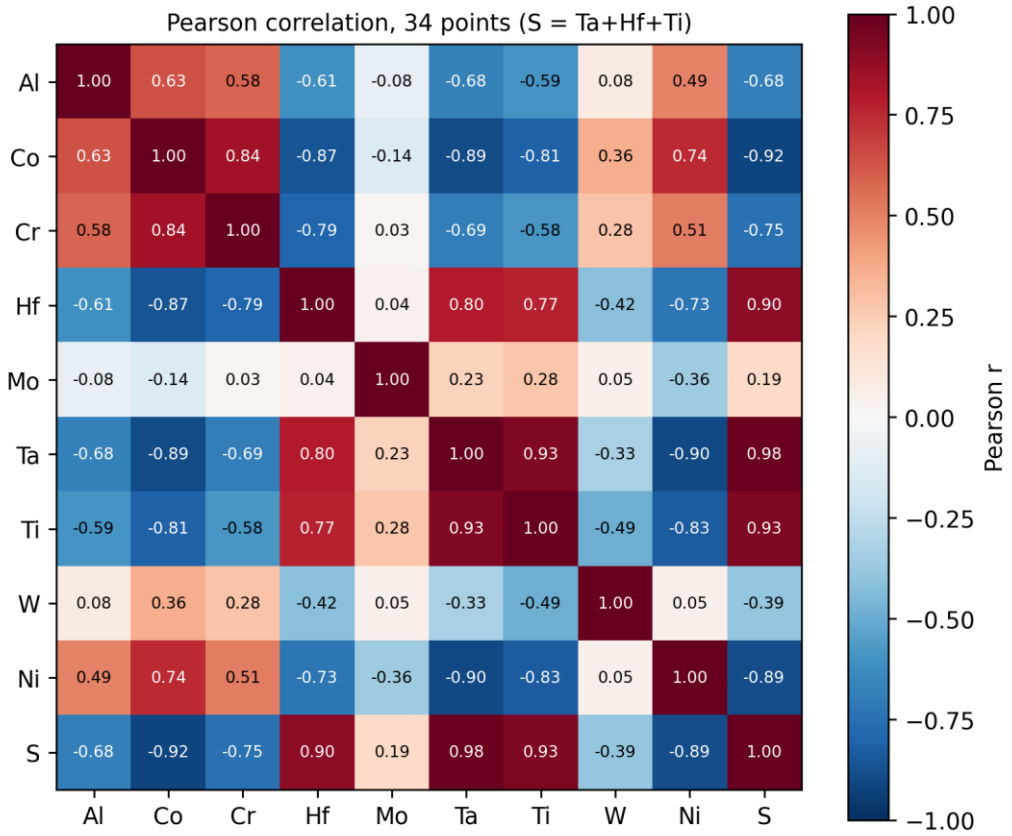


Figure 4.3 Pearson correlation heat map of the 34-point axial compositions (S = Ta+Hf+Ti)

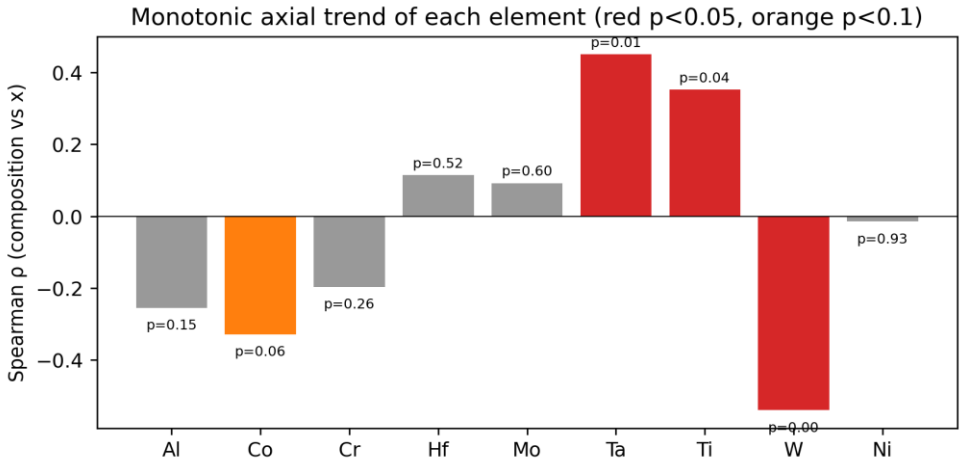


Figure. 4.4 Spearman monotonic-trend coefficient of each element versus axial position (red $p < 0.05$, orange $p < 0.1$)

To test whether each element changes monotonically along the bar, the Spearman rank correlation between the element content and the distance x from the chill face was computed over the 34 measuring points:

$$\rho s = 1 - 6\sum di^2/[n(n^2-1)] \tag{4.2}$$

where di is the difference between the composition rank and the position rank of point i, and n = 34. Its significance is tested with the t distribution with n − 2 degrees of freedom:

$$t = \rho s\sqrt{[(n-2)/(1-\rho s^2)]} \tag{4.3}$$

As shown in Fig. 4.4, W decreases significantly with x ($\rho s = -0.54$, $p = 0.001$), Ta and Ti increase ($\rho s = +0.45$, $p = 0.008$; $\rho s = +0.35$, $p = 0.04$), Co tends to decrease ($\rho s = -0.33$, $p = 0.06$), and Al, Cr, Hf, Mo and Ni show no significant monotonic trend (absolute ρs not above 0.25, $p > 0.1$). The p-values treat the points as independent and ignore the autocorrelation of the composition series, so they are optimistic. These directions agree only partly with the Pearson-based classification: Ta and Ti of Category 2 rise as solidification proceeds, Co of Category 1 and W of Category 3 fall, and the trends of Hf, Mo, Al and Cr are not statistically significant.

JMatPro-calculated phase diagram data for CM247LC were used to analyze segregation behavior. The Lever microsegregation model performs single-point equilibrium calculations at each temperature step, assuming complete diffusion in both liquid and solid phases, allowing calculation continuation after liquid phase disappearance to determine solid-phase changes. The Scheil model neglects solid-phase diffusion and assumes sufficient liquid-phase diffusion; at

each temperature step, it determines liquid composition, solid fraction, and liquid temperature, using the prior step's liquid composition in subsequent Lever calculations.

Liquidus slope m_L and solid-liquid partition coefficients k_Lever and k_Scheil calculated from JMatPro phase diagrams are shown in Figure 4.5. Elements with positive liquidus slope (both Lever and Scheil) include W, Co, and Al; those with negative slope include Cr, Mo, Ta, Ti, and Hf. Elements with partition coefficient k > 1 (both models) are W, Co, and Al; those with k < 1 are Mo, Ta, Ti, and Hf. For Cr, k_Scheil > 1 while k_Lever < 1 but approaches unity. Based on JMatPro phase diagram calculations, Mo, Ta, Ti, and Hf preferentially partition into the liquid phase during solidification, while W, Cr, Co, and Al partition into the solid phase.

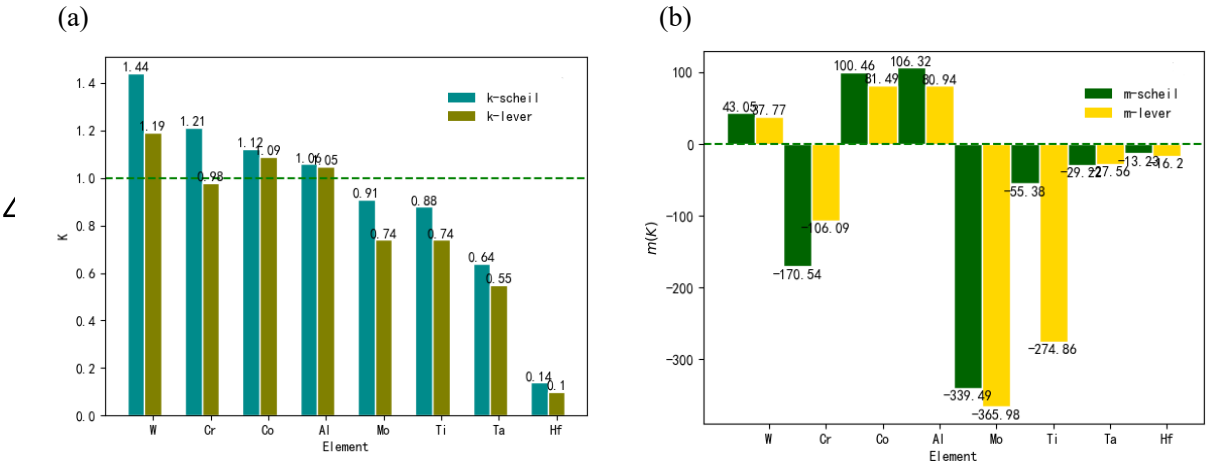


Figure 4.5 k and m calculated using JMatPro:
(a) Solid-liquid partition coefficient; (b) Liquidus slope

4.3 Macrosegregation Analysis

4.3.1 Log-linear Regression Model

The macroscopic segregation curve for unidirectional solidification under a steady-state temperature field follows an exponential function: `$C_s=k_EC_0(1-f_s)^{k_E-1}$ `. In this section, a logarithmic linear regression model is employed to fit the composition data obtained from directional solidification experiments.

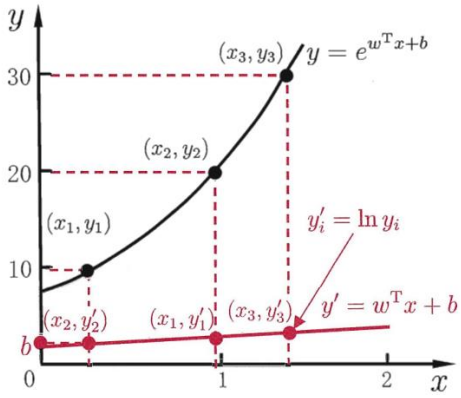


Figure 4.6 Schematic diagram of logarithmic linear regression

The linear model is simple in form and highly interpretable; the prediction function is the product of the attribute and the weight, plus a bias term:

$$f(x)=\omega^T x+b \tag{4.4}$$

As shown in Figure. 4.6, logarithmic linear regression can map the exponential model to a linear regression model. By converting the output corresponding to $iny=e^{w^Tx+b}$ into changes on an exponential scale, the model’s predicted values can be output as the logarithm of the target approximated by the linear model $lny=w^Tx+b$. For the univariate exponential function $y=a(1-x)^b$, we have

$$\log y = \log a + b \cdot \log(1 - x) \tag{4.5}$$

whose linear form is $z = A + b \cdot w$, where $A = \log a$； $w = \log(1 - x)$； $z = \log y$

For the macroscopic segregation curve of the solute, we have $k_EC_0=a$,$k_E-1=$ b.

The goodness-of-fit statistic is a measure employed in linear regression analysis to evaluate how well a regression equation fits the observed sample data; it quantifies the extent to which the sample observations cluster around the fitted regression line. In the context of a derived regression equation, the goodness of fit partitions the total variation in the dependent

variable *y* into two distinct components: variation attributable to changes in the independent variable *x*, and variation attributable to random error. The regression equation captures only the first of these components. The sum of squares corresponding to the variation explained by the independent variable *x* is termed the regression sum of squares (SSR); the sum of squares corresponding to the variation due to random factors is termed the residual sum of squares (SSE); and the total sum of squares (SST) represents the overall variation in the dependent variable, where

$$SST=\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^2=SSR+SSE=\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 \tag{4.6}$$

The higher the proportion of the regression sum of squares (SSR) relative to the total sum of squares (SST), the better the goodness of fit of the regression equation. For simple linear regression, the most commonly used goodness-of-fit statistic is the coefficient of determination,

$$R^2=\frac{SSR}{SST}=1-\frac{SSE}{SST}=1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^2} \tag{4.7}$$

R^2, which takes values in the interval [0, 1]. A value closer to 1 indicates a better fit, meaning the regression equation explains a greater proportion of the total variation in the dependent variable. Specifically, R^2 is defined as the ratio of the regression sum of squares to the total sum of squares.

4.3.2 Gradient Descent Algorithm

To further improve the regression model, the gradient descent algorithm is employed during the curve-fitting process. In machine learning, learning refers to the process of deriving a model from data — also known as training — which is accomplished by executing a learning algorithm. Gradient descent is used to find the local minimum of a loss function; in the univariate case, this is analogous to finding the derivative. The algorithm iteratively moves in the direction of steepest descent to locate the minimum of the function, progressively reducing the value of the loss function throughout the training process. For a simple linear regression model with a single attribute, the loss function is defined as:

$$cost=\frac{1}{n}SSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2\,\frac{1}{n}\sum_{i=1}^{n}(y_i-\omega*x)^2 \tag{4.8}$$

The partial derivative of cost(ω) with respect to each parameter is given by:

$$\frac{\mathrm{dcost}(\omega)}{\mathrm{d}\omega_i}=\frac{2}{\mathrm{n}}\sum_{\mathrm{i}=1}^{\mathrm{n}}(\hat{\mathrm{y}}_\mathrm{i}-\mathrm{y}_\mathrm{i})\,\mathrm{x}_\mathrm{i} \tag{4.9}$$

The linear regression model uses the loss function to evaluate model performance, where the mean squared error is defined as the sum of squared deviations caused by random factors, divided by the total sample size. Here, y_i denotes the actual observed value and $\hat{y}_i$ denotes the predicted value; as the loss decreases, the predicted values converge toward the actual values. Since the loss function is a quadratic function of ω, gradient descent updates the slope at each iteration with the objective of minimizing the loss. An initial slope value is first defined, and the loss function is differentiated with respect to this value as the independent variable. At each parameter update step, the value is reduced by $\alpha \cdot \frac{d\,\mathrm{cost}(\omega)}{d\,\omega_i}$, where α is the learning rate, which controls the speed at which the model parameters are updated.

The portion of the full dataset used for training is referred to as the training set, where each individual sample constitutes a single training instance, and the collection of all training instances forms the complete training set. Testing refers to the process of evaluating the learned model on new data to assess its generalization ability; the samples used for this purpose constitute the test set. The full dataset is commonly partitioned into a training set and a test set using methods such as the holdout method or cross-validation. The holdout method divides the entire dataset into two mutually exclusive subsets — one serving as the training set and the other as the test set — whereby a model is trained on the training set and its generalization error is subsequently estimated using the test set.

4.3.3 Fitting Results of Segregation Curves

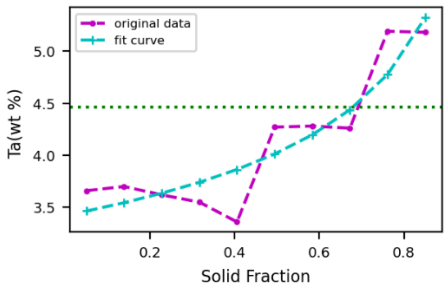
This section presents the fitting results for macroscopic segregation of solute elements during directional solidification, obtained via leave-one-out cross-validation. The figures below show the regression curves and solidification parameters calculated for the Ta element using the gradient descent method, with a learning rate of 0.01 and 4,000 iterations.

Figure 4.7(a) shows the 10-point fitting curve for Ta; Figure 4.7(b) shows the 17-point fitting curve; Figure 4.7(c) shows the 34-point fitting curve.

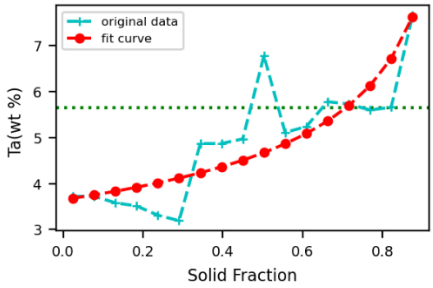
The table below the figure lists the loss values, goodness-of-fit (R^2), effective partition coefficients for macroscopic segregation (k_E), and initial compositions (C_0) obtained from

the machine learning regression fitting. The nominal composition of Ta is 3.2 wt%.

(a)



(b)



(c)

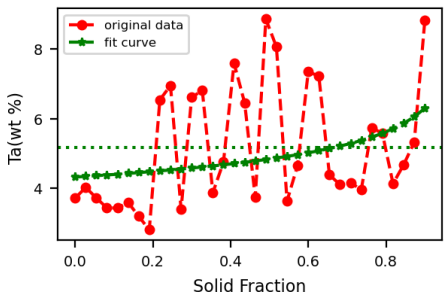


Figure 4.7 Macro-segregation curve fitting of Ta
(a) 10-point fit; (b) 17-point fit; (c) 34-point fit

Table 4.1 Macro-segregation parameters of Ta

	cost	R^2	k_E	C_0
10-point	0.002	0.832	0.768	4.459
17-point	0.01	0.666	0.645	5.648
34-point	0.047	0.056	0.836	5.163

Figure 4.8 shows the regression curves and solidification parameters calculated using the gradient descent method for the elements , Co, and . Figure 4.8(a) shows the 10-point fitting curve for Co; Figure 4.8(b) shows the 17-point fitting curve; and Figure 4.8(c) shows the 34-point fitting curve.The table below the figure lists the loss values, goodness-of-fit (R^2),

effective partition coefficients for macrosegregation (k_E), and initial compositions (C_0) obtained from the machine learning regression fit. The nominal composition of Co is 9.2 wt%.

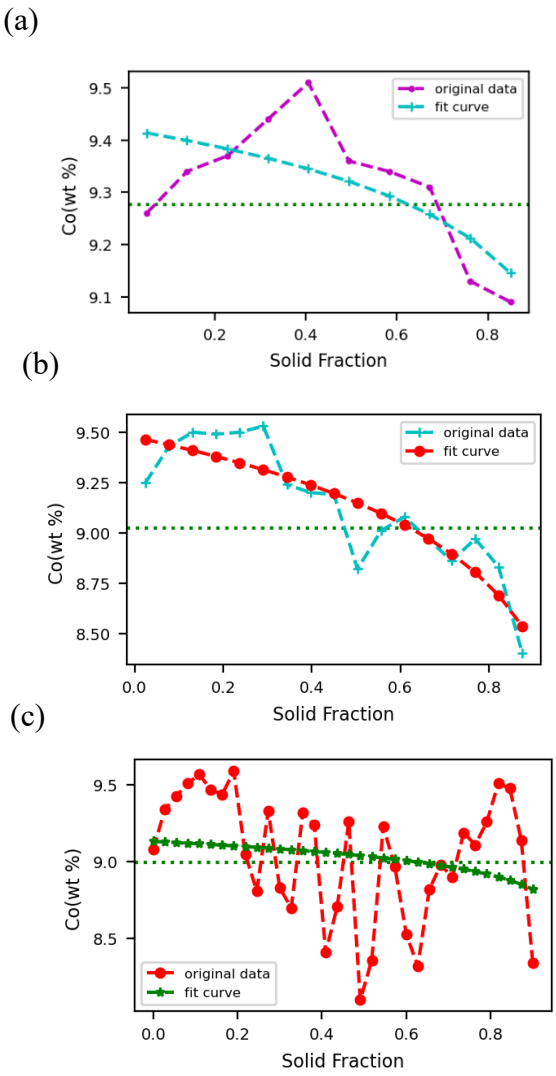


Figure. 4.8 Curve fitting of Co macro-segregation: (a) 10-point fit; (b) 17-point fit; (c) 34-point fit

Table 4.2 Macro-segregation parameters of Co

	cost	R^2	k_E	C_0
10-point	0.00	0.483	1.016	9.276
17-point	0.00	0.791	1.05	9.02
34-point	0.001	0.043	1.015	8.997

Table 4.3 shows the macroscopic segregation parameters for all alloying elements, as well as the fitted values for the initial compositionC_0 .

Table 4.3 Macro-segregation parameters of all elements

	kE_10	kE_17	kE_34	C0_10	C0_17	C0_34	kE_34_lo95	kE_34_hi95	p_slope_34	R2_34	p_slope_17	p_slope_10
Al	1.0268	1.0373	1.0189	7.1324	6.9543	6.8239	0.9762	1.0617	0.3738	0.0248	0.0307	0.0178
Co	1.0162	1.0506	1.0152	9.2757	9.0223	8.9975	0.9892	1.0412	0.2414	0.0426	0	0.0241
Cr	1.0063	1.0343	1.0056	8.4749	8.3023	8.2881	0.9837	1.0274	0.6074	0.0083	0	0.2364
Hf	0.5743	0.5097	0.9644	1.2911	1.6738	1.1333	0.5273	1.4015	0.8694	0.0009	0.0006	0.0005
Mo	0.879	0.803	0.9746	0.6736	0.763	0.6728	0.8437	1.1055	0.6954	0.0049	0	0
Ta	0.7679	0.6456	0.8363	4.4591	5.648	5.1626	0.658	1.0146	0.0707	0.0985	0	0.0004
Ti	0.7803	0.75	0.8811	0.9476	1.0622	1.0147	0.7575	1.0048	0.059	0.107	0.0002	0.0001
W	1.0599	1.0448	1.0569	9.5296	9.4238	9.1852	1.0211	1.0928	0.0028	0.2467	0.0081	0
Ni	1.0134	1.0233	1.0032	58.4522	57.6877	58.1356	0.9864	1.0201	0.7003	0.0047	0.0003	0.0014

Normalisation closure alone produces an apparent $k_E \approx 1.010$ for the matrix elements, so $k_E>1$ for Ni, Cr, Co and Al cannot be distinguished from closure.

Table 4.4 Solid-liquid partition coefficients from phase diagram JMatPro

	Ni	W	Ta	Al	Co	Cr	Hf	Mo	Ti
k_J	-	1.232	0.57	1.056	1.097	1.012	0.107	0.764	0.756

NOTE:Table 4.4 lists representative JMatPro Scheil-path values over 1370.9–1256 °C; at the liquidus they are W 1.122, Co 1.093, Ta 0.481, Ti 0.729, Al 0.910, Cr 0.941, Mo 0.775, Hf 0.095

Comparing the phase diagram calculation results (JMatPro) in Table 4.4 with the experimental measurement fitting results in Table 4.3 reveals that for alloying elements with $k<1$, such as Ta, Hf, Mo, and Ti, the fitting values in k_E are higher than the theoretical calculations; for alloying elements with $k>1$, such as W、Al、Co、Cr, the fitting values in k_E are lower than the theoretical calculations.This discrepancy indicates that the degree of macroscopic segregation under actual solidification conditions is lower than that predicted by theoretical calculations.

4.3.4 Elemental Microsegregation

The first solid phase formed during solidification is the γ phase dendrites . The solubility of certain alloying elements in the γ phase dendrites is much lower than their concentration in the liquid alloy, so they become enriched in the liquid phase during solidification. The uneven distribution of alloying elements between the early-solidifying dendrite stems and the late-solidifying dendrite interstices leads to microsegregation.The degree of microsegregation in a

material can be characterized by the element segregation coefficient (k'). For a given element (x), the segregation coefficient represents the ratio of the composition in the interstitial space to that in the dendrite, as shown in the following equation:

$$k'_x = C_{x,core}/C_{x,inter} \tag{4.10}$$

An element segregation coefficient (k') of 1 indicates that no segregation exists for that element, meaning the composition is the same in the interstitial regions and the dendrites.k' also indicates the tendency toward segregation; a value of k'_x <1 indicates that the element tends to segregate into the interstitial regions.

(a) (b)

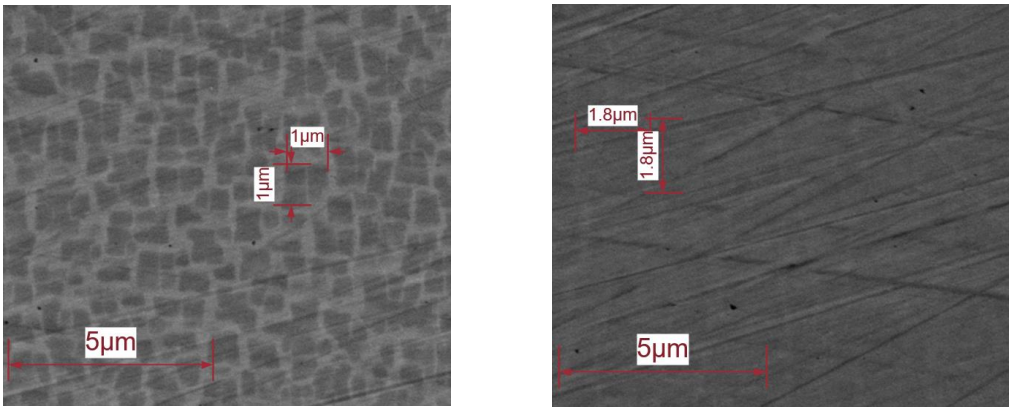


Figure. 4.9 Comparison of the microstructure of the dendrite core and the -dendrite
(a) Dendrite core, (b) Inter-dendrite

Table 4.5 Elemental Segregation Coefficients

	Al	Ti	Ta	Cr	Co	Mo	W
$C_{x,inter}$	8.01	0.88	4.9	8.38	8.87	0.4	8.18
$C_{x,core}$	6.89	0.46	4.29	7.89	9.2	1.15	12.31
k'_x	0.86	0.52	0.88	0.94	1.04	2.87	1.5

Figure 4.9 (a) shows the microstructure of the dendrite core region, while Figure 4.9 (b) shows the microstructure of the interdendritic region. A comparison reveals significant differences in the size and shape of the γ between the two regions; the microstructure in the dendrite core region is smaller and more uniformly distributed. Table 4.5 presents the composition data obtained from the two regions shown in Figure 4.9 and the calculated microsegregation coefficients for each element. Based on the k'_x values, the following

conclusions can be drawn: For the CM247LC alloy, Al, Ti, and Ta tend to segregate in the interdendritic regions; Mo and W tend to segregate in the dendritic regions; while Co and Cr show no obvious segregation tendencies. A comparison of this subsection with the previous one reveals that the macroscopic segregation tendencies of alloying elements do not align with their microsegregation tendencies.

4.3.5 Phase Diagram-Assisted Analysis

In this subsection, the JMatPro software was used to calculate the trends in the composition of the main phases of CM247LC and the trends in the composition of each alloying element in each phase using the Scheil-Gulliver model.

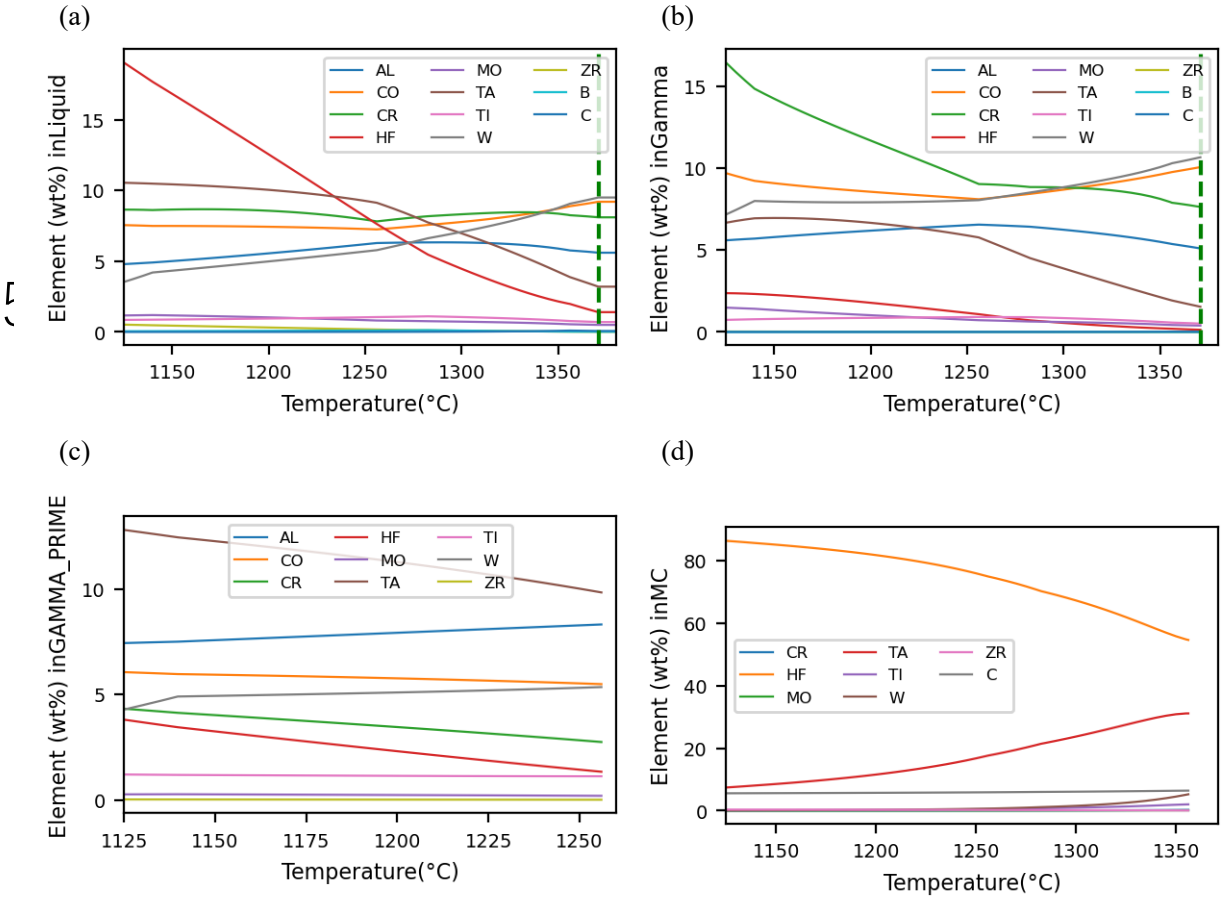
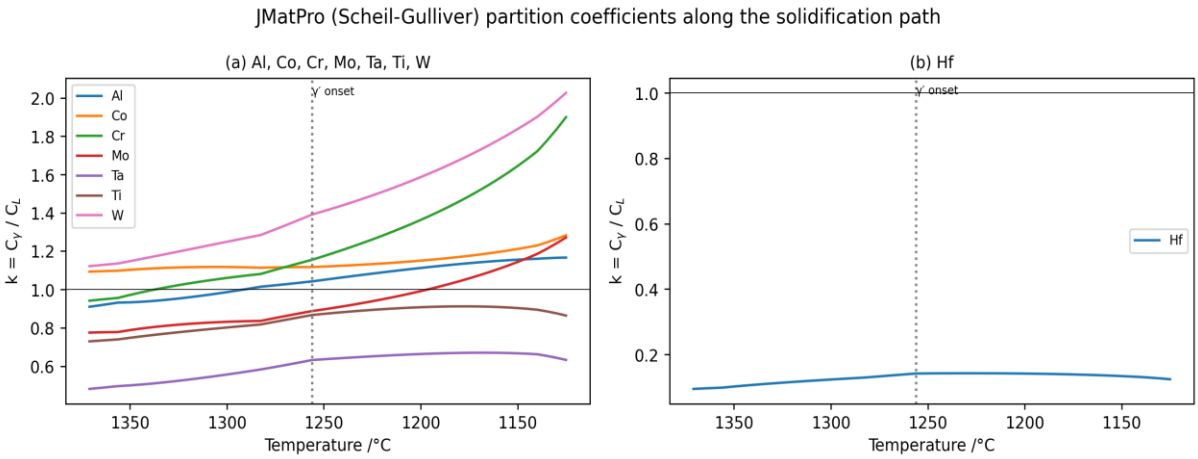


Figure 4.10 Composition trends of the major phases in CM247LC: (a) liquid phase; (b) γ phase; (c)γ' phase; (d) MC carbide



F16 ·JMatPro Scheil-path partition coefficients k(T) = Cγ/C_L

As shown in Figure. 4.10, the primary solute elements in the γ phase are Cr, Co, and W; the primary alloying elements in the γ' phase are W, Al, Ta, and Ti; Ta and Hf are the primary elements forming MC carbides and are enriched in the liquid phase during the late solidification stage.

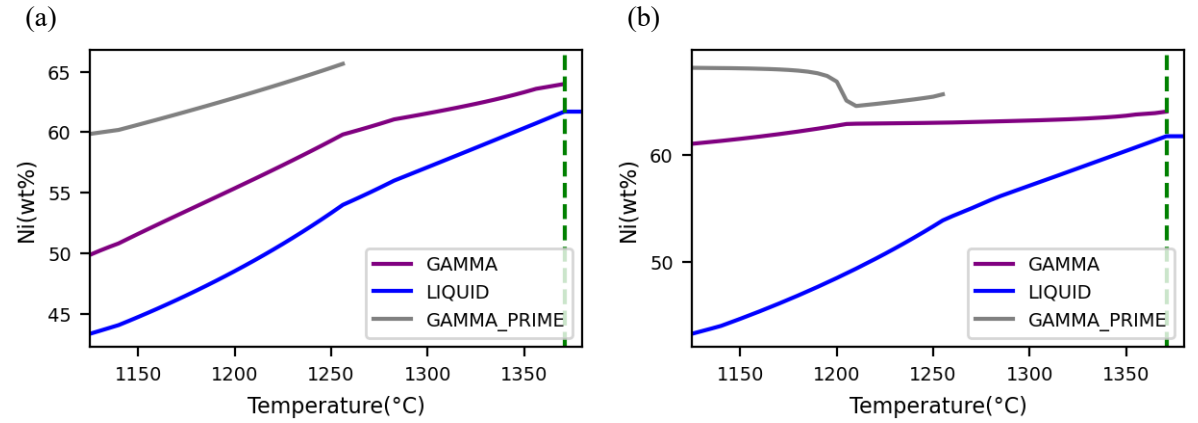


Figure. 4.11 Concentration of Ni in each phase (a) Instant concentration, (b) Average concentration

The nominal Ni content in CM247LC is 61.5 wt%. Under non-equilibrium solidification conditions, Ni in CM247LC primarily exists in the γ' and γ phases. The Ni content in each phase decreases monotonically as the solidification process progresses, and the solid-liquid partition coefficient k > 1.

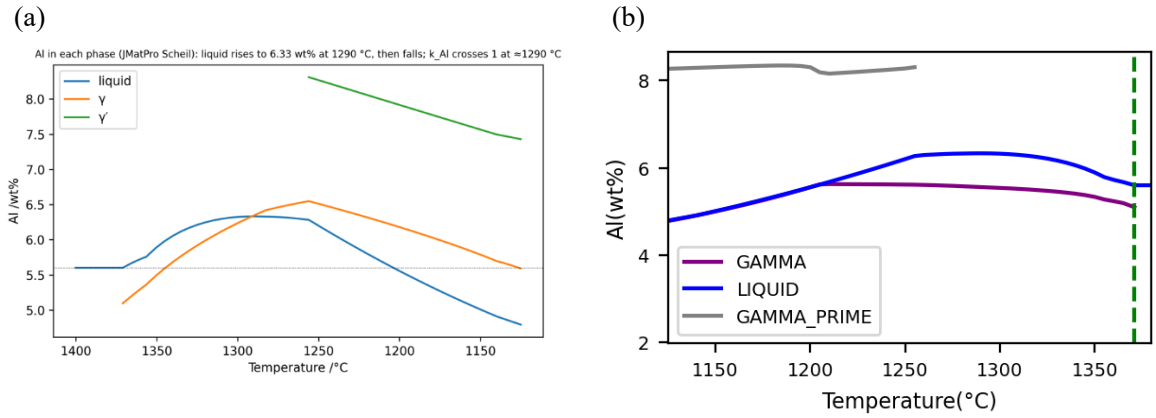


Figure. 4.12 Concentration of Al in each phase
(a) Instantaneous concentration (liquid Al peaks at 6.33 wt% at 1290 ℃)
, (b) Average concentration

The nominal Al content in CM247LC is 5.6 wt%, making it a phase-forming element forγ' .Under non-equilibrium solidification conditions, Al in CM247LC primarily exists in the γ' and γ phases. Liquid Al rises from 5.60 to 6.33 wt% at 1290 ℃ and then falls; k_Al is 0.91 at the liquidus and crosses 1 at about 1290 ℃; γ' appears at 1256 ℃.

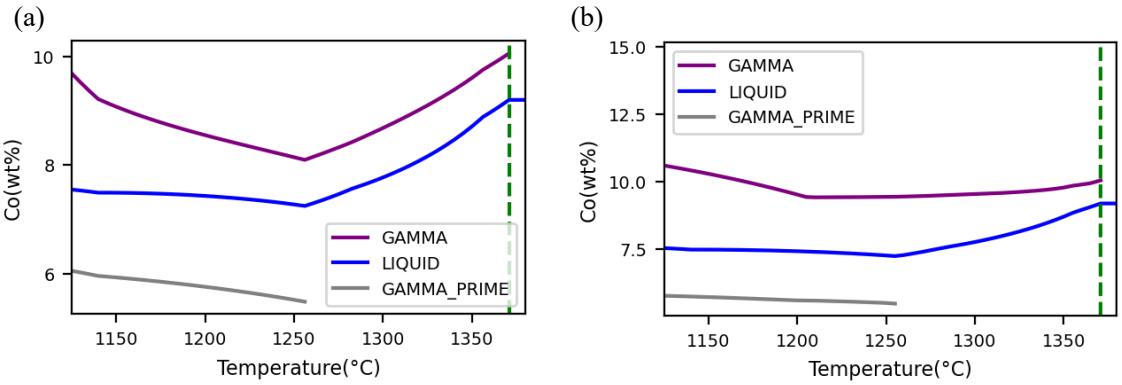


Figure. 4.13 Concentration of Co in each phase
(a) Instantaneous concentration, (b) Average concentration

The nominal Co content in CM247LC is 9.2 wt%, making it a phase-forming element (γ). Under non-equilibrium solidification conditions, Co in CM247LC primarily exists in the γ' andγ phases; during alloy solidification,$C_{Co}^{\gamma'}<C_{Co}^{Liquid}<C_{Co}^{\gamma}$. The liquid-phase Co content C_{Co}^{Liquid} first decreases and then increases during solidification; ultimately, Co in the as-cast microstructure tends to enrich in the dendrite stems and deplete in the interdendritic regions.At the end of solidification, the Co content in the liquid phase C_{Co}^{Liquid} is significantly lower than

the nominal composition; over the entire solidification process, Co tends to be distributed into the solid phase, with a solid-liquid partition coefficient k > 1.

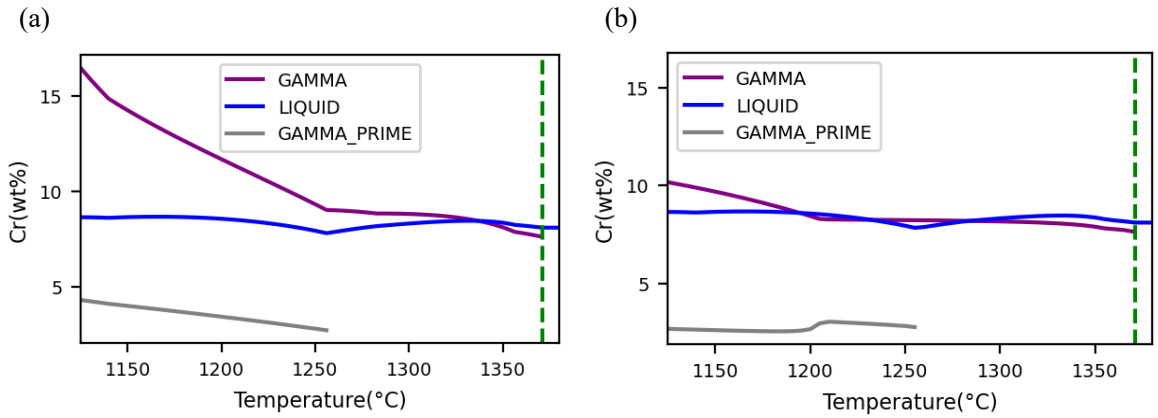


Figure. 4.14 Concentration of Cr in each phase:
(a) Instantaneous concentration, (b) Average concentration

The nominal Cr content in CM247LC is 8.1 wt%, making it a phase-forming element inγ .Under non-equilibrium solidification conditions, Cr in CM247LC primarily exists in the γ' and γ phases; during solidification, $C_{Cr}^{\gamma'} \ll C_{Cr}^{Liquid} < C_{Cr}^{\gamma}$. During solidification, C_{Cr}^{γ} is relatively close to C_{Cr}^{Liquid} ; C_{Cr}^{γ} generally shows an upward trend as the solidification temperature decreases, and Cr tends to be distributed between dendrites. Throughout the solidification process, C_{Cr}^{Liquid} does not exhibit a distinct rising or falling trend and differs little from the nominal composition; therefore, the solid-liquid distribution ratio of Cr is close to 1.

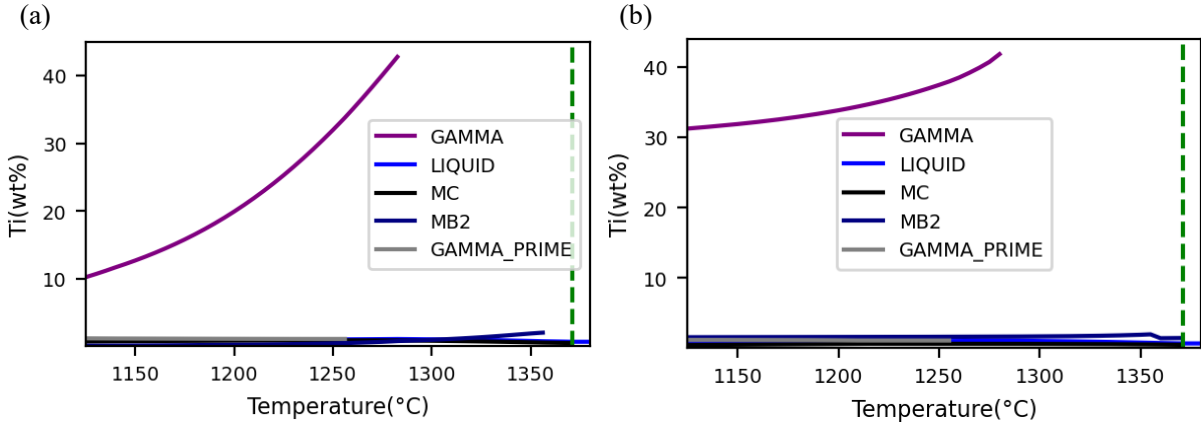


Figure. 4.15 Concentration of Ti in each phase (a) Instantaneous concentration, (b) Average concentration

The nominal Ti content in CM247LC is 0.7 wt%. In CM247LC, Ti primarily exists in the γ' and γ phases and serves as a constituent of the γ' phase. During solidification,$C_{Ti}^{\gamma} < C_{Ti}^{Liquid} < C_{Ti}^{\gamma'}$. During solidification,C_{Ti}^{Liquid} first increases and then decreases;

in the as-cast microstructure, Ti tends to be distributed in the interdendritic regions.At the end of solidification, the Ti concentration in the liquid phase C_{Ti}^{Liquid} is higher than the nominal composition. Over the entire solidification process, Ti tends to be distributed into the liquid phase, with a solid-liquid partition coefficient k < 1.

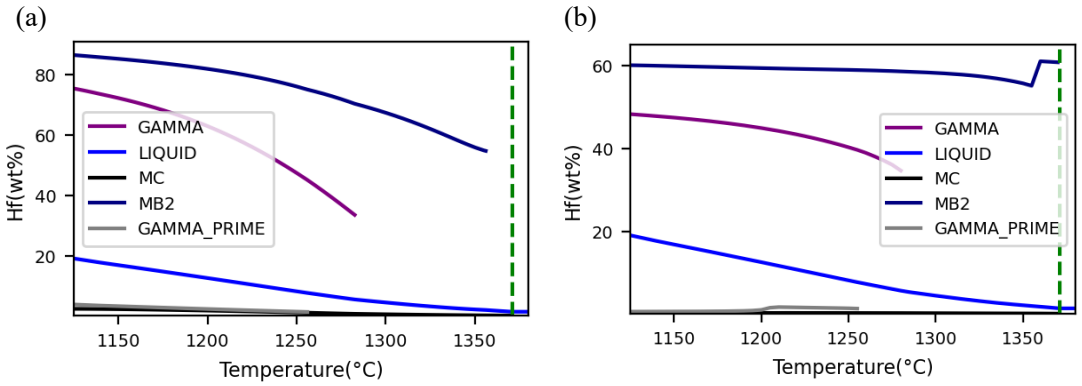


Figure. 4.16 Hf concentration in each phase: (a) Instantaneous concentration, (b) Average concentration

The nominal Hf content in CM247LC is 1.4 wt%. Under non-equilibrium solidification conditions, Hf in CM247LC primarily exists in MC carbides,γ' phases,γ phases, and MB_2 borides. During solidification,$C_{Hf}^{\gamma} \ll C_{Hf}^{Liquid} < C_{Hf}^{MB_2} < C_{Hf}^{MC}$, Hf tends to be present in carbides and borides in the as-cast microstructure and serves as a γ' phase-forming element.During alloy solidification,C_{Hf}^{Liquid} rises significantly; Hf shows a marked tendency to partition into the liquid phase, with a solid-liquid partition coefficient k < 1.

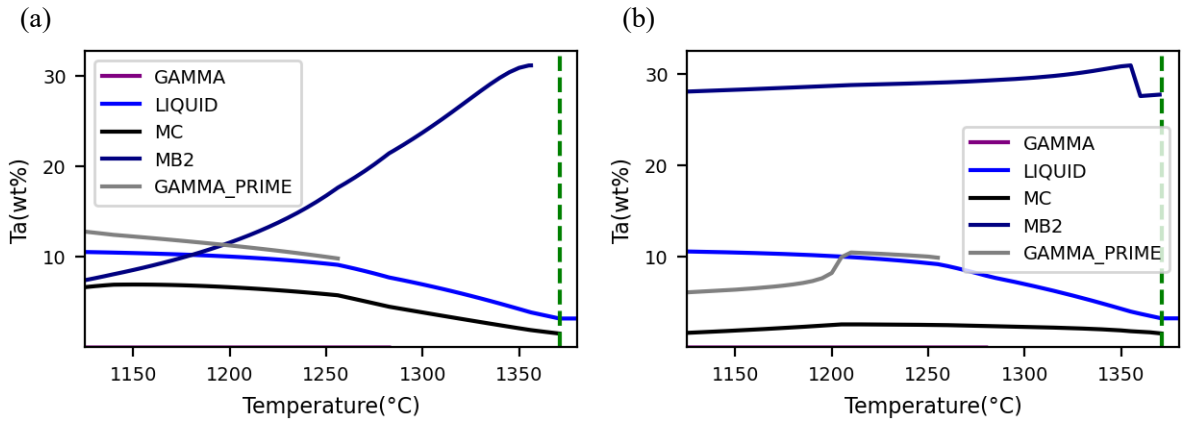


Figure. 4.17 Concentration of Ta in each phase (a) Instantaneous concentration, (b) Average concentration

The nominal concentration of Ta in CM247LC is 3.2 wt%, making it a phase-forming element in γ' .Under non-equilibrium solidification conditions, Ta in CM247LC primarily exists in the γ' and γ phases and is also a key constituent of MC carbides; during solidification,$C_{Ta}^{\gamma}<C_{Ta}^{\gamma'}<C_{Ta}^{Liquid}<C_{Ta}^{MC}$. Ta has low solubility in the γ phase and tends to be distributed in the interdendritic regions. C_{Ta}^{Liquid} monotonically increases with decreasing temperature during solidification; Ta accumulates in the residual liquid phase at the end of solidification, with a solid-liquid partition coefficient k < 1.

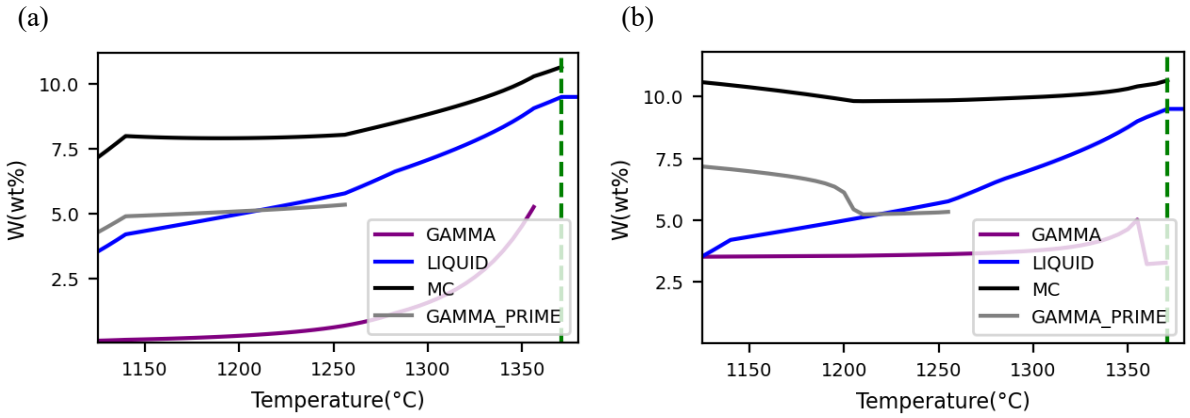


Figure. 4.18 Concentration of W in each phase (a) Instantaneous concentration, (b) Average concentration

The nominal composition of W in CM247LC is 9.5 wt%, making it a phase-forming element in γ .Under non-equilibrium solidification conditions, W in CM247LC primarily exists in the γ' and γ phases; during solidification, $C_{W}^{Liquid}<C_{W}^{\gamma'}<C_{W}^{\gamma}$. During solidification,C_{W}^{Liquid} monotonically decreases, and W in the as-cast microstructure tends to be distributed to the dendrite stems. At the end of solidification, the W content in the liquid phase C_{W}^{Liquid} is significantly lower than the nominal composition; for the overall solidification process, W tends to be distributed to the solid phase, with a solid-liquid distribution coefficient k > 1.

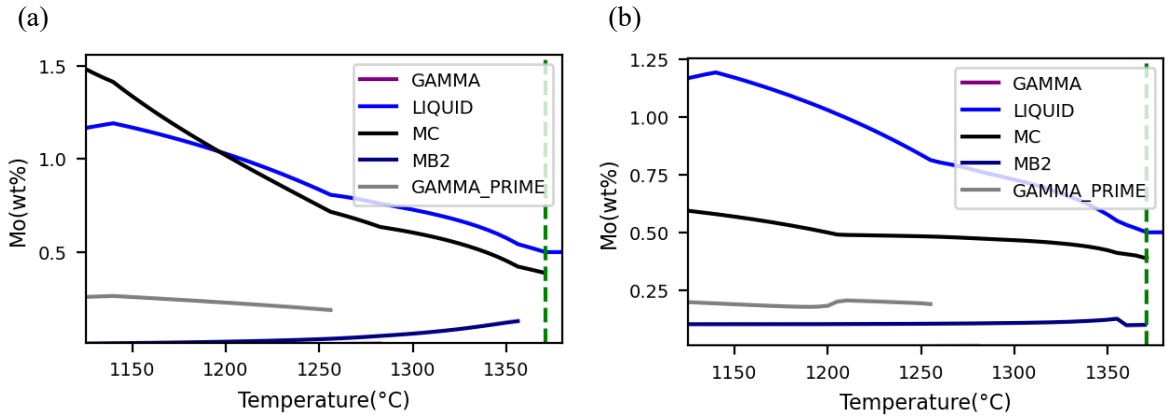


Figure. 4.19 Concentration of Mo in each phase (a) Instantaneous concentration, (b) Average concentration

The nominal concentration of Mo in CM247LC is 0.5 wt%; it is a phase-forming element of γ , primarily present in the γ' and γ phases. During solidification, $C_{Mo}^{\gamma'} < C_{Mo}^{\gamma} < C_{Mo}^{Liquid}$. At the end of solidification, the concentration of Mo in the liquid phase C_{Mo}^{Liquid} is significantly higher than the nominal concentration. For the overall solidification process, Mo tends to be distributed into the liquid phase, and the solid-liquid distribution coefficient k < 1.

4.3.6 Phase Stability Analysis

Phase stability assessment refers to the evaluation of the tendency for harmful phases to precipitate in topologically dense clusters. The methods for phase stability assessment include the average electron vacancy theory[77] and the d-electron theory[78] . The theoretical foundations of both are derived from the electronic structure of nickel-based alloys. For multi-component nickel-based high-temperature alloys containing various alloying elements , the calculation formula for the average electron vacancy $\overline{N_v}$ is as follows:

$$\overline{N_v} = \sum_i m_i \times (N_{v_i}) \quad (4.11)$$

Where m_i and N_{v_i} denote the atomic mole fraction and the number of electronic vacancies, respectively, of alloying element i in the alloy under evaluation. When N_v exceeds the critical value N_v^c , the alloy microstructure tends to precipitate the TCP phase; when N_v is less than the critical value N_v^c , the alloy microstructure is stable and does not precipitate the TCP phase.

The d-electron theory, also known as the New Phase Composition Method (New PHACOMP), is a method for evaluating the phase stability of high-temperature alloys

developed based on DV-Xα cluster molecular orbital calculations. For multi-component nickel-based high-temperature alloys, the average value $\overline{M_d}$ can be calculated using the following formula:

$$\overline{M_d}=\sum_i m_i \times(M_{d_i}) \quad (4.12)$$

According to the New Phase Composition Method, the lower the M_d value of the alloy, the more favorable it is for phase stability.

The phase stability of castings is closely related to their local chemical composition. Many researchers have demonstrated that the segregation of solute elements increases the tendency for the TCP phase to form in nickel-based high-temperature alloys. Yue Xiaoda et al.[79] found that the supersaturation of refractory elements in the γ phase is the primary cause of TCP phase precipitation in the γ phase; increased segregation of refractory elements enhances the tendency for TCP phase precipitation. Caron [1]proposed that phase stability criteria vary depending on the properties of the TCP phase and the as-cast microstructure; inter-dendritic segregation results in lower N_v^c values for cast alloys compared to forged alloys. Cieslak [80] found through welding experiments that segregation behavior during solidification leads to solute element enrichment in the interdendritic regions, resulting in higher M_d values in these regions compared to the dendrite trunks. Directional solidification investment casting is a key method for forming nickel-based high-temperature alloy blades; macrosegregation induced by the solidification sequence causes differences in phase stability at the macroscopic scale.

Table 4.6 N_v and M_d values of various elements[77,78]

Element		N_v	M_d,eV
	Ti	6.66	2.271
3d	Cr	4.66	1.142
	Co	1.71	0.777
	Ni	0.66	0.717
4d	Zr	6.66	2.944
	Mo	4.66	1.550
	Hf	6.66	3.020
5d	Ta	5.66	2.224
	W	4.66	1.655
	Al	7.66	1.900

According to the standards[83], the composition of the γ phase in nickel-based alloys consists of borides, carbides, and the remaining solid solution after the precipitation of the γ' phase. Calculations assume that Ni, Cr, Ti, and Mo form M_3B_2 borides first; subsequently, Hf, Ta, Ti, and Zr form MC carbides with C, while Ni, Mo, W, and Co form M_6C carbides with C; finally, all remaining Al, Ti, Hf, Ta, a portion of Cr, and three times the amount of Ni relative to the sum of the above alloying elements form the γ' phase. Figure 4.20 shows the axial macrosegregation curve of the γ solid solution composition. The equilibrium single γ phase consists of Ni, Cr, Co, Mo, and W. The proportions of each alloying element in the γ phase solid solution differ significantly from the initial composition; the atomic mole fraction of Ni in the γ phase of the CM247LC alloy is approximately 50%;the atomic fractions of Cr and Co are similar, with a combined proportion of approximately 45%; the atomic fraction of W is about 6%, and that of Mo is about 1%.

The elemental compositions of the γ solid solution differ considerably from the overall alloy composition, as shown in Fig. 4.20(a). The atomic fraction of Ni in the γ phase is approximately 47%, representing a substantial decrease compared with its value of approximately 63% in the overall composition. Notably, the minima in the Ni concentration profile within the γ phase correspond to the positions where Ti, Ta, and Hf exhibit concentration peaks in the overall composition. This behavior can be attributed to the precipitation of the γ′ phase, which preferentially incorporates Ni together with γ′-forming elements such as Ti, Ta, and Hf. Since CM247LC is a low-carbon and low-boron Ni-based superalloy, the amount of MC carbides formed is relatively limited. Consequently, γ′ precipitation plays a major role in the redistribution of these alloying elements. The formation of the γ′ phase consumes a substantial amount of Ni, resulting in a reduced Ni atomic fraction in the remaining γ matrix, particularly in regions where the overall concentrations of Ti, Ta, and Hf are relatively high.

The measured wt% were rescaled element-wise before the γ-phase calculation (Ta ÷1.39, Al ÷1.27, Ti ÷1.35, Mo ÷1.35, Cr ÷1.05, Ni (−0.092) ÷0.945, …) so that the means approach the nominal composition; factors listed in Appendix B5.The compositional distributions of Co, Cr, W, and Mo in the γ phase are presented in Figs. 4.20(b), (c), and (d). As Al, Ti, Hf, and Ta are preferentially incorporated into the γ′ phase together with Ni, the relative atomic fractions

of the remaining alloying elements in the γ matrix increase significantly. In particular, the atomic fractions of Cr and Co in the γ phase are approximately equal, and their combined atomic fraction increases from approximately 20% in the overall composition to 45% in the γ phase. Similarly, the atomic fraction of W increases from approximately 3% to 6%, while that of Mo increases from approximately 0.3% to 0.6%. Furthermore, compared with the overall composition profiles, the elemental distributions in the γ phase exhibit more pronounced macroscopic compositional inhomogeneity, characterized by substantially increased fluctuation amplitudes. These fluctuations generally follow trends similar to those of the alloying elements with $k^m < 1$in the overall composition.

Overall, the Ni concentration profile in the γ phase exhibits pronounced minima at positions corresponding to the concentration maxima of Ti, Ta, and Hf in the overall composition. In contrast, the concentration profiles of Cr, Co, Mo, and W in the γ phase exhibit maxima at these positions. These results indicate that γ′ precipitation and the associated elemental partitioning significantly influence the compositional distribution and amplify the apparent macrosegregation of alloying elements in the remaining γ matrix.

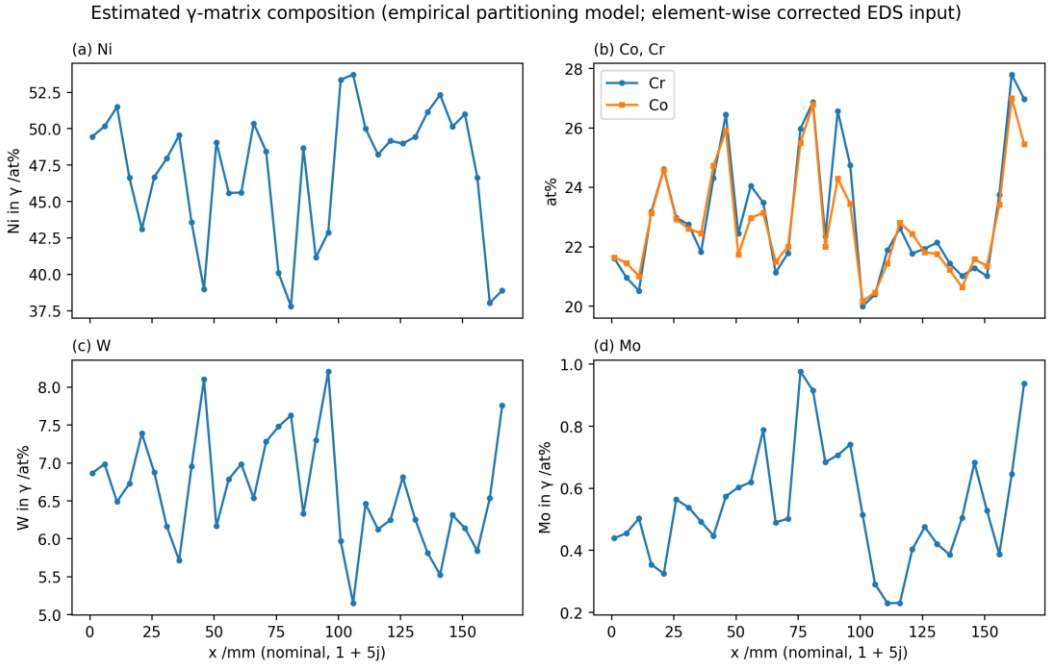


Figure. 4.20 Estimated γ-matrix composition along the bar:
(a) Ni; (b) Co, Cr; (c) W; (d) Mo

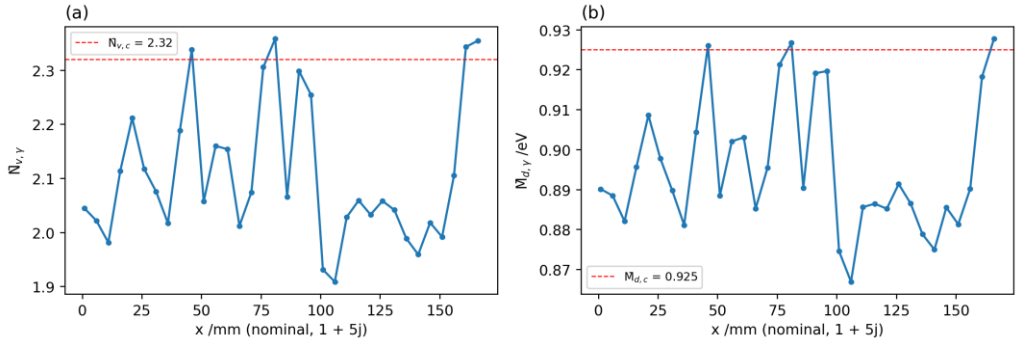


Figure. 4.21 Average electron-vacancy number N̄v (a)
and average d-orbital energy M̄d (b) of the γ phase along the bar

The phase stability of Ni-based superalloys can be evaluated using two parameters: the average electron vacancy number ($\bar{N}_{v,\gamma}$), derived from electron vacancy theory, and the average d-orbital energy level ($\bar{M}_{d,\gamma}$), derived from d-electron theory. Figure 4.21 presents the phase stability parameters calculated from the γ-phase compositions shown in Fig. 4.20, using the elemental N_v and M_d values listed in Table 4.6. Both parameters exhibit generally consistent variation trends along the specimen.

The variations in these phase stability parameters closely follow the compositional fluctuations of the alloying elements in the γ phase. As shown in Table 4.6, Ni has the lowest N_v and M_d values among the constituent elements, at 0.66 and 0.717, respectively. Consequently, regions with a lower Ni atomic fraction and higher fractions of other alloying elements tend to exhibit higher values of $\bar{N}_{v,\gamma}$ and $\bar{M}_{d,\gamma}$, indicating reduced phase stability and an increased tendency toward the precipitation of topologically close-packed (TCP) phases.

Figure 4.21(a) shows the distribution of the average electron vacancy number, $\bar{N}_{v,\gamma}$, which ranges from approximately 1.90 to 2.35. At several local maxima, $\bar{N}_{v,\gamma}$ exceeds the critical value of $\bar{N}_{v,c} = 2.32$[81], suggesting a potential tendency for TCP phase precipitation in these regions.

Figure 4.21(b) presents the distribution of the average d-orbital energy level, $\bar{M}_{d,\gamma}$, which varies between approximately 0.870 and 0.928. According to the critical criterion reported in Ref. [78], the threshold for TCP phase formation is $\bar{M}_{d,\gamma} = 0.925$. The calculated values at the local maxima approach this critical threshold, with a few peaks slightly exceeding it. These results indicate that certain regions of the γ matrix may be susceptible to TCP phase precipitation.

Overall, the two phase stability criteria exhibit consistent trends and identify similar regions of potentially reduced phase stability. The local enrichment of alloying elements in the γ matrix, accompanied by a decrease in the Ni atomic fraction, increases the calculated phase stability parameters and may consequently enhance the susceptibility to TCP phase formation.

4.4 Analysis of Thermal-Field and Compositional Fluctuations

During alloy solidification, the evolution of the solute concentration field is closely coupled with the thermal and fluid flow fields. As discussed previously, convection promotes solute transport within the melt, thereby influencing the axial and radial compositional distributions in the solidified specimens. ProCAST enables coupled simulations of heat transfer and fluid flow and allows the extraction of numerical results at arbitrary locations within the finite element model, facilitating detailed quantitative analysis. To systematically investigate the macrosegregation and compositional fluctuations of alloying elements during the solidification of Ni-based superalloys, this section analyzes the relationship between thermal-field evolution and elemental compositional fluctuations using temperature-field data extracted from the numerical simulations.

4.4.1 Multiphysics Coupling

The theoretical basis for fluid calculations in the ProCAST software is the multiphysics coupling model proposed by SCHNEIDER in 1994[65-68] . In the two-way coupling calculation of the temperature field and the flow field, changes in melt density obey the following conservation relationship:

$$\rho^{m}(T)=\rho_{ref}^{m}+\left(\frac{\partial\rho}{\partial T}\right)_{m}(T-T_{ref}^{m}) \tag{4.13}$$

4.4.2 Temperature Field Simulation Results

To investigate the evolution of the temperature field within the specimen during directional solidification, two parallel axial paths were defined in the finite element model for data extraction. As shown in Fig. 4.22, Axis 2 coincides with the geometric centerline of the specimen and serves as the primary location for thermal-field analysis, whereas Axis 1 is located 5 mm radially away from the centerline and serves as a reference for comparing

thermal-field characteristics at different radial positions. Along both axes, data extraction points were arranged at uniform intervals of 5 mm in the solidification direction, consistent with the axial sampling intervals used for experimental compositional measurements. This arrangement facilitates subsequent correlation analysis between the simulated temperature field and the measured elemental compositional fluctuations. The multiphysics coupled simulations were performed using the thermal boundary conditions established in Chapter 3. Temperature–time data were extracted along both axes from the ProCAST finite element simulation results, with subsequent quantitative analyses primarily focusing on the centerline data (Axis 2). The temperature evolution curves at the selected points along Axis 2 are presented in Fig. 4.23. Although the sampling points are equally spaced along the axial direction, the time intervals between adjacent points reaching the liquidus temperature are not uniform but exhibit noticeable fluctuations, indicating that the advancement of the solidification front is characterized by a certain degree of unsteadiness.

By processing the extracted temperature-field data, several local thermal parameters were obtained, including the cooling rate, solid–liquid interface advancement velocity, local undercooling, and local temperature gradient. Their variations with solid fraction are presented in Fig. 4.24. Specifically, V1 and V2 represent the local cooling rates along Axes 1 and 2, respectively; R1 and R2 denote the corresponding solid–liquid interface advancement velocities; $\Delta T1$ and $\Delta T2$ indicate the local undercooling; and G1 and G2 represent the local temperature gradients. Subscripts 1 and 2 correspond to Axis 1, located 5 mm from the specimen centerline, and Axis 2, coinciding with the specimen centerline, respectively. In Fig. 4.24, the cooling rate, interface advancement velocity, and local undercooling are plotted against the left vertical axis, whereas the local temperature gradient is plotted against the right vertical axis.

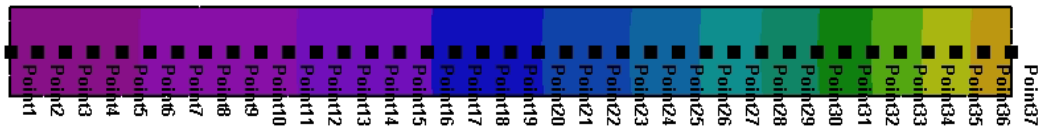


Figure. 4.22 Schematic of reference points along Axes 2

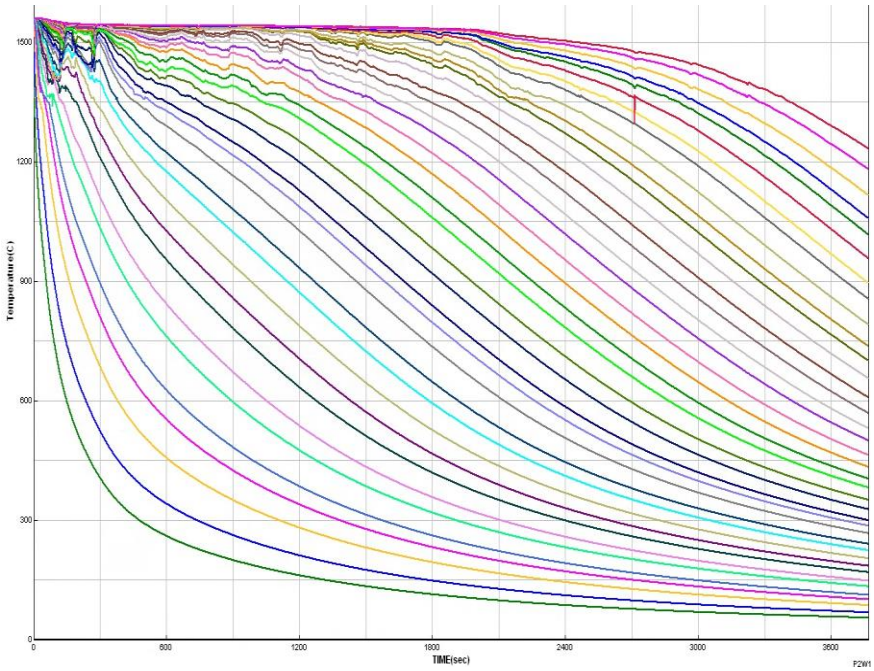


Figure. 4.23 Thermal histories at reference points along Axes 2

To quantitatively characterize the evolution of local thermal conditions during directional solidification, the temperature–time and solid fraction–time data extracted from the ProCAST finite element simulations were processed using a Python program. Four local thermal parameters were calculated, including the cooling rate (V), solid–liquid interface advancement velocity (R), temperature gradient (G), and undercooling (ΔT). A total of 37 sampling points were uniformly distributed along the 180 mm axial length of the specimen, with a spacing of 5 mm between adjacent points.

The local cooling rate, V, was calculated from the temperature–time history at each individual sampling point using a finite-difference method. For each point, the temperature records within the range of 1200–1400 °C were first identified. The first two valid time records within this interval were then selected, and the negative ratio of the corresponding temperature difference to the time difference was used to determine the local cooling rate, expressed in K/s. This approach characterizes the local rate of temperature decrease within the selected high-temperature interval.

The solid–liquid interface advancement velocity, R, was estimated from the differences in characteristic arrival times between adjacent axial sampling points. For each point, the first recorded time at which the temperature entered the range of 1200–1400 °C was identified. The axial spacing of 5 mm was divided by the time difference between two adjacent points, with

the appropriate unit conversion, to obtain the characteristic isotherm advancement velocity in mm/min. Because this procedure uses the first valid time record within a prescribed temperature interval rather than the exact crossing time of the liquidus temperature, the calculated velocity represents an approximation of the solidification-front advancement velocity.

The local temperature gradient, G, was estimated using an axial interval spanning two adjacent point spacings, corresponding to a distance of 10 mm. For each sampling point, the temperature and time associated with the first entry into the range of 600–1400 °C were identified. The temperature at the same spatial location was then extracted at the corresponding reference time of the point located two sampling intervals downstream. The difference between these two temperatures was divided by the 10 mm axial distance to obtain an effective local temperature gradient in K/mm. This procedure estimates the temperature gradient from the temporal temperature variation at a fixed location during the axial advancement of the solidification front and therefore represents a characteristic-time-based finite-difference approximation.

The local undercooling, ΔT, was determined from the evolution of the solid fraction. For each sampling point, the first recorded time at which the solid fraction reached or exceeded 5% was identified from the solid fraction–time data. The corresponding temperature was then extracted from the temperature–time data, and the local undercooling was calculated as the difference between the alloy liquidus temperature and the temperature at that time. A liquidus temperature of 1370.9 °C was adopted in this study, and the resulting undercooling was expressed in K.

The calculated thermal parameters provide quantitative information on the axial evolution of the solidification conditions. These results form the basis for subsequent comparisons between the initial and steady solidification stages and for investigating the relationship between local temperature-gradient fluctuations and elemental compositional fluctuations.

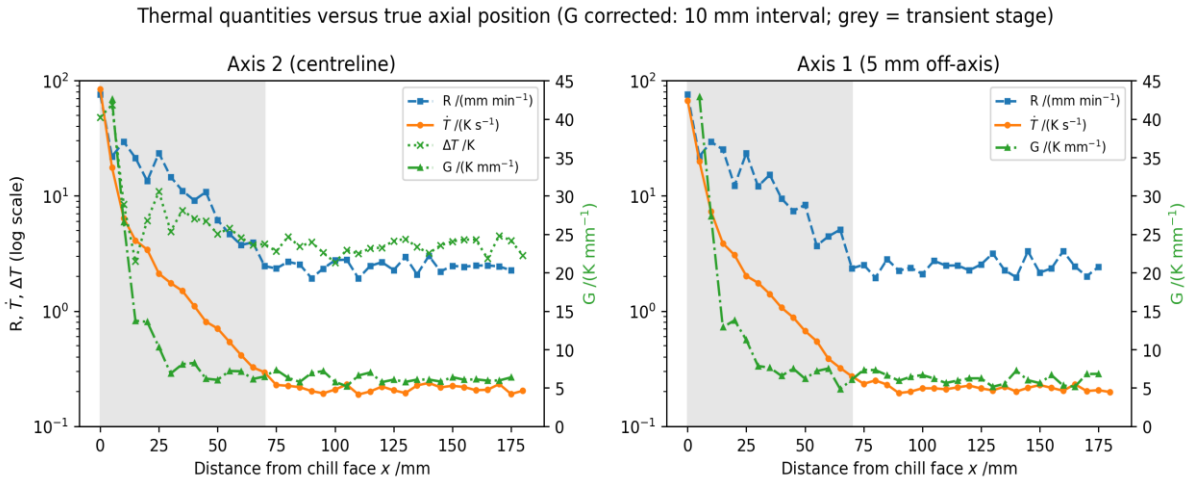


Figure. 4.24 Cooling rate, interface velocity, undercooling and temperature gradient on the two axes versus true axial position (G corrected to the 10 mm interval; grey = transient stage)

4.4.3 Temperature-field fluctuation and composition fluctuation

Fig. 4.25 compares the normalized experimental compositions with the local temperature gradient obtained from the simulation of §4.4.2, the blue polyline being the local temperature gradient G ahead of the solid–liquid interface. Apart from the initial stage that has a high solidification rate, a clear correspondence exists between the composition fluctuation and the local-temperature-gradient fluctuation: the concentration peaks of the rejecting elements (k_E<1) are in phase with the G peaks, whereas the concentration peaks of the absorbing elements (k_E>1) are in anti-phase with the G peaks.

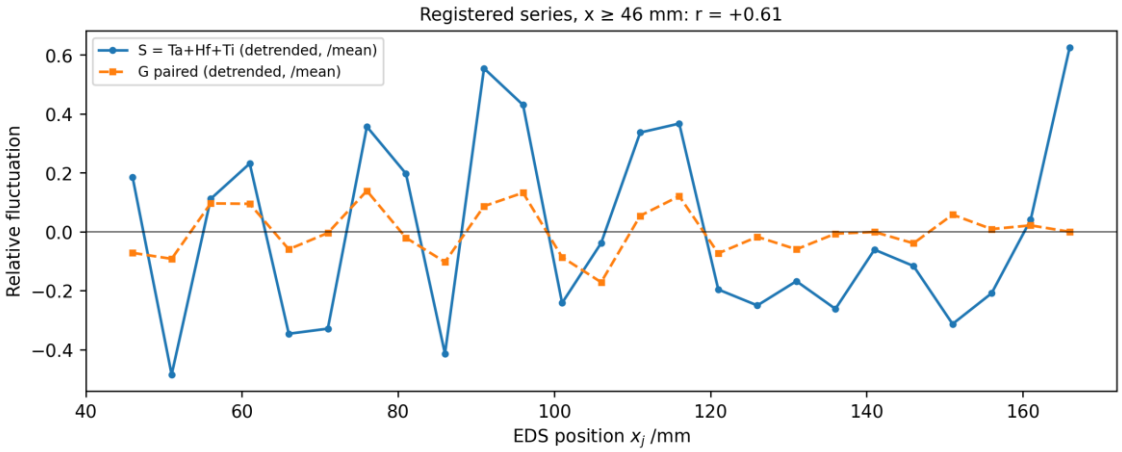


Fig4.25 Registered detrended series of S = Ta+Hf+Ti and centreline G (x ≥ 46 mm, r = +0.61)

To resolve this correspondence more clearly in the position domain, the simulated temperature gradient and the measured compositions are re-plotted along Axes 2 (solidification

direction): Fig. 4.26 is the position-domain local temperature gradient $G(x)$ with a magnified steady stage, and Fig. 4.27 is the normalized composition fluctuation grouped by segregation character.

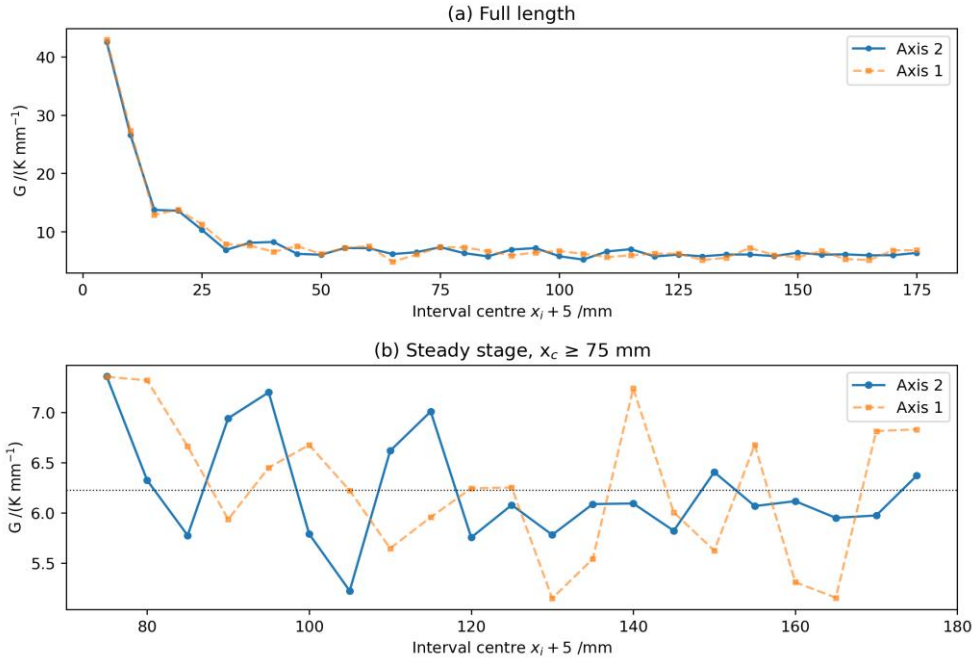
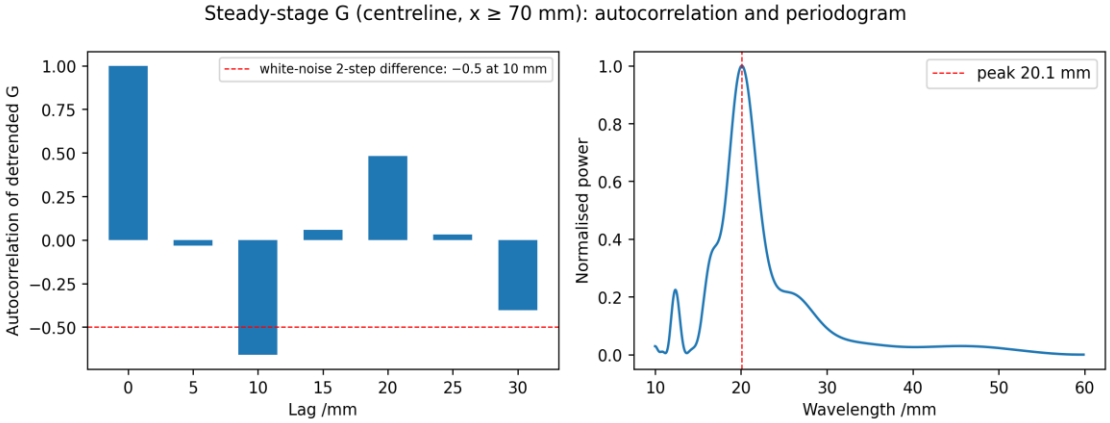
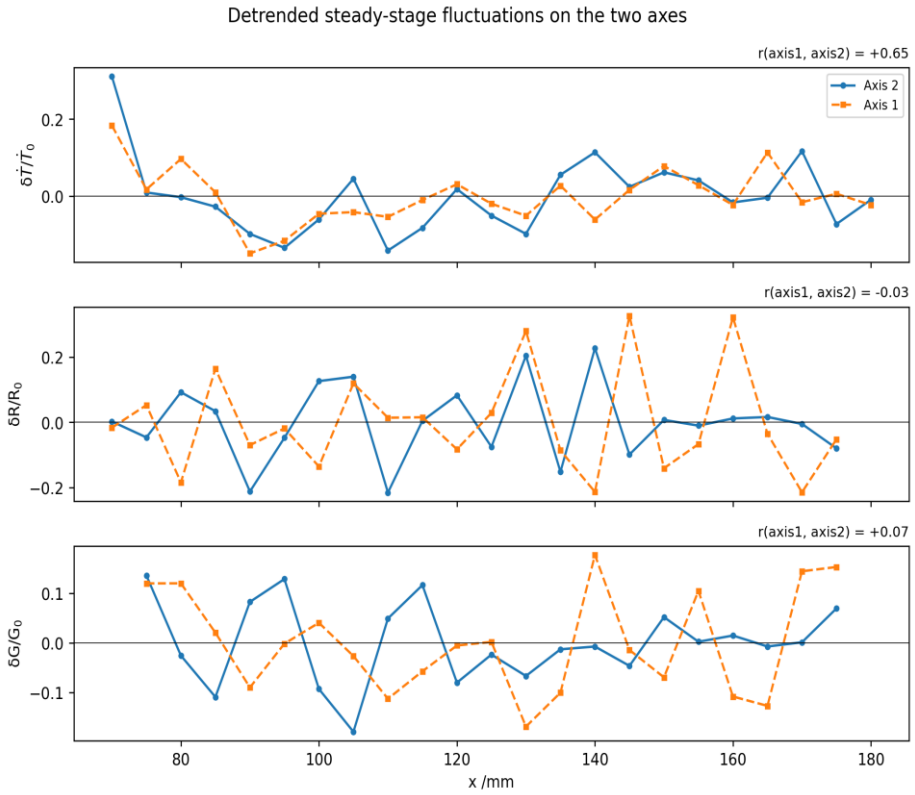


Fig. 4.26 Position-domain local temperature gradient G(x):
(a) full length; (b) steady stage magnified (x_c ≥ 75 mm)



Autocorrelation and periodogram of the detrended steady-stage centreline G
(dashed: −0.5 expected for a two-step difference of white noise)



Detrended steady-stage fluctuations of Ṫ, R and G on the two axes with their correlation coefficients

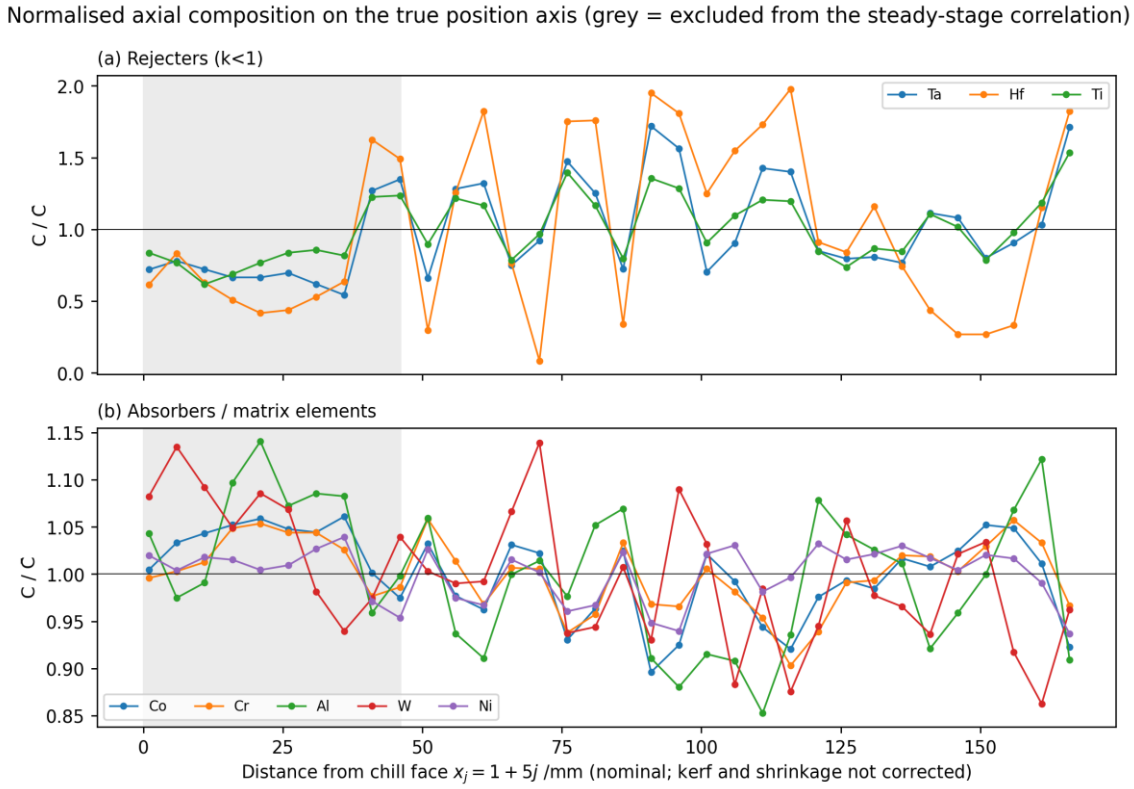


Fig. 4.27 Normalised axial composition on the true position axis:
(a) rejecters Ta, Hf, Ti; (b) absorbers/matrix elements Co, Cr, Al, W, Ni.
(grey = excluded from the correlation window)

According to growth-layer theory[84], the temperature undulation caused by melt flow is an important origin of the composition undulation.

$$C_s(t) = k_0\{ C^{(0)} + \left|C_\omega^{(1)}\right|\cos(\omega t + \phi) \} \tag{4.14}$$

In a gravity field, the density difference of the liquid is the driving force of natural convection. Under upward directional solidification, if the interdendritic liquid at some depth of the mushy zone is less dense than the bulk liquid above it, the lighter lower liquid tends to rise under buoyancy, perturbing the original flow; once the flow becomes unstable it induces temperature undulation.

For CM247LC this density inversion has a definite compositional origin: the strongly absorbed heavy refractory element W is depleted in the interdendritic liquid and the light element Ti (and Cr) is enriched there, both of which make the interdendritic liquid lighter[89]; Ta and Hf are also enriched in the interdendritic liquid, but as heavy elements they make it denser and partly offset the instability — which is precisely why a higher Ta/W ratio suppresses freckling[92]; Al, whose partition coefficient about 0.91 at the liquidus (JMatPro), crossing 1 at about 1290 ℃, has only a weak effect on the density of the upper mushy zone. With Ta/W≈0.34 in CM247LC, the W depletion dominates and the interdendritic residual liquid becomes lighter than the bulk liquid above it. As shown in Fig. 4.28(b), the liquid density of CM247LC computed by JMatPro along the solidification path reaches a minimum at about 1345–1350 ℃, i.e. a density inversion of the mushy-zone liquid occurs during solidification, providing the driving force for thermosolutal convection; with the steady-stage G and $\dot{T}$, the inverted layer is about 3.4–4.2 mm thick and material resides in it for about 100–120 s. This mechanism agrees with the consensus of current studies on freckles/segregation channels in Ni-based superalloys: the light interdendritic liquid rises and forms solute plumes at the solidification front, and after mutual competition a few develop into stable segregation channels[90,92].

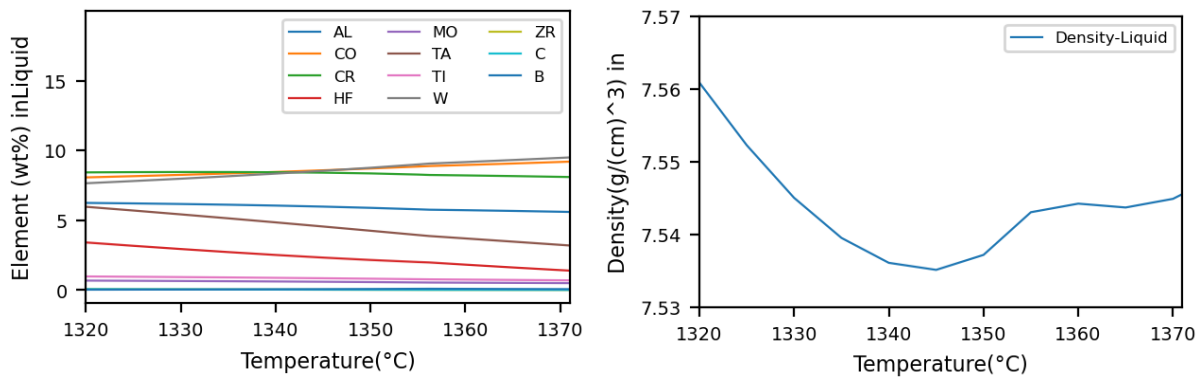


Fig 4.28 Changes in liquid phase concertration and density of CM247LC
(a)Concertration,(b) Density

Convection makes the local temperature gradient $G(t)$ oscillate at high frequency (Fig. 4.24(b) and the magnified steady stage in Fig. 4.26). Note that Eq. (4.13) is a linear thermal density model: if $(\partial\rho/\partial T)_m$ is taken as a constant in the simulation, the simulated convection is purely thermal — in the Bridgman furnace the outer surface of the bar loses heat faster by radiation, the radial temperature difference tilts the isopycnals, no hydrostatic equilibrium exists and thermal convection arises without any threshold; the oscillation of G still originates from convection, but the simulation does not explicitly contain the solutal density inversion described above. It should also be noted that the curves labelled V_1, V_2 in Fig. 4.24 are cooling rates, which are nearly constant in the steady stage (about 0.21 K/s, coefficient of variation about 7%, see Table 4.11); the true solid–liquid interface advancing velocity is labelled R_1, R_2 and oscillates regularly by about ±22% in the steady stage (mean 2.46 mm/min, i.e. 41 μm/s, equal to the pulling speed). The three are linked by the kinematic identity of the cooling rate (Eq. (2.15) of this thesis):

$$\dot{T} = G \cdot R \Rightarrow R(t) = \frac{\dot{T}}{G(t)} \tag{4.15}$$

Since the cooling rate $\dot{T}$ is nearly constant in the steady stage, linearizing Eq. (4.15) about the operating point (R_0, G_0) gives the relative-fluctuation relation between the interface advancing velocity and the temperature gradient:

$$\frac{\delta R}{R_0} = -\frac{\delta G}{G_0} \tag{4.16}$$

i.e. R and G oscillate 180° out of phase: R falls to a trough when G reaches a peak. This is consistent with the in-situ synchrotron observation by Reinhart et al.[87] that solutal buoyant

flow makes the dendrite-tip growth velocity oscillate during directional solidification of CMSX-4, and is precisely the physical origin of the R_1, R_2 oscillation in Fig. 4.24(b).

4.4.3.1 Compositional response — effective partition coefficient: static effect and fluctuation correction

The interface advancing velocity R acts on the solid composition through the effective partition coefficient. By Eq. (2.5), under partial mixing of the liquid the effective partition coefficient is

$$k_E(R) = \frac{k_0}{k_0 + (1 - k_0)\exp(-P_c)}, \qquad P_c = \frac{R\,\delta}{D_L}$$

where δ is the thickness of the effective solute-exchange layer at the interface. The first-order sensitivity of k_E to P_c is

$$\frac{\partial k_E}{\partial P_c} = \frac{k_0(1-k_0)\exp(-P_c)}{[\,k_0+(1-k_0)\exp(-P_c)\,]^2} \tag{4.17}$$

From Eq. (4.17), the sign of $\partial k_E/\partial P_c$ always equals sgn$(1-k_0)$ (the denominator is always positive), so k_E moves monotonically from k_0 toward 1 as P_c increases. A static conclusion follows at once: as long as a finite effective exchange layer exists at the interface ($P_c>0$), k_E must lie between k_0 and 1, i.e. $|k_E - 1| \le |k_0 - 1|$. This is the main reason why the measured fitted k_E (Table 4.3) is generally closer to 1 than the phase-diagram k_J (Table 4.4): the measured k_E of $k<1$ elements exceeds k_J, that of $k>1$ elements is below k_J, and the measured macrosegregation is weaker than the pure phase-diagram prediction. Inverting Eq. (2.5) gives the effective Péclet number of each element directly from the two tables:

$$P_c = \ln\frac{k_E\,(1 - k_0)}{k_0\,(1 - k_E)}$$

The closer k_E is to unity, the more sensitive P_c becomes to errors in k_E. By substituting the values of k_0(Table 4.4) and k_E(Table 4.3), the results are obtained as presented in Table 4.7.

Table 4.7. Effective Péclet numbers obtained by inversion and the corresponding sensitivity magnitudes $\left|\frac{\partial P_c}{\partial k_E}\right|$

Element	k0_T4_4	k0_liquidus	kE_10	kE_17	kE_34	Pc_34	DL_34_1e9	$\left\|\frac{\partial P_c}{\partial k_E}\right\|$
W	1.232	1.1217	1.0599	1.0448	1.0569	1.2514	3.141	16.6144
Co	1.097	1.0929	1.0162	1.0506	1.0152	1.7734	3.2453	64.6348
Ta	0.57	0.4808	0.7679	0.6456	0.8363	1.3494	2.267	7.3061
Ti	0.756	0.7291	0.7803	0.75	0.8811	0.8722	1.8316	9.5468
Al	1.056	0.9098	1.0268	1.0373	1.0189	1.0492	2.2033	51.8623
Cr	1.012	0.9412	1.0063	1.0343	1.0056	0.762	1.2116	178.698
Mo	0.764	0.7748	0.879	0.803	0.9746	2.4731	3.9569	40.4173
Hf	0.107	0.0949	0.5743	0.5097	0.9644	5.4218	7.4279	29.1519

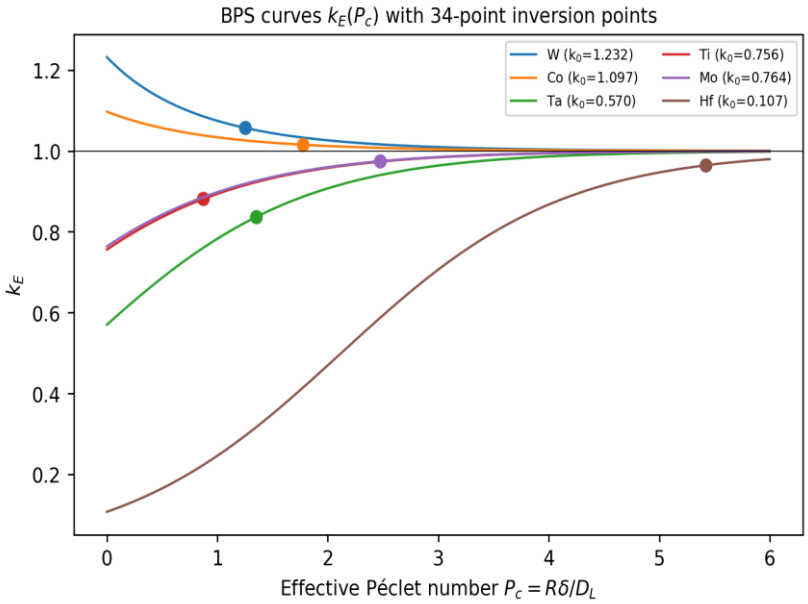


Figure 4.29. BPS effective partition coefficient curves $k_E(P_c)$ and element-specific inversion points

For W, Co, Ta and Ti, whose $|k_0-1|$ is large and whose fits are fairly window-stable, the 34-point fits give P_c=1.25, 1.77, 1.35 and 0.87, i.e. P_c≈0.9–1.8. If all elements share the same exchange layer, $P_cD_L=R\delta$ should be element-independent: the measured $R\delta$=(2.6±0.7)×10⁻⁹ m²/s scatters by about ±23% and corresponds to an effective exchange-layer thickness δ≈64±17

μm, of the same order as the solute diffusion length D_L/R_0 (about 45 μm) and the secondary dendrite-arm spacing (43–92 μm).

The physical picture is that most of the solute rejected during dendritic solidification is trapped between the dendrites, and only the part crossing an exchange layer about one diffusion length thick enters the bulk liquid and forms macrosegregation. The P_c values inverted for Hf and Mo are too large and vary strongly with the fitting window, consistent with their carbide sampling and EDS-precision problems; for Al and Cr $|k_0-1|$ is too small and the inversion is ill-conditioned, so they are excluded from this test.

On top of this static effect, consider the correction to the time-averaged partition coefficient of the solid over one period due to the periodic oscillation of R. The thesis, following Ref. [87], points out that the crystal layer grown in the fast-growth half period is thicker and has a larger k_E, so the total solute incorporated in one period increases; its quantitative form is as follows. Let the interface advancing velocity oscillate sinusoidally, $R(t)=R_0(1+\varepsilon\sin\omega t)$; weighting k_E by the crystal-layer thickness (proportional to R) and expanding to order ε^2:

$$\langle k_E\rangle = \frac{\langle k_E(R)\,R\rangle}{\langle R\rangle} \approx k_E(R_0) + \frac{\sigma_R^2}{R_0}\,k_E{}'(R_0) + \frac{1}{2}\,\sigma_R^2\,k_E{}''(R_0) \qquad (4.18)$$

where $\sigma_R^2 = 1/2\,R_0^2\varepsilon^2$ is the variance of the interface advancing velocity. Since $P_c \propto R$, Eq. (4.18) can be written as $\Delta\langle k_E\rangle = (\sigma_R/R_0)^2\,[\,P_c\;\partial k_E/\,\partial P_c + 1/2\,P_c^2\;\partial^2 k_E/\,\partial P_c^2\,]$. For $P_c \lesssim 2$ the first-order term dominates and has the sign of Eq. (4.17), so k_E of both element types drifts toward 1; thickness weighting is essential for this direction (the naive unweighted average even moves Ta and W away from 1); for $P_c>2$ (Mo, Hf) the second-order term takes over and the direction reverses. In magnitude, an exact period average with $\sigma_R/R_0=0.117$ (Table 4.11) and the above P_c gives a correction of only 10^{-4} – 10^{-3}, i.e.0.1% – 0.7% of the measured k_E-k_J. The systematic difference between Tables 4.3 and 4.4 is therefore explained by the static effect of Eq. (2.5); Eq. (4.18) is only a second-order correction acting in the same direction (for $P_c \lesssim 2$).

4.4.3.2 Quasi-static response of the solute boundary layer

Whether the above "instantaneous $k_E(R)$" treatment holds depends on how fast the solute boundary layer responds relative to the oscillation period. Taking the solute diffusion length $l_D=D_L/R_0$ as the characteristic scale (note that l_D differs physically from the exchange-layer

thickness δ of Eq. (2.5); their ratio is P_c), the build-up/relaxation time of the boundary layer is

$$\tau \approx \frac{l_D}{R_0} = \frac{D_L}{R_0^2} \tag{4.19}$$

Substituting the present parameters (D_L taken as the mean of Table 3.2 of the thesis, 1.85×10^{-9} m²/s, and R_0≈41 μm/s) gives l_D≈45 μm and τ≈1.1 s. Autocorrelation and spectral analysis of the steady stage of Fig. 4.26 give a spatial period λ≈15–20 mm for the temperature-gradient oscillation (positive autocorrelation peak of 0.58 at a lag of 20 mm, spectral peak at 19.3 mm, mean detrended peak spacing 15 mm), corresponding to a temporal period $T=\lambda/R_0$≈370–490 s, so the dimensionless criterion is

$$\omega\tau = \frac{2\pi D_L}{R_0^2 T} \approx 0.016 \ll 1 \tag{4.20}$$

i.e. the solute boundary layer responds quasi-statically to the oscillation of the interface advancing velocity: the phase lag of the concentration response is arctan($\omega\tau$)≈1°, i.e. a spatial displacement of about 0.05 mm ($\omega\tau$=0.014–0.019 for λ=15–20 mm); even with the most conservative planar-front initial-transient time $D_L/(k_0R_0^2)$ (about 7.6 s for Hf, which has the smallest k_0), $\omega\tau$ does not exceed 0.13 and the spatial lag does not exceed 0.31 mm, both far smaller than the 5 mm sampling interval. This also rules out the strong-hysteresis picture of a "severely lagged solute cloud compressed into an extremely narrow grid causing an exponential concentration blow-up" (which does not hold when $\omega\tau\ll1$). It should be stressed that the quasi-static criterion only shows that the growth-rate channel acting through the solute diffusion layer introduces no phase lag; it does not determine the sign of the alignment.

4.4.3.3 Sign of the alignment — export/retention phase of the convection.

According to growth-layer theory[84,85], if the temperature undulation at the boundary layer in the melt is $\Delta T=\Delta T_0\exp(i\omega t)$, the concentration response in the crystal is

$$C_s(t) = k_0\{\, C^{(0)} + \left|C_\omega^{(1)}\right|\cos(\omega t + \phi)\,\} \tag{4.14}$$

where $C^{(0)}$ is the solute concentration in the melt at the interface without temperature disturbance, $C_\omega^{(1)}$ is the amplitude of the concentration undulation at the interface when a disturbance of frequency ω is present, and ϕ is the phase difference between the concentration response and the temperature disturbance[85].

Eq. (4.14) fixes the form of the linear response; the phase ϕ (in phase or anti-phase) depends on the physical coupling channel. If the coupling acts through the growth rate — i.e. the oscillation of R modulates k_E through Eq. (2.5) — Eqs. (4.16) and (4.17) give $\partial \ln C_s / \partial \ln G = -P_c \, (\partial k_E / \partial P_c)/k_E$, which is negative for $k_0<1$: this channel predicts the rejecters to be in anti-phase with G, opposite to the measured sign in Fig. 4.25 and Table 4.12, and its magnitude (about −0.22 for Ta) is only about one tenth of the measured elasticity (+2.46). A direct test with R registered to the compositions gives r(Ta,R)=−0.39 and r(Ni,R)=+0.38 in the steady stage(for the x ≥ 71 mm window (n=20) with R taken at the nearest interval centre), again opposite to the prediction of this channel. The growth-rate channel therefore contributes only a small term of opposite sign and is not the driver of the alignment.What agrees with all measured signs is the export/retention phase of the convection. The upflow branch of a convection cell near the front (a solute plume) carries interdendritic liquid at near-liquidus temperature upward, cooling the liquid above the front and widening the isotherm spacing (G decreases), while exporting the rejected strong rejecters (Ta, Hf, Ti) from the mushy zone, so these elements are low in the local solid. The return branch pushes hot bulk liquid toward the front, compressing the isotherms (G increases); solute export is suppressed and the solute is retained between the dendrites, so Ta, Hf and Ti are locally high and MC carbides — which precipitate from the interdendritic liquid early in solidification[91] — and γ/γ′ eutectic become more frequent. In terms of scaling, the height of the density-inverted layer is inversely proportional to G and the mushy-zone Rayleigh number decreases accordingly[92], so convection is weaker and export smaller where G is high, in the same direction; the anti-phase of the matrix elements is transmitted mainly through the normalization closure (§4.4.4). This mechanism is consistent with the signs of Table 4.12, the carbide/eutectic pool test (§4.4.5) and the closure dilution, but it remains a hypothesis at present.

4.4.4 Element grouping and amplitude law of the composition fluctuation

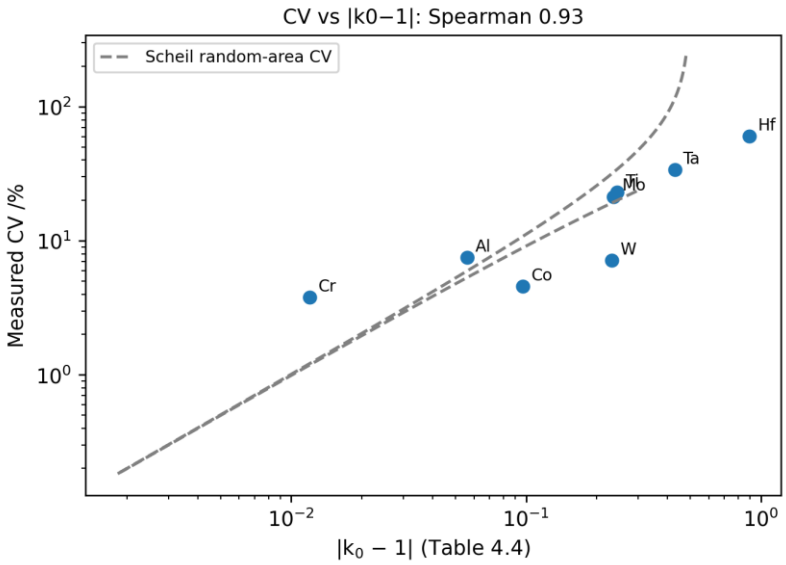
4.4.4.1 Fluctuation amplitude

For the 34 axial measurement points, the coefficient of variation CV=$\sigma/\bar{C}$ of each element is used as the measure of fluctuation amplitude and is compared, one by one, with the atomic

number Z, the partition-coefficient deviation $|k_0-1|$ (Fig. 4.5), the absolute liquidus slope $|m|$ (Fig. 4.5) and the liquid diffusion coefficient D_L (Table 3.2); the results are given in Table 4.8.

Table 4.8 Fluctuation amplitude of each element versus candidate criteria

Element	CV/%	Z	$\|k_0-1\|$ T4.4	$\|k_0-1\|$ liquidus	$\|k_0-1\|$ Procast	$\|m\|$/(K·wt%$^{-1}$)	D_L/(10^{-9} m^2·s^{-1})
Hf	60.2	72	0.89	0.905	0.905	16	1.37
Ta	33.9	73	0.43	0.519	0.174	27	1.68
Ti	23.0	22	0.24	0.271	0.192	274	2.10
Mo	21.4	42	0.24	0.225	0.361	200	1.60
Al	7.5	13	0.06	0.09	0.065	81	2.10
W	7.2	74	0.23	0.122	0.03	38	2.51
Co	4.6	27	0.10	0.093	0.14	82	1.83
Cr	3.8	24	0.01	0.059	0.19	106	1.59
Spearman (CV,·)	—	0.26	0.93	0.929	0.548	−0.38	0.24 (vs 1/D_L)



Measured coefficient of variation CV versus the absolute value of k0−1 with the Scheil random-area-sampling CV curve

From Table 4.8, the fluctuation amplitude has a Spearman rank correlation of 0.93 with $|k_0-1|$, but no significant correlation with atomic number (0.26), $1/D_L$ (0.24) or $|m|$ (−0.38). A first-order amplitude law follows. The fine scan (single-point scan area 100 μm×100 μm, smaller than the primary dendrite arm spacing) resolves dendrite core and interdendritic region.

By the Scheil equation (Eq. (2.3)), the first-solidified dendrite core ($f_s \to 0$) has solid

composition $C_{s,core}=k_0C_0$, so the normalized departure of the core from the nominal composition is

$$A_i \equiv \left|\frac{C_{s,\mathrm{core}}}{C_0} - 1\right| = |k_{0,i} - 1| \tag{4.21}$$

i.e. the microsegregation amplitude is proportional to the deviation of the partition coefficient from 1. Eq. (4.21) is highly consistent with the CV ranking of Table 4.8 (Hf>Ta>Ti>Mo>…>Cr).The only element that deviates from Eq. (4.21) is W: its $|k_0-1|=0.23$ is comparable to Ti and Mo, yet its CV is only 7.2%, less than one-third of theirs. The reason is the asymmetry between absorbers and rejecters in "the area over which the contrast is diluted". Let the interdendritic (last-to-solidify) area fraction be φ ($\varphi \ll 1$); a measurement point falls randomly in an interdendritic pool or in a dendrite core, and its composition variance is

$$\mathrm{Var} \propto \varphi(1-\varphi)\left(C_{\mathrm{pool}} - C_{\mathrm{core}}\right)^2 \tag{4.22}$$

For rejecters ($k<1$) the solute is squeezed into the small interdendritic pool, whose enrichment is about $(1/k_0-1)$ (up to 8 for Hf, about 0.75 for Ta), far larger than the core deficit $(1-k_0)$; for absorbers ($k>1$) the excess solute is spread over the dendrite core that occupies most of the area, and the pool deficit is only about $(1-1/k_0)$ (about 0.19 for W). Hence the frame-to-frame contrast of rejecters is dominated by $(1/k_0-1)$ and is large, whereas that of absorbers is dominated by (k_0-1) and is small; the low amplitude of W is thereby explained by this geometric asymmetry. The ranking by (1/k0−1) is identical to that by the absolute value of k0−1.

More rigorously, if the measuring frames sample the dendritic pattern at random, the stereological principle that area fraction equals volume fraction[94] implies that the solid fraction at which the sampled material solidified is uniformly distributed on [0,1]; Eq. (2.3) then gives the exact coefficient of variation of the Scheil distribution, $\mathrm{CV} = |1-k_0|/\sqrt{2k_0-1}$ ($k_0>1/2$). Its zeroth-order term is Eq. (4.21); its first-order term amplifies rejecters and compresses absorbers, which is the continuous form of the geometric asymmetry above; for $k_0 \le 1/2$ (Hf) the variance diverges and is truncated by eutectic/carbide precipitation, which is the mathematical origin of the bimodal distributions of Hf and Ta.

4.4.4.2 Element grouping — correlation matrix

The pairwise Pearson correlation coefficients of the composition sequences are given in Table 4.9.

Table 4.9 Pearson correlation matrix of the axial compositions

	Al	Co	Cr	Hf	Mo	Ta	Ti	W	Ni	S
Al	1.00	0.63	0.58	−0.61	−0.08	−0.68	−0.59	0.08	0.49	−0.68
Co		1.00	0.84	−0.87	−0.14	−0.89	−0.81	0.36	0.74	-0.92
Cr			1.00	−0.79	0.03	−0.69	−0.58	0.28	0.51	-0.75
Hf				1.00	0.04	0.80	0.77	−0.42	−0.73	0.90
Mo					1.00	0.23	0.28	0.05	−0.36	0.19
Ta						1.00	0.93	−0.33	−0.90	0.98
Ti							1.00	−0.49	−0.83	0.93
W								1.00	0.05	-0.39
Ni									1.00	-0.89
S										1.00

NOTE: S represents the sum of Ta, Hf, and Ti concentrations at each axial sampling position.

From Table 4.9 the elements fall into three groups plus two “outliers”:

Group A (rejecters + MC-carbide formers): Hf, Ta, Ti. $k_0<1$, $m<0$ (Fig. 4.5). Strong intra-group positive correlations (Ta–Ti 0.93, Ta–Hf 0.80, Ti–Hf 0.77); the largest amplitude, ranked by $|k_0-1|$ and carbide-forming tendency. This group is bimodally distributed (Table 4.10 below): the 100 μm frame sometimes falls into an interdendritic (Ta, Hf, Ti)C carbide + γ/γ′ eutectic pool and sometimes into a dendrite core, consistent with §4.3.5 (Hf mainly enters the MC/MB_2 phase and enriches strongly at the end of solidification) and with microstructural observations[88] (script-like interdendritic MC carbide, radial γ/γ′ eutectic and associated porosity). That the MC carbide in CM247LC is composed mainly of Ta, Hf and Ti and co-grows with the γ/γ′ eutectic is also confirmed by dedicated studies[91].

Group B (absorbers / matrix elements): Co, Cr, Al, Ni. $k_0\geq1$ (Cr≈1; Co and Al slightly above 1). Positive within the group; their negative correlation with Group A arises mainly from normalization closure rather than from an independent flow signal. Since $\Sigma C_i=100$ wt%, the fluctuations satisfy $\Sigma\delta C_i=0$. Dividing the solutes into the strongly-segregating MC-former set S={Ta, Hf, Ti} and the remaining matrix set M, a local enrichment $\delta_S=\delta(C_{Ta}+C_{Hf}+C_{Ti})$ forces

the matrix elements to share a compensating dilution in proportion to their abundance:

$$\delta C_j \approx -\frac{w_j}{\sum_{m\in M} w_m}\,\delta_S, \qquad j \in M \tag{4.23}$$

Eq. (4.23) predicts a strong negative correlation between the matrix elements and *S*, in agreement with *r*(Ni,*S*)=−0.89, *r*(Co,*S*)=−0.92 and *r*(Cr,*S*)=−0.75 in Table 4.9, showing that a large part of the "anti-phase" of the *k*>1 group is a closure-dilution effect; Cr (*k*≈1, CV 3.8%) is almost purely dilution.

Attribution of Al — adjudicated by the data. Fig. 4.5 gives *k*=1.05–1.06 for Al, and §4.3.5 notes that liquid Al first rises (to 6.33 wt% at 1290 ℃) and then falls; k_Al increases from 0.91 at the liquidus and crosses 1 at about 1290 ℃, so Al is weakly enriched in the solid macroscopically. The measured data support this treatment: *r*(Al,Ni)=+0.49, *r*(Al,Co)=+0.63, *r*(Al,*S*)=−0.68, and Al is actually lower in the carbide-pool frames (Table 4.10). That is, the true microsegregation of Al is cancelled to k_E≈1 by the "deep-V reversal", and what the fine scan sees for Al is mainly passive dilution by the carbide/eutectic pools, so it follows the matrix group.

W — a class of its own. W is the strongest counterexample to the "atomic-number view": small amplitude (geometric asymmetry of Eq. (4.22)) but carrying a genuine mesoscale depletion signal. From Table 4.10 the carbide-pool frames have W=9.00 wt% versus 9.58 wt% for the matrix frames (0.58 lower); after removing the closure effect represented by Ni, *r*(W,Ta) changes from −0.33 to −0.66 (the closure effect had been masking it), i.e. W is indeed depleted where Ta is enriched. This agrees with §1.1.2 that W solid-solution-strengthens the dendrite core and that the Ta/W ratio (3.2/9.5=0.34 for this alloy) governs the tendency of casting convective instability[92].

Mo — unresolved. *r* with all elements is near 0 (*r*(Mo,Ni)=−0.36, *r*(Mo,*S*)=+0.19), and at 0.5 wt% it is at the edge of EDS accuracy; §1.1.2 that Mo isomorphously substitutes for W and reduces W segregation also implies that its behaviour is governed by W, so no separate conclusion is drawn for Mo here.

Table 4.10 Carbide/eutectic-pool frames versus matrix frames (framed by Hf content; wt%)

Frame class	Hf	Ta	Ti	Al	W	Co	Cr	Ni
Carbide-pool frames (Hf>1.5, n=15)	2.27	6.60	1.21	6.55	9.00	8.69	8.09	56.89
Matrix frames (Hf≤1.5, n=19)	0.74	4.01	0.84	7.13	9.58	9.31	8.48	59.24
Difference	+1.54	+2.59	+0.36	−0.58	−0.58	−0.62	−0.39	−2.35

From Table 4.10, in the pool frames the Group-A elements (Ta, Hf, Ti) rise markedly, Ni and the Group-B elements are diluted, and W is depleted — the typical compositional signature of an interdendritic MC carbide + γ/γ′ eutectic pool. This also shows that the "fluctuation" of Group A is not a smooth sinusoid but a discrete sampling of interdendritic carbide/eutectic pools superposed on a slowly varying background.

It should be noted that the amplitude ranking depends on the EDS scan area. The fine scan of this thesis (100 μm×100 μm, smaller than the primary dendrite arm spacing) resolves the core/interdendritic region of a single dendrite, so the amplitude is dominated by $|k_0-1|$, Group-A spikes stand out and W is smoothed away; a larger-area scan (e.g. 1500 μm×500 μm, spanning about 3 cells) averages out single-dendrite microsegregation and leaves the mesoscale channel/regional signal, in which the Al/Ta-rich, W-poor interdendritic flow (segregation channel) produces a large mesoscale depletion of W where it passes[88]. The two scales are not contradictory; both point to the same mechanism — the light interdendritic liquid rises to form solute plumes and segregation channels[90,92], and where a channel passes more (Ta, Hf, Ti)C carbide and γ/γ′ eutectic are deposited[91], which is the origin of the high cluster of the Group-A bimodal distribution.

Accordingly, the composition fluctuation in §4.4 should be understood as the superposition of two scales and two origins: on the fine scale, a discrete sampling of interdendritic carbide/eutectic pools with amplitude ordered by $|k_0-1|$; on the mesoscale, a channel/plume modulation driven by density inversion (signalled by the depletion of W).

4.4.5 Quantitative alignment of fluctuation

The correspondence in §4.4.3 came from a visual comparison of normalized curves. This

section cross-correlates, point by point, the centerline thermal quantities derived in §4.4.2 with the measured compositions at the 34 points.

Table 4.11 gives the steady-stage (x≥70 mm) statistics of the cooling rate $\dot{T}$, the liquidus-isotherm advancing velocity R, the local temperature gradient G, and the undercooling ΔT below the liquidus (1370.9 ℃) at which the solid fraction reaches 5%, all at the centerline. The mean R of 2.46 mm/min equals 41 μm/s, the pulling speed, which anchors the units and the point spacing; the extreme fluctuation of R, ±22%, matches Fig. 4.24(b); after removing the transient tail point at x=70 mm, $\dot{T}$=0.213±0.015 K/s (coefficient of variation 6.9%), fluctuating less than G (8.6%) and R (11.7%), so the "nearly constant cooling rate" assumption used in Eq. (4.16) holds as a first-order approximation.

Table 4.11 Steady-stage statistics of the centerline thermal quantities (x≥70 mm)

Quantity	n	Mean	S.D.	CV/%	Range	Extremes vs mean
Cooling rate $\dot{T}$/(K·s^{-1})	23	0.216	0.022	10.3	0.189–0.295	−13%/+36%
Interface velocity R/(mm·min^{-1})	22	2.46	0.29	11.7	1.93–3.01	−22%/+23%
Temperature gradient G/(K·mm^{-1})	21	6.23	0.54	8.6	5.22–7.36	−16%/+18%
Undercooling ΔT/K	23	3.67	0.51	14.0	2.65–4.52	−28%/+23%
Cooling rate, x>70 mm (transient point removed)	22	0.22	0.015	6.9	0.19-0.24	−11.3%/+11.9%

NOTE: G is a 10 mm interval average centred at x_i+5

Correlating the detrended R and G in the steady stage: the centres of their finite differences are 2.5 mm apart, and at this registration $r(R,G)$=−0.44 to −0.50; when the registration is shifted to 7.5 mm (about 0.4λ) the correlation reverses to +0.56, as expected for a signal of period 15–20 mm. The regression slope is $\delta R/R_0$=0.63×($-\delta G/G_0$); corrected for the misregistration attenuation cos(2π×2.5/17.5)≈0.62 of Eq. (4.24) it is about 0.92, consistent with the slope of 1 predicted by Eq. (4.16). It should be noted that R and G are both derived from the same sequence of isotherm crossing times, so their anti-phase contains a constructional coupling; it is therefore used here only as a consistency check, not as independent evidence. If the fluctuation of $\dot{T}$ itself is included, the logarithmic differential of Eq. (4.15) implies that the relative variance of

R is approximately the sum of those of $\dot{T}$ and G: $\sqrt{6.9^2 + 8.6^2} \approx 11.0\%$, in agreement with the measured 11.7%.

Axis 1, 5 mm off the centerline, and the centerline differ by less than 0.5% in the steady-stage means of $\dot{T}$, R and G; the detrended cooling-rate fluctuations of the two axes have a correlation coefficient $r(V_1,V_2)$=0.65, 0.39 (p=0.07) without the x=70 mm transition point; r(R1,R2)=−0.03, r(G1,G2)=0.07.

The G of derived point i is obtained from the temperature drop between its own liquidus-isotherm crossing time and that of the derived point 10 mm downstream, and represents an interval average centred at x_i+5 mm; EDS point j lies 1+5j mm from the chill face. The j-th composition point should therefore be paired with the G of derived point j−1, with a residual misregistration of 1 mm. Pairing by equal index would give a misregistration of 4 mm≈λ/4; for a quasi-periodic signal the correlation under a misregistration Δx decays as

$$\rho(\Delta x) \approx \cos(2\pi\, \Delta x/\lambda) \tag{4.24}$$

And a 4 mm misregistration suppresses it to within ±0.1 (measured $r(S,G)$=−0.13). The registration is fixed uniquely by the finite-difference construction and the sampling geometry; it is not a fitted parameter.

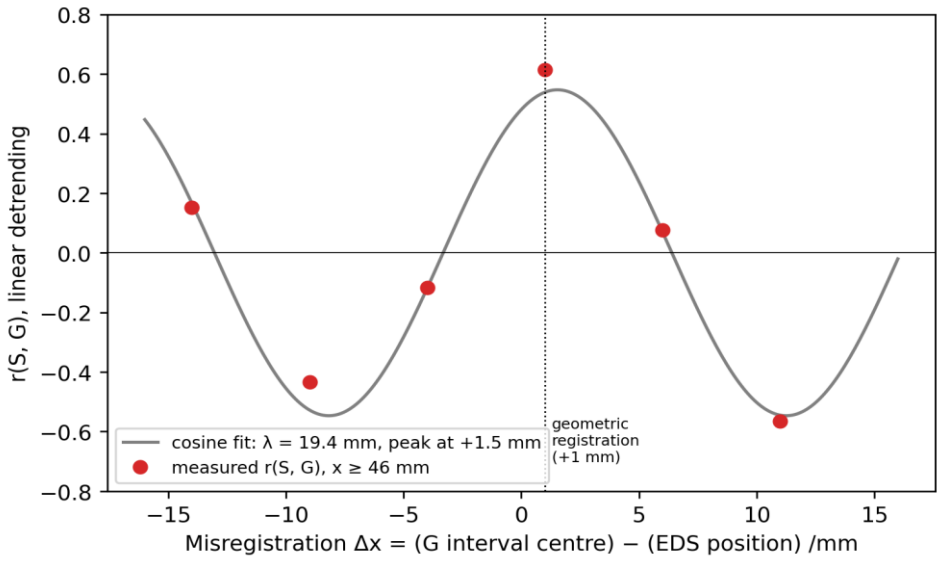


Figure 4.30 Variation of r(S,G) with registration misfit and cosine fit
(x ≥ 46 mm; λ = 19.4 mm, peak at +1.5 mm)

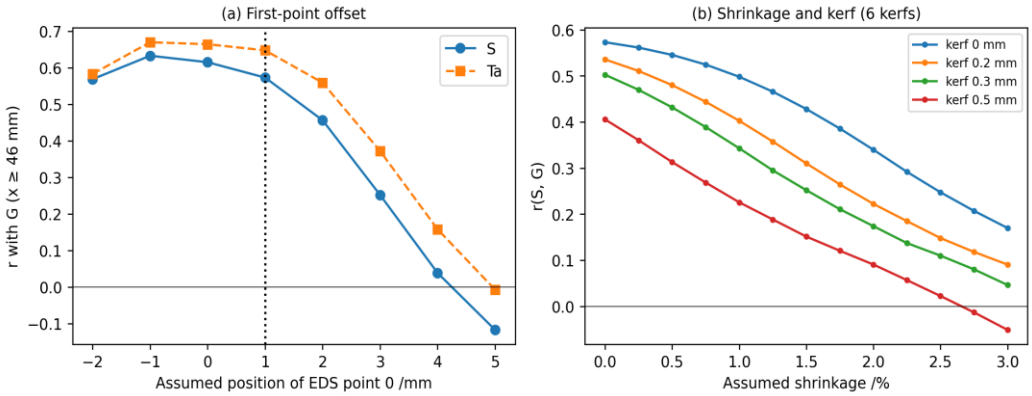


Figure 4.31 Sensitivity of r(S,G) to the first-point position (a) and to shrinkage and kerf (b)

Registration sensitivity: r(S,G) ≥ 0.57 for a first-point position between −2 and +1 mm, about 0 at +4 mm; 0.40 with 1 % shrinkage + 0.2 mm kerf; 0.17 with 2 % + 0.3 mm.

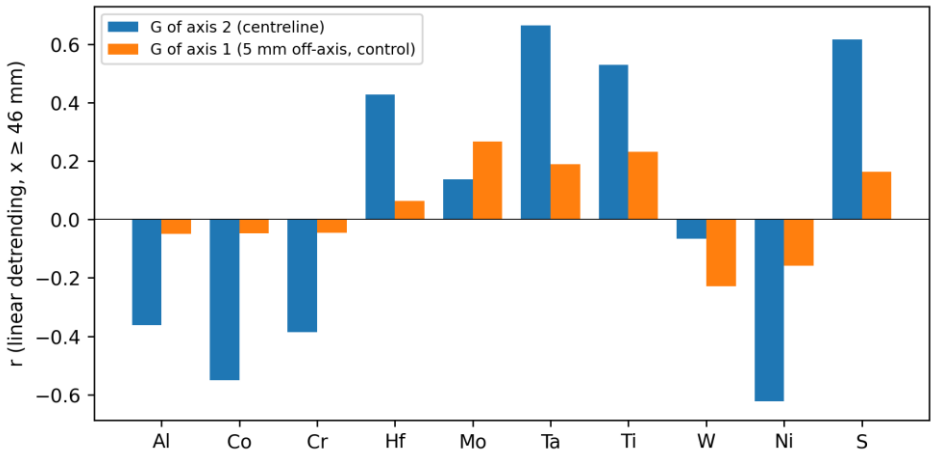


Figure 4.32·Correlation of each element with the centreline G and with the off-axis (axis 1) G as a control

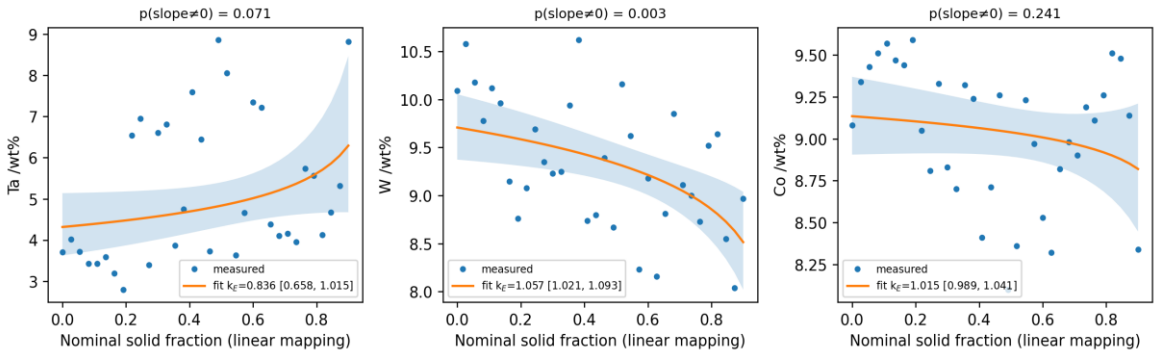


Figure 4.33 34-point Scheil log-linear fits of Ta, W and Co with 95 % confidence bands

Table 4.12 Point-by-point correlation of each element's composition with the local temperature gradient G ($x \geq 46$ mm (n=25))

Element	Linear r	Difference r	Moving-avg r	p (uncorrected)	$n_{eff_Bartlett}$	p (autocorr.-corrected)	R2pct
Al	-0.362	-0.5624	-0.7231	0.0758	18.8888	0.1296	13.067
Co	-0.550	-0.6941	-0.8274	0.0044	12.9877	0.0516	30.265
Cr	-0.386	-0.5396	-0.6638	0.0566	18.1101	0.1123	14.907
Hf	0.4273	0.595	0.6738	0.0331	13.2327	0.1409	18.258
Mo	0.1365	0.0538	0.0792	0.5152	23.6378	0.5282	1.864
Ta	0.6642	0.725	0.8513	0.0003	11.1976	0.0242	44.111
Ti	0.5286	0.6376	0.7622	0.0066	13.1784	0.0611	27.94
W	-0.066	-0.056	-0.0981	0.7549	23.7137	0.7618	0.432
Ni	-0.622	-0.7283	-0.8565	0.0009	11.5012	0.0354	38.723
S	0.6149	0.7099	0.831	0.0011	11.3326	0.0402	37.808

Note: the uncorrected p assumes independent samples; since both series are autocorrelated, the corrected p uses the Bartlett effective sample size[93] $n_{\text{eff}} = n/[1 + 2\sum_k \rho_C(k)\rho_G(k)]$ ($k \leq n/2$).

At the correct registration, Pearson correlations between the linearly detrended compositions and G in the steady stage (n=25) were computed and verified with two further detrending schemes (first differences; residuals from a 5-point moving average); the results are given in Table 4.12 (S=Ta+Hf+Ti is the Group-A sum).

Table 4.12 quantifies the visual alignment of §4.4.3: the rejecters correlate positively with G, the matrix elements negatively, W is near zero and Mo unresolved, in full agreement with the grouping of §4.4.4; the negative correlation of Al with G is consistent in direction with its attribution to the matrix group. The fluctuation of G explains 44% of the axial variance of Ta and 39% of that of Ni. After correcting for autocorrelation, Ta, S, Ni and Co remain significant at the 0.05 level (p=0.022–0.048), Ti is marginal (0.063), and Hf, Cr and Al agree in direction but are not individually significant (0.09–0.13); because of the normalization closure the elements are not mutually independent, so their sign agreement cannot be combined as independent events.

Four structural pieces of evidence support the result: all nine elements show the expected sign simultaneously at the same registration; the signs reverse as a whole when the registration

is shifted by half a period (S: −0.60/−0.43); a cosine fit (Eq. (4.24)) to the correlations of S at six registration offsets (−11 to +14 mm) gives λ=19.3 mm, identical to the spectral peak of G itself (19.3 mm), with the fitted peak less than 0.5 mm from the geometric registration point; and the values are essentially unchanged when the steady window is tightened to x≥75 mm (n=19; Ta +0.67, Ni −0.65, S +0.62). The detrended peak positions of S (60, 75, 90, 115, 130, 140 mm) and of G (55, 75, 95, 115, 125, 140,150,160 mm) differ by no more than one sampling point each.

Using the framing of §4.4.4 (Hf>1.5 wt%), the mean detrended G at the pool frames (n=14) in the steady stage is 0.34 K/mm (about 0.6 standard deviations) higher than at the matrix frames (n=11), one-sided Mann–Whitney p=0.047: interdendritic carbide/eutectic pools tend to occur where G is locally high, consistent with the expectation of the convective export/retention mechanism of §4.4.3 (return flow, locally high G, solute retention, more pools). The sample is small, so this is stated as a tendency.

In summary, the alignment of the composition fluctuation with the temperature-gradient fluctuation is no longer a visual impression but a quantitative relation verified by several schemes within a one-sampling-point registration tolerance; together with the quasi-static criterion of §4.4.3 and the amplitude law of §4.4.4 it forms the chain of evidence from density inversion to composition alignment, in which the convective export/retention mechanism for the sign of the alignment still awaits a direct test with flow-field data.

4.4.6 Limitations of the present analysis

This section has moved the correspondence between the composition and temperature-field fluctuations from a visual comparison to a quantitative registration, but five limitations are inherent to the study, of which the convective mechanism is the principal one.

(1) The convective interpretation is theoretical and has no quantitative support at the flow-field level. §4.4.3 proposes two possible drivers: baroclinic thermal convection caused by the radial temperature difference, and a solutal density inversion of the interdendritic liquid. The former necessarily exists in the simulation; the latter is given by the JMatPro liquid-density curve along the solidification path (an inversion of about 0.14 % near 1345–1350 °C). It therefore cannot be decided whether the oscillation of G originates from the radial asymmetry

of the thermal field or from solute-driven interdendritic convection, nor whether the real specimen experienced a convection mode of the same pattern as the simulation. The export/retention mechanism agrees with the signs of the correlations only and remains a hypothesis. ProCAST is commercial software whose mushy-zone flow model, permeability relation and the thermophysical data actually invoked are not open, which limits the traceability of the simulation. This is an intrinsic limitation of the study.

(2) The nature of the temperature-field oscillation itself needs care. The cooling-rate fluctuations on the two axes correlate at 0.65, but at 0.39 ($p = 0.07$) once the transition point at $x = 70$ mm is removed; the gradient and the interface velocity are not coherent between the axes ($r = 0.07$ and -0.03). In the steady stage the residuals of the isotherm arrival times are close to white noise (s.d. ≈ 11 s), while G is a two-step finite difference; such a difference of white noise theoretically has an autocorrelation of -0.5 at 10 mm and a spectral peak near 20 mm. The measured G has -0.66 and a peak at 19–20 mm, and about 81 % of its variance is explained by the arrival-time differences. Part of the 15–20 mm “quasi-period” is therefore contributed by the difference operator and cannot be attributed entirely to a physical convection mode.

(3) The composition–G correlation is sensitive to registration. The registration follows from the difference construction and the sampling geometry without any fitted parameter, and at that registration the autocorrelation-corrected p of Ta, S, Ni and Co is 0.02–0.05. But the 1 mm position of the first EDS point is an estimate, the kerf was not subtracted, and the real bar is shorter than the 180 mm model because of shell shrinkage.With the G of axis 1, 5 mm off-centre, the correlation vanishes ($r(S,G1) = 0.16$, pool-frame test $p = 0.42$); under convection the off-axis G need not correlate with the centreline composition, so this control only shows that the correlation is not a generic property of the thermal field, and statistically suggestive at the pre-specified registration, mechanism not yet established.

(4) The macrosegregation fits are uncertain. In the 34-point Scheil fits only W has a slope significantly different from zero ($p = 0.003$); Ta and Ti are marginal ($p \approx 0.06$–0.07) and all others have $R^2 < 0.05$; the 95 % interval of k_E for Ta is [0.66, 1.01]. The $k_E > 1$ of the matrix elements cannot be separated from the normalisation closure (apparent $k_E \approx 1.010$). The inverted Pc and δ are order-of-magnitude estimates, and the Pc intervals of Co, Ta and Ti have

no upper bound.

(5) Limits of the EDS data. The 100 μm × 100 μm scan area is smaller than the primary dendrite-arm spacing, so the composition series is a random sampling of the dendritic structure; the measured means of Ta, Ti, Mo and Al are 1.61, 1.43, 1.37 and 1.23 times the nominal values, i.e. the absolute values carry a systematic bias and all conclusions here rest on relative fluctuations; the γ-phase and phase-stability calculations used element-wise corrected values (Appendix B5).

4.5 Python-Based Processing and Analysis of Data

To facilitate quantitative analysis of experimental measurements, numerical simulation results, and thermodynamic parameters during the directional solidification of the Ni-based superalloy CM247LC, a series of data processing and numerical calculation programs were developed and debugged using Python. Scientific computing libraries, including NumPy, Pandas, Matplotlib, and Scikit-learn, were employed to integrate experimental measurements, finite element simulation results, and theoretical models of alloy solidification. These programs enabled batch data extraction, physical parameter calculation, model fitting, and scientific visualization. The main computational procedures are described below.

(1) Thermal-field data processing during directional solidification. A numerical post-processing program was developed to analyze the temperature–time and solid fraction–time data exported from ProCAST finite element simulations. By identifying characteristic temperatures and corresponding times at different axial sampling points, local thermal parameters, including cooling rates, solid–liquid interface advancement velocities, temperature gradients, and undercooling, were calculated using finite-difference approximations. The resulting data were visualized to examine the evolution of thermal conditions during different solidification stages, providing a quantitative basis for investigating the relationship between thermal-field variations and elemental compositional fluctuations.

(2) Automated extraction and organization of experimental compositional data. A batch-processing program was developed for the elemental composition data obtained from energy-dispersive X-ray spectroscopy (EDS) measurements at multiple regions and sampling points. The program was designed to read measurement files, identify measurement regions and

sampling positions, extract elemental weight fractions, and perform data classification, rearrangement, conditional filtering, descriptive statistical analysis, and visualization. This procedure reduced repetitive data preparation and provided structured compositional datasets for subsequent investigations of elemental segregation behavior.

(3) Calculation of alloy phase stability parameters. A numerical calculation program was implemented to evaluate phase stability based on the elemental compositions of the Ni-based superalloy. Elemental weight fractions were converted into atomic fractions, and the composition of the γ phase was estimated using prescribed elemental partitioning relationships. Based on electron vacancy theory and d-electron theory, the average electron vacancy number, average d-orbital energy level, and average bond order were calculated. These parameters provided a theoretical basis for assessing variations in phase stability and the potential susceptibility to topologically close-packed (TCP) phase precipitation at different positions within the alloy.

(4) Inversion of effective partition coefficients using a gradient descent algorithm. Based on the Scheil solidification model, the relationship between elemental concentration and solid fraction was transformed into a logarithmic linear form. A gradient descent optimization procedure was independently implemented by defining a mean squared error loss function and iteratively updating the regression parameters. The effective partition coefficients of alloying elements were subsequently obtained from the fitted regression coefficients. The goodness of fit was evaluated using the coefficient of determination, and the calculated concentration profiles were compared with experimental measurements to support quantitative analysis of macrosegregation behavior.

(5) Calculation of elemental partitioning parameters from thermodynamic phase diagram data. A computational program was developed to process thermodynamic data obtained from Scheil and lever-rule calculations. Elemental partition coefficients were determined from the corresponding solid- and liquid-phase compositions, and their variations with temperature were analyzed. Linear regression was also employed to estimate the slopes of temperature–composition relationships. These calculations provided theoretical parameters for investigating solidification segregation and comparing experimentally inferred partition coefficients with

thermodynamic predictions.

(6) Registration and correlation program. The registration between the composition points and the derived thermal points is fixed from the finite-difference construction and the sampling geometry; Pearson correlations with the local temperature gradient are computed under three detrending schemes, significance is corrected with the Bartlett effective sample size, and sensitivity to registration offset, steady window, first-point position and shrinkage/kerf is examined, with the off-axis gradient as a control.

Together, these programs established a complementary computational workflow covering experimental data extraction, simulation post-processing, theoretical parameter calculation, and model fitting. The developed procedures improved the efficiency of processing data from different sources and provided computational support for investigating thermal-field evolution, segregation, and phase stability during the directional solidification of Ni-based superalloys.

4.6 Chapter Summary

The main content of this chapter is the analysis of the segregation behaviour of alloying elements, including trend analysis and fluctuation analysis of the experimentally measured compositions.

(1) The composition of each alloying element on the central axis of the cylindrical specimen was measured by EDS. Pearson coefficients were computed to analyse the linear relationships among the element compositions, and a Spearman trend test was used to determine the rising or falling trend of each element with the progress of solidification: W decreases, Ta and Ti increase, Co tends to decrease, and the other elements show no significant monotonic trend.

(2) Based on the Scheil model, the measured values of each element were log-linearly fitted by gradient descent to obtain the effective partition coefficient k_E at a pulling speed of 41 μm/s. Combined with JMatPro phase-diagram calculation and the Scheil model, the phase fractions and segregation behaviour of each element during solidification of CM247LC were obtained.

(3) Phase-stability calculations based on the measured compositions were performed to judge the risk of TCP-phase precipitation.

(4) The simulated temperature-field data were processed to obtain key data such as the local temperature gradient at the liquidus of each test point. After normalization, the composition fluctuation of the $k_E<1$ elements is highly similar to and positively correlated with the local-temperature-gradient fluctuation, and the $k_E>1$ elements are negatively correlated.

(5) From the cooling-rate identity $\dot{T} = G \cdot R$, the interface advancing velocity R oscillates in anti-phase with G in the steady stage; from the quasi-static criterion of the solute boundary layer $\omega\tau \approx 0.016 \ll 1$, the concentration response via the growth rate has no phase lag; however, that channel predicts the opposite sign, and the sign of the alignment should be explained by the export/retention phase of the convection — upflow lowers G and exports the rejected solute, while return flow raises G and retains it. Point-by-point cross-correlation quantifies this alignment: the correlation coefficients of Ta, Ti and Hf with G are +0.66, +0.53 and +0.43, those of Ni, Co, Cr and Al are −0.62, −0.55, −0.39 and −0.36, and W is near zero; after correcting for autocorrelation, p=0.022 and 0.031 for Ta and Ni; fitting against the registration offset gives a fluctuation period of 19.3 mm, identical to the spectral peak of G itself; carbide/eutectic pools tend to lie where G is locally high (p=0.047); the cooling-rate fluctuations of two axes 5 mm apart correlate at 0.65, so the oscillation is a radially coherent, genuine field phenomenon.

(6) A quantitative element-grouping and amplitude law of the composition fluctuation is further established: the fluctuation amplitude has a rank correlation of 0.93 with the partition-coefficient deviation $|k_0-1|$, far higher than its correlation with atomic number or liquid diffusion coefficient; the geometric asymmetry between $k>1$ and $k<1$ explains the small amplitude of W, its exact form being the random-area-sampling coefficient of variation of the Scheil distribution, $|1-k_0|/\sqrt{2k_0-1}$; the normalization-closure relation identifies the dilution component of the "anti-phase" of the $k>1$ group. That the measured k_E is closer to 1 than k_J is a static effect of Eq. (2.5): inverting the two tables gives an effective Péclet number of about 0.9–1.5 and an effective exchange-layer thickness of about 60 μm; the thickness-weighted fluctuation correction acts in the same direction but only at the 10^{-4}–10^{-3} level. Overall, the composition fluctuation is the superposition of a fine-scale sampling of interdendritic carbide/eutectic pools with amplitude ordered by $|k_0-1|$ and a mesoscale channel/plume modulation driven by density inversion.

Conclusion

(1) Directional-solidification simulation and experiment of the Ni-based superalloy CM247LC were carried out. The simulated dendrite morphology agrees with the experimental result, verifying the feasibility of the simulation model and the accuracy of the boundary-condition parameters, and the temperature boundary conditions and heat-transfer parameters suitable for this directional-solidification experiment were determined.

(2) According to the rules of superalloy directional solidification, a macro/micro directional-solidification simulation model of CM247LC was built in ProCAST, and the nucleation and growth of the microstructure were simulated with the CAFE module.

(3) The alloying-element compositions of the CM247LC specimen were measured and analysed. In terms of overall trend, $k_E>1$ is significant for W; for Al, Co, Cr and Ni k_E is slightly above 1 and indistinguishable from the normalisation closure, whereas Ta, Ti, Hf and Mo are the opposite ($k_E<1$); the measured effective partition coefficients k_E are generally closer to 1 than the phase-diagram values k_J, and the macrosegregation is weaker than the pure phase-diagram prediction; inverting the effective-partition-coefficient relation gives an effective Péclet number of about 0.9–1.5 and an effective exchange-layer thickness of about 60 μm, showing that this is a static effect of the limited solute exchange between the mushy zone and the bulk liquid. Phase-stability calculations based on the measured compositions show a high tendency of TCP-phase precipitation at the composition peaks of Ta and Ti. JMatPro phase-by-phase analysis shows that liquid Al first rises (to 6.33 wt% at 1290 ℃) and then falls; k_Al increases from 0.91 at the liquidus and crosses 1 at about 1290 ℃, so Al is weakly enriched in the solid macroscopically.

(4) Using the temperature boundary conditions verified by the microstructure simulation, a flow-field–temperature-field coupled simulation of the directional-solidification process of CM247LC was performed. The simulation shows that the interdendritic liquid density inverts at about 1350 ℃ owing to W depletion and Ti enrichment, driving thermosolutal convection and imposing high-frequency oscillation on the local temperature gradient G; from the cooling-rate identity $\dot{T} = G \cdot R$, the interface advancing velocity R oscillates in anti-phase with G in the steady stage, in agreement with the dendrite-tip velocity oscillation observed by in-situ

synchrotron radiography. The simulated temperature field exhibits fluctuations similar to the measured composition fluctuations; the $k_E<1$ elements are in phase with G and the $k_E>1$ elements in anti-phase; the quasi-static criterion of the solute boundary layer $\omega\tau\approx0.016\ll1$ shows that the concentration response via the growth rate has no phase lag, but its predicted sign is opposite to the measured one, so the sign of the alignment should be explained by the export/retention phase of the convection (to be tested with flow-field data). Point-by-point cross-correlation confirms this quantitatively: Ta, Ti and Hf correlate positively with G (+0.66, +0.53, +0.43), Ni, Co, Cr and Al negatively (−0.62, −0.55, −0.39, −0.36), and W is near zero, with autocorrelation-corrected p=0.022 and 0.031 for Ta and Ni; the cooling-rate fluctuations of the two axes correlate at 0.65 (0.39 without the transition point) while G and R do not.

(5) A quantitative element-grouping and amplitude law of the composition fluctuation was established: the fluctuation amplitude has a rank correlation of 0.93 with the partition-coefficient deviation $|k_0-1|$, far higher than its correlation with atomic number or liquid diffusion coefficient; the geometric asymmetry between $k>1$ and $k<1$ explains the small amplitude of the strong absorber W; the normalization-closure relation identifies the dilution component of the “anti-phase” of the matrix elements. Thickness-weighted averaging over the velocity oscillation drives k_E slightly further toward 1, but only by about 10^{-3}, leaving the static conclusion of (3) unchanged. Overall, the composition fluctuation is the superposition of a fine-scale sampling of interdendritic carbide/eutectic pools with amplitude ordered by $|k_0-1|$ and a mesoscale channel/plume modulation driven by density inversion.

Note on the 2026 supplementary analysis

This master's thesis was completed in 2023. At that time the relation between the composition fluctuation and the temperature-field fluctuation in Chapter 4 had only been established at the level of a theoretical link: a visual comparison of normalised curves (Fig. 4.25) and the general form of the compositional response from growth-layer theory [84,85] (Eq. 4.14), without specific quantification.

The journal paper by the same first author in 2024 [88], which reported the microstructural evolution of the same specimen and the mesoscale W depletion seen in large-area scans, prompted a supplementary analysis of Chapter 4 in 2026. It comprises: the static BPS effect and effective-Péclet-number inversion, the quasi-static criterion of the solute boundary layer and the discussion of the alignment sign in §4.4.3.1–4.4.3.3; the element grouping, closure dilution and amplitude law in §4.4.4; the registration rule (Eq. 4.24), point-by-point cross-correlation (Table 4.12), autocorrelation-corrected significance, registration scan and cosine fit (Fig. 4.30) in §4.4.5; Tables 4.7–4.12 and Figs 4.26–4.30; and the limitations in §4.4.6. The mathematical treatment rests on fixing the interval centre of G at $x_i + 5$ mm from the difference construction, fixing the pairing $j \leftrightarrow i = j - 1$ from the EDS sampling geometry ($x_j = 1 + 5j$ mm), computing Pearson correlations under three detrending schemes and correcting p-values with the Bartlett effective sample size.

All experimental and simulated data used in the supplement are the original 2023 data; no new experiments were performed. Data and programs are given in Appendices B and C. References added at this stage are [88]–[94] and Burton–Prim–Slichter (1953), numbered [95]; references up to [87] belong to the 2023 original.

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Appendix A:Symbols

Symbols	Definition	Expression	Unit
Part I	**Symbols of solidification theory**		
C'	Length of the eutectic line		at.%, wt.%
C^*	Composition at the solid–liquid interface		at.%, wt.%
D_l	Diffusion coefficient in the liquid phase	$-J/G_c$	m²/s
D_0	Pre-exponential factor of the diffusion coefficient		m²/s
D_s	Solid-phase diffusion coefficient		m²/s
E_l	Exponential integral function		—
G	Interfacial temperature gradient	$\frac{dT}{dz}$	K/m
$\bar{G}$	Average temperature gradient	$\frac{k_s G_s + k_l G_l}{k_s + k_l}$	K/m
G_c	Concentration gradient of the liquid phase at the interface	$\left(\frac{dC_l}{dz}\right)_{z=0}$	at.%/m, wt.%/m
G_l	Temperature gradient in the liquid phase	$\left(\frac{dT_l}{dz}\right)_{z=0}$	K/m
G_s	Temperature gradient in the solid phase	$\left(\frac{dT_s}{dz}\right)_{z=0}$	K/m
G^*	Effective temperature gradient	$\frac{k_s G_s \xi_l + k_l G_l \xi_s}{k_l + k_s}$	K/m
I	Nucleation rate		1/(m³·s)
J	Mass flux		1/(m²·s)
K	Curvature	$1/r_1 + 1/r_2$	1/m
N	Quantity		—
P	Péclet number	L/δ	—
P_c	Solute Péclet number	$VR/2D$	—
P_t	Thermal Péclet number	$VR/2a$	—
R	Radius		m
T	Temperature		K
$\dot{T}$	Cooling rate	dT/dt	K/s
T_f	Melting point of a pure substance		K
T_l	Liquidus temperature		K
T_s	Solidus temperature		K
V	Interfacial migration rate		m/s

Symbols	Definition	Expression	Unit
V_a	Critical growth rate for absolute stability		m/s
V_c	Critical growth rate for constitutional undercooling		m/s
a	Thermal diffusivity	κ/c	m²/s
c	Volumetric specific heat		J/(m³·K)
f_l	Liquid-phase volume fraction		—
f_s	Solid-phase volume fraction	$1-f_l$	—
k	Equilibrium partition coefficient	C_s/C_l	—
m	Liquidus slope	dT_l/dC	K/at.%, K/wt.%
n	Grain number density		
p	Complementary distribution coefficient	$1-k$	—
s_c	Solute capillary length	$\Gamma/[mC^*(k-1)]$	m
s_t	Thermal capillary length	$\Gamma c/\Delta h_f$	m
t	Time		s
t_f	Local solidification time		s
α	Dimensionless back-diffusion coefficient	$D_s t_f/L^2$	—
α'	Dimensionless back-diffusion coefficient between dendrites	$\alpha[1-\exp(-1/\alpha)]$ $-0.5\exp[-1/(2\alpha)]$	—
Γ	Gibbs–Thomson coefficient	$\sigma/\Delta s_f$	K·m
δ_c	Solute boundary layer thickness	$2D/V$	m
δ_c	Characteristic diffusion length	D/V	m
ΔC_0	Concentration difference between the liquid and solid phases of the alloy at the liquidus temperature	$C_0(1-k)/k$	at.%, wt.%
Δh_f	Latent heat of fusion per unit volume		J/m³
ΔT	Undercooling	$T_f - T$	K
$\Delta T'$	Temperature difference between the tip and root of a dendrite		K
ΔT^*	Critical undercooling		K
ΔT_0	Solidification temperature range of the alloy of composition C_0		K
ΔT_r	Curvature undercooling		K
ΔT_c	Solute diffusion undercooling		K
ΔT_t	Thermal diffusion undercooling		K
ε	Perturbation amplitude		m

Symbols	Definition	Expression	Unit
$\dot{\varepsilon}$	Amplitude growth rate		m/s
κ	Thermal conductivity		W/(K·m)
ξ	Stability parameter		—
ρ	Density		kg/m³
σ^*	Stability constant		—
Φ	Constitutional undercooling	$mG_c - G$	K/m
Ω_t	Dimensionless solute supersaturation		—
Part II	**Symbols of the statistical and numerical analysis in Chapter 4**		
II.1	**Sampling geometry and composition data (§4.1, §4.4.2, §4.4.4)**		
x	Axial distance from the chill face		mm
j	Index of the EDS measuring point, $j = 0,1,\ldots,33$		—
x_j	Nominal position of EDS point j (first point about 1 mm from the chill face; kerf and shrinkage not corrected)	$1 + 5j$	mm
i	Index of the thermal export point along an axis, $i = 0,1,\ldots,36$; also used as element index		—
x_i	Position of thermal export point i	$5i$	mm
Δ	Axial sampling interval of the EDS points and the thermal export points	5	mm
x_c	Centre of the 10 mm support interval of G_i	$x_i + 5$	mm
$C,\ C_i$	Mass fraction of an element (of element i) at a measuring point		wt%
$\bar{C}$	Mean composition over the measuring points	$\sum C/n$	wt%
C_0	Nominal (initial) composition; in Tables 4.1–4.3, the initial composition returned by the fit	a/k_E	wt%
S	Group-A sum of the rejecting MC-carbide formers	$C_{\mathrm{Ta}} + C_{\mathrm{Hf}} + C_{\mathrm{Ti}}$	wt%
$S,\ M$	Set of the strongly segregating MC formers {Ta, Hf, Ti}; set of the remaining matrix elements (Eq. 4.23)		—
II.2	**Correlation, trend and significance tests (§4.2.1, §4.4.4.2, §4.4.5)**		
$X,\ Y$	Paired series (variables) in a correlation analysis		unit of X, Y

Symbols	Definition	Expression	Unit
n	Sample size, i.e. number of paired points ($n = 34$ for the axial EDS series)		—
$\mathrm{cov}(X, Y)$	Covariance of X and Y	$\sum(X_i - \bar{X})(Y_i - \bar{Y})/(n-1)$	unit of XY
σ, σ_X, σ_Y	Standard deviation (S.D.) of a series, of X and of Y	$\sqrt{\sum(X_i - \bar{X})^2/(n-1)}$	unit of X, Y
ρ, r	Pearson linear correlation coefficient (expanded form: Eq. 4.1)	$\mathrm{cov}(X, Y)/(\sigma_X \sigma_Y)$	—
$r(X, Y)$	Pearson coefficient between two named quantities, e.g. $r(S, G)$, $r(\mathrm{Ta}, R)$, $r(V_1, V_2)$		—
$r_{XY \cdot Z}$	Partial correlation of X and Y with Z held fixed (Z = Ni removes the closure effect, §4.4.4.2)	$\dfrac{r_{XY} - r_{XZ} r_{YZ}}{\sqrt{(1 - r_{XZ}^2)(1 - r_{YZ}^2)}}$	—
ρ_s	Spearman rank correlation coefficient between composition and axial position (Eq. 4.2, without tied ranks)	$1 - \dfrac{6\sum d_i^2}{n(n^2-1)}$	—
d_i	Difference between the composition rank and the position rank of point i	$\mathrm{rank}(C_i) - \mathrm{rank}(x_i)$	—
t	Test statistic following the t distribution with $n-2$ degrees of freedom (Eq. 4.3)	$\rho_s \sqrt{(n-2)/(1-\rho_s^2)}$	—
$t_{0.975, n-2}$	Two-sided 95 % critical value of the t distribution with $n-2$ degrees of freedom		—
p	p-value (significance probability); the "uncorrected" p assumes independent samples		—
U	Mann–Whitney U statistic comparing pool frames with matrix frames (one-sided test, §4.4.5)	$\sum \mathrm{rank}_1 - n_1(n_1+1)/2$	—
n_1, n_2	Numbers of carbide/eutectic-pool frames and of matrix frames in the U test		—
R^2_{pct}	Percentage of the axial variance of an element explained by G (Table 4.12)	$100 r^2$	%
II.3	**Regression fitting of the macrosegregation curves (§4.3.1–§4.3.3, Table 4.3, Fig. F15)**		
C_s	Solid composition on the macrosegregation (Scheil-type) curve	$k_E C_0 (1 - f_s)^{k_E - 1}$	wt%
f_s	Nominal solid fraction assigned to the measuring points by linear mapping (34 points: 0–0.9); not a simulated solid fraction		—

Symbols	Definition	Expression	Unit
k_E	Effective partition coefficient of macrosegregation obtained from the fit	$b + 1$	—
$f(x)$	Prediction function of the linear model (Eq. 4.4)	$\omega^{\mathrm{T}} x + b$	—
$x,\ \omega$	Attribute (input) and weight of the linear model (Eq. 4.4)		—
b	Bias term in Eq. (4.4); exponent (slope) of the linearized model in Eq. (4.5)	$b = k_E - 1$ (Eq. 4.5)	—
a	Pre-factor of the univariate exponential model $y = a(1-x)^b$ (Eq. 4.5)	$k_E C_0$	wt%
$A,\ w,\ z$	Intercept, regressor and response of the linearized model $z = A + bw$ (Eq. 4.5)	$A = \log a,\ w = \log(1-x),$ $z = \log y$	—
$x_i,\ y_i$	Regressor and observed response of sample i in Eqs. (4.6)–(4.9)		—
$\hat{y}_i,\ \bar{y}$	Predicted value of sample i; mean of the observed responses		—
SST	Total sum of squares (Eq. 4.6)	$\sum(y_i - \bar{y})^2 = \mathrm{SSR} + \mathrm{SSE}$	—
SSR	Regression sum of squares	$\sum(\hat{y}_i - \bar{y})^2$	—
SSE	Residual sum of squares	$\sum(y_i - \hat{y}_i)^2$	—
R^2	Coefficient of determination (Eq. 4.7)	$\mathrm{SSR}/\mathrm{SST} = 1 - \mathrm{SSE}/\mathrm{SST}$	—
$\mathrm{cost}(\omega)$	Loss function (mean squared error) minimized by gradient descent (Eq. 4.8)	SSE/n	—
α	Learning rate of gradient descent (Eq. 4.9)	$\omega \leftarrow \omega - \alpha\, d\,\mathrm{cost}(\omega)/d\omega$	—
s	Residual standard deviation of the fit	$\sqrt{\mathrm{SSE}/(n-2)}$	—
S_{xx}	Sum of squared deviations of the regressor	$\sum(x_i - \bar{x})^2$	—
$\mathrm{SE}(b)$	Standard error of the fitted slope	$s/\sqrt{S_{xx}}$	—
$k_{E,\mathrm{lo95}},\ k_{E,\mathrm{hi95}}$	Lower and upper bounds of the 95 % confidence interval of k_E (Table 4.3)	$\hat{b} \mp t_{0.975,\,n-2}\mathrm{SE}(b) + 1$	—
p_{slope}	p-value of the slope test against $b = 0$ (i.e. $k_E = 1$, no macrosegregation)	$t = \hat{b}/\mathrm{SE}(b);\ t^2 = (n-2)R^2/(1-R^2)$	—
$\mathrm{SE}(\hat{y}\|x)$	Standard error of the fitted mean at x; the 95 % confidence band (Fig. F15) is $\hat{y} \pm t_{0.975,\,n-2}\mathrm{SE}(\hat{y}\|x)$ (the prediction band replaces $1/n$ by $1 + 1/n$)	$s\sqrt{1/n + (x-\bar{x})^2/S_{xx}}$	—
subscripts 10, 17, 34	Results of the 10-, 17- and 34-point data sets, e.g. $k_{E,34},\ C_{0,10},\ P_{c,34}$ (the 34-point set is the reference)		—

Symbols	Definition	Expression	Unit
II.4	**Partition coefficients, microsegregation and phase stability (§4.2.1, §4.3.4–§4.3.6)**		
$k_0,\ k_{0,i}$	Equilibrium partition coefficient (of element i) used in Chapter 4; Table 4.4 values unless liquidus values are stated		—
k_J	Solid–liquid partition coefficient from the JMatPro phase diagram (Table 4.4)		—
k_P	Partition coefficient adopted in the ProCAST simulation (Table 3.2)		—
$k_{\text{Lever}},\ k_{\text{Scheil}}$	Partition coefficients calculated with the lever-rule and Scheil models in JMatPro (Fig. 4.5)		—
$k(T)$	JMatPro Scheil-path partition coefficient as a function of temperature (Fig. F16)	C_γ/C_L	—
m_L	Liquidus slope calculated with JMatPro (Fig. 4.5); $\lvert m \rvert$ denotes its absolute value (Table 4.8)		K/wt%
C_i^{ph}	Concentration of element i in phase ph = Liquid, γ, γ', MC or MB_2 (§4.3.5), e.g. $C_{\text{Co}}^{\gamma'}$		wt%
k'_x	Microsegregation coefficient of element x (Eq. 4.10)	$C_{x,\text{core}}/C_{x,\text{inter}}$	—
$C_{x,\text{core}},\ C_{x,\text{inter}}$	Concentration of element x in the dendrite core and in the interdendritic region (Table 4.5)		wt%
Z	Atomic number (Table 4.8)		—
m_i	Atomic (mole) fraction of element i (Eqs. 4.11, 4.12)		—
N_v	Average electron vacancy number (Eq. 4.11)	$\sum_i m_i\,(N_v)_i$	—
$(N_v)_i$	Electron vacancy number of element i (Table 4.6)		—
$N_{cv},\ \bar{N}_{v,c}$	Critical average electron vacancy number for TCP-phase precipitation ($\bar{N}_{v,c} = 2.32$)		—
$\bar{N}_{v,\gamma}$	Average electron vacancy number of the γ phase (Fig. 4.21a)		—
M_d	Average d-orbital energy level (Eq. 4.12)	$\sum_i m_i\,(M_d)_i$	eV

Symbols	Definition	Expression	Unit
$(M_d)_i$	d-orbital energy level of element i (Table 4.6)		eV
$\bar{M}_{d,\gamma}$	Average d-orbital energy level of the γ phase; TCP threshold 0.925 (Fig. 4.21b)		eV
II.5 Thermal-field quantities (§4.4.1–§4.4.3, §4.4.5)			
$\rho_m(T)$	Melt density in the two-way coupled thermal–flow calculation (Eq. 4.13)	$\rho_{\mathrm{ref}}^m + (\partial\rho/\partial T)_m \times (T - T_{\mathrm{ref}}^m)$	kg/m³
ρ_{ref}^m, T_{ref}^m	Reference melt density and reference temperature		kg/m³, K
$(\partial\rho/\partial T)_m$	Thermal coefficient of the melt density		kg/(m³·K)
V, $\dot{T}$	Local cooling rate at a sampling point (Fig. 4.24, Table 4.11)	$\dot{T} = GR$ (Eq. 4.15)	K/s
R	Solid–liquid interface (liquidus-isotherm) advancing velocity	$R_i = \Delta/(t_{i+1} - t_i)$	mm/min
G	Local temperature gradient, averaged over a 10 mm interval centred at $x_i + 5$	$G_i = \dfrac{T_i(t_i) - T_i(t_{i+2})}{2\Delta}$	K/mm
ΔT	Local undercooling at which the solid fraction first reaches 5 %	$T_L - T(f_s = 0.05)$	K
T_L	Liquidus temperature of the alloy (1370.9 °C)		°C
t_i	Characteristic arrival time of the isotherm at point i		s
$T_i(t)$	Temperature at point i at time t		K
$t_{i+2} - t_i$	Two-step difference of the arrival times; $G_i \approx \dot{T}_i(t_{i+2} - t_i)/(2\Delta)$, and for white-noise arrival times its autocorrelation is −0.5 at a lag of 10 mm (§4.4.6)		s
subscripts 1, 2	Axis 1 (5 mm off the centreline) and Axis 2 (centreline), e.g. V_1, R_2, G_1		—
R_0, G_0	Steady-stage operating-point (mean) values of R and G		mm/min, K/mm
$\delta R/R_0$, $\delta G/G_0$	Relative fluctuations of the interface velocity and the temperature gradient (Eq. 4.16)	$\delta R/R_0 = -\delta G/G_0$	—
u_x	Vertical flow velocity at the liquidus in the ProCAST results		m/s
II.6 Effective partition coefficient and solute boundary layer (§4.4.3)			

Symbols	Definition	Expression	Unit
$k_E(R)$	Effective partition coefficient under partial liquid mixing (BPS model, Eq. 2.5)	$\frac{k_0}{k_0 + (1-k_0)\exp(-P_c)}$	—
P_c	Effective Péclet number of the solute-exchange layer; inverted from the measured k_E as $P_c = \ln\frac{k_E(1-k_0)}{k_0(1-k_E)}$ (Table 4.7)	$R\delta/D_L$	—
δ	Thickness of the effective solute-exchange layer at the interface		μm
D_L	Solute diffusion coefficient in the liquid (Table 3.2)		m²/s
P_cD_L	Element-independent product used to test a common exchange-layer thickness	$R\delta$	m²/s
$\partial k_E/\partial P_c$	First-order sensitivity of k_E to P_c (Eq. 4.17)	$\frac{k_0(1-k_0)\exp(-P_c)}{[k_0 + (1-k_0)\exp(-P_c)]^2}$	—
$\lvert\partial P_c/\partial k_E\rvert$	Sensitivity magnitude of the inverted P_c to errors in k_E (Table 4.7)	$1/\lvert k_E(1-k_E)\rvert$	—
$\langle k_E\rangle$	Growth-layer-thickness-weighted period average of k_E (Eq. 4.18)	$\langle k_E(R)R\rangle/\langle R\rangle$	—
$\langle\cdot\rangle$	Time average over one oscillation period		—
ε	Relative amplitude of the sinusoidal oscillation of R	$R(t) = R_0(1+\varepsilon\sin\omega t)$	—
σ_R	Standard deviation of the interface advancing velocity	$\sigma_R^2 = R_0^2\varepsilon^2/2$	mm/min
k_E', k_E''	First and second derivatives of k_E with respect to R		min/mm, (min/mm)²
$\Delta\langle k_E\rangle$	Correction of the period-averaged k_E caused by the oscillation of R	$(\sigma_R/R_0)^2[P_c\,\partial k_E/\partial P_c + P_c^2\,\partial^2 k_E/\partial P_c^2/2]$	—
l_D	Solute diffusion length	D_L/R_0	μm
τ	Build-up/relaxation time of the solute boundary layer (Eq. 4.19)	$l_D/R_0 = D_L/R_0^2$	s
λ	Spatial period (wavelength) of the temperature-gradient fluctuation		mm
T	Temporal period of the oscillation (§4.4.3.2)	λ/R_0	s
ω	Angular frequency of the oscillation (temperature disturbance)	$2\pi/T$	rad/s
$\omega\tau$	Dimensionless quasi-static criterion of the boundary-layer response (Eq. 4.20)	$2\pi D_L/(R_0^2T)$	—

Symbols	Definition	Expression	Unit
ΔT_0	Amplitude of the temperature undulation at the boundary layer (i: imaginary unit)	$\Delta T = \Delta T_0 \exp(\mathrm{i}\omega t)$	K
$C^{(0)}$	Solute concentration in the melt at the interface without temperature disturbance (Eq. 4.14)		wt%
$C_\omega^{(1)}$	Amplitude of the interfacial concentration undulation at frequency ω (Eq. 4.14)		wt%
ϕ	Phase difference between the concentration response and the temperature disturbance		rad
$\partial \ln C_s / \partial \ln G$	Elasticity of the solid composition with respect to G through the growth-rate channel (§4.4.3.3)	$-P_c(\partial k_E / \partial P_c)/k_E$	—
II.7	**Fluctuation amplitude, random sampling and normalization closure (§4.4.4)**		
CV	Coefficient of variation of the 34-point composition series (measure of fluctuation amplitude)	$\sigma/\bar{C}$	%
A_i	Normalized departure of the dendrite-core composition from the nominal composition (microsegregation amplitude) of element i (Eq. 4.21)	$\lvert C_{s,\mathrm{core}}/C_0 - 1\rvert = \lvert k_{0,i} - 1\rvert$	—
$C_{s,\mathrm{core}}$	Solid composition of the first-solidified dendrite core ($f_s \to 0$)	$k_0 C_0$	wt%
φ	Area fraction of the interdendritic (last-to-solidify) region, $\varphi \ll 1$		—
C_{pool}, C_{core}	Composition of an interdendritic pool and of a dendrite core		wt%
Var	Composition variance when a measuring frame falls randomly in a pool or in a core (Eq. 4.22)	$\propto \varphi(1-\varphi) \times (C_{\mathrm{pool}} - C_{\mathrm{core}})^2$	(wt%)²
$\mathrm{CV}_{\mathrm{Scheil}}$	Exact CV of the Scheil distribution under random area sampling (f_s uniform on [0, 1]); defined for $k_0 > 1/2$ and divergent for $k_0 \le 1/2$	$\lvert 1 - k_0\rvert / \sqrt{2k_0 - 1}$	—
$\sum C_i$	Normalization (closure) condition of the EDS data	$\sum C_i = 100$ wt%, $\sum \delta C_i = 0$	wt%
δC_j	Compensating dilution of matrix element $j \in M$ caused by a local Group-A enrichment δS (Eq. 4.23)	$-w_j\, \delta S / \sum_{m \in M} w_m$	wt%

Symbols	Definition	Expression	Unit
$w_j,\ w_m$	Abundance (mean mass fraction) of the matrix elements $j,\ m \in M$		wt%
II.8	**Registration, detrending, autocorrelation and effective sample size (§4.4.5, §4.4.6, Table 4.12)**		
Δx	Misregistration between an EDS point and the centre of the paired G interval; EDS point j is paired with G_{j-1}, giving $\Delta x = +1$ mm	$x_j - x_c$	mm
$\rho(\Delta x)$	Correlation of a quasi-periodic signal under misregistration Δx (Eq. 4.24)	$\approx \cos(2\pi\Delta x/\lambda)$	—
$A_r,\ \Delta x_0$	Amplitude and peak position of the cosine fit $r(S,G) = A_r\cos[2\pi(\Delta x - \Delta x_0)/\lambda]$ (Fig. 4.30)		—, mm
x_{model}	Model coordinate of EDS point j allowing for the first-point position x_{first}, the kerf loss per cut and the shell shrinkage (registration sensitivity, Fig. F12)	$\dfrac{x_{\text{first}} + 5j + \text{kerf}\cdot\lfloor j/5\rfloor}{1 - \text{shrinkage}}$	mm
δX	Detrended fluctuation of a quantity X (e.g. $\delta C_i,\ \delta S,\ \delta G,\ \delta R$); by default the residual from a least-squares straight line in x (first scheme of Table 4.12)	$X_i - (\hat{a} + \hat{b}x_i)$	unit of X
∇X_i	First difference (second detrending scheme of Table 4.12)	$X_{i+1} - X_i$	unit of X
$\text{MA}_5(X)_i$	Five-point centred moving average; the residual $X_i - \text{MA}_5(X)_i$ is the third detrending scheme, and two points are lost at each end ($n = 21$ when $n = 25$)	$\dfrac{1}{5}\sum_{l=-2}^{2} X_{i+l}$	unit of X
$\rho(k)$	Sample autocorrelation function of a series y at lag k	$\dfrac{\sum_{i=1}^{n-k}(y_i - \bar{y})\,(y_{i+k} - \bar{y})}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	—
k	Lag in sampling steps in the autocorrelation analysis (one step = Δ = 5 mm)		—
$\rho_C(k),\ \rho_G(k)$	Autocorrelation of the detrended composition series and of the detrended G series at lag k		—
n_{eff}	Bartlett effective sample size (Table 4.12)	$\dfrac{n}{1 + 2\sum_{k=1}^{K}\rho_C(k)\rho_G(k)}$	—
K	Lag truncation of the Bartlett sum ($k \le n/2$)	$\lfloor n/2\rfloor$	—

Symbols	Definition	Expression	Unit
p_{corr}	Autocorrelation-corrected p-value, from the t test with $n_{\mathrm{eff}} - 2$ degrees of freedom	$t = r\sqrt{(n_{\mathrm{eff}} - 2)/(1 - r^2)}$	—
f	Spatial frequency in the periodogram (spectral) analysis	$1/\lambda$	mm^{-1}
$\lvert H(f)\rvert$	Gain of the two-step difference operator that constructs G; maximum at $\lambda = 4\Delta =$ 20 mm, zero at $\lambda = 2\Delta$	$2\lvert\sin(2\pi f\Delta)\rvert$	—

Notes:

(1) "—" denotes a dimensionless quantity, and log denotes the natural logarithm.

(2) In Part II, equation, section, table and figure numbers refer to Chapter 4 unless stated otherwise (e.g. Eq. 2.5, Table 3.2).

(3) The following symbols carry a Chapter-4-specific meaning that differs from Part I; within Chapter 4 the meaning given in Part II applies: V (local cooling rate), R (interface advancing velocity), G (finite-difference gradient in K/mm), ΔT (undercooling at f_s = 5 %), ΔT_0 (undulation amplitude), P_c ($R\delta/D_L$), ρ (correlation coefficient; the melt density is written ρ_m), ε (relative amplitude of R), α (learning rate), n (sample size), m_i (atomic fraction), p (p-value), t (test statistic of Eq. 4.3), T (oscillation period, §4.4.3.2), k (lag in autocorrelation) and σ (standard deviation).

(4) Within Chapter 4, b is the bias in Eq. (4.4) but the slope in Eq. (4.5); w is the regressor in Eq. (4.5) but the abundance weight in Eq. (4.23); S denotes both the Group-A sum and the set {Ta, Hf, Ti}; x denotes the axial position and, in Eqs. (4.4)–(4.5), the regression input.

Appendix B Data

Appendix B consists of five tables. B1 and B2/B3 are given in full below; for B4, B5 and the correlation coefficients, the CSV files in the package folder results/ are authoritative and only characteristic values are listed here.

B1 Derived thermal-field quantities

Definitions: Ṫ is the local cooling rate from the first two records within 1200–1400 °C; R = 5 mm ÷ the difference between the times at which adjacent points first enter 1200–1400 °C; G is the temperature difference between the crossing time of point i itself and the crossing time of point i+2, ÷ 10 mm (the original program divided by 5 mm; the values here have been divided by 2), and represents the average over the interval [x_i, x_i+10]; ΔT = 1370.9 °C − the temperature at which the solid fraction first reaches 5 %. The ΔT of axis 1 was computed in the

original program with the solid fraction of axis 2; it is invalid and is not listed. Means over the steady stage x ≥ 70 mm: $\dot{T}_2$ 0.216 K/s, R_2 2.46 mm/min, G_2 6.23 K/mm, ΔT_2 3.67 K.

i	x /mm	$\dot{T}_2$ /K s^{-1}	R_2 /mm min^{-1}	G_2 /K mm^{-1}	ΔT_2 /K	$\dot{T}_1$	R_1	G_1
0	0	84.313	76.19	42.59	48.21	67.448	76.19	42.92
1	5	17.704	22.02	26.58	61.80	20.006	22.02	27.33
2	10	6.411	29.45	13.74	8.49	7.330	29.45	12.90
3	15	4.110	21.38	13.60	2.70	3.881	25.26	13.80
4	20	3.419	13.46	10.31	6.14	3.064	12.28	11.25
5	25	2.115	23.53	6.88	10.90	2.020	23.53	7.85
6	30	1.754	14.63	8.10	4.93	1.759	12.12	7.64
7	35	1.497	11.09	8.25	7.51	1.407	15.25	6.56
8	40	1.111	9.16	6.21	6.33	1.071	9.49	7.52
9	45	0.810	10.81	6.05	6.06	0.881	7.41	6.20
10	50	0.706	6.17	7.21	4.67	0.673	8.36	7.20
11	55	0.542	4.69	7.17	5.23	0.548	3.70	7.53
12	60	0.415	3.74	6.16	4.34	0.385	4.45	4.85
13	65	0.325	3.96	6.49	3.74	0.321	5.10	6.08
14	70	0.295	2.46	7.36	3.82	0.272	2.35	7.35
15	75	0.229	2.34	6.32	3.30	0.234	2.53	7.32
16	80	0.225	2.68	5.78	4.42	0.250	1.95	6.66
17	85	0.219	2.54	6.94	3.64	0.230	2.82	5.93
18	90	0.202	1.93	7.20	3.96	0.194	2.25	6.45
19	95	0.193	2.34	5.79	3.21	0.200	2.38	6.68
20	100	0.208	2.77	5.22	2.65	0.214	2.10	6.22
21	105	0.230	2.80	6.62	3.39	0.214	2.73	5.65
22	110	0.189	1.93	7.01	3.15	0.210	2.48	5.96
23	115	0.201	2.47	5.75	3.51	0.218	2.49	6.24
24	120	0.221	2.66	6.08	3.55	0.226	2.26	6.25
25	125	0.206	2.27	5.78	4.07	0.213	2.54	5.15
26	130	0.194	2.96	6.09	4.23	0.205	3.17	5.54
27	135	0.226	2.08	6.09	3.60	0.221	2.27	7.24
28	140	0.238	3.01	5.82	3.18	0.200	1.97	6.01
29	145	0.218	2.21	6.40	3.72	0.216	3.30	5.62
30	150	0.225	2.47	6.07	4.03	0.228	2.16	6.68
31	155	0.219	2.43	6.12	4.18	0.216	2.35	5.31
32	160	0.206	2.48	5.95	4.17	0.204	3.31	5.16
33	165	0.208	2.49	5.97	2.86	0.232	2.44	6.81
34	170	0.233	2.44	6.37	4.52	0.203	2.01	6.83
35	175	0.191	2.26	—	4.11	0.206	2.41	—
36	180	0.203	—	—	3.04	0.198	—	—

B2/B3 34-point EDS composition (wt%) and registered centreline thermal parameters

x_j = 1 + 5j mm is the nominal position; S = Ta + Hf + Ti; i is the paired derived point (i = j − 1), and G and R are taken from axis 2 in B1 (the R correlation in §4.4.3.3 instead uses i = j, i.e. the derived point whose interval centre is closest to the measuring point). The steady window of Table 4.12 is j ≥ 9 (x ≥ 46 mm, n = 25). The pool frame (Hf > 1.5 wt%) comprises 15 points.

j	x /mm	Ta	Hf	Ti	S	Al	Co	Cr	W	Mo	Ni	i	G /K mm⁻¹	R /mm min⁻¹
0	1	3.71	0.87	0.84	5.42	7.17	9.08	8.27	10.09	0.61	59.36	—	—	—
1	6	4.02	1.18	0.77	5.97	6.70	9.34	8.33	10.58	0.64	58.43	0	42.59	76.19
2	11	3.72	0.89	0.62	5.23	6.81	9.43	8.41	10.18	0.71	59.25	1	26.58	22.02
3	16	3.43	0.72	0.69	4.84	7.54	9.51	8.71	9.78	0.52	59.09	2	13.74	29.45
4	21	3.43	0.59	0.77	4.79	7.84	9.57	8.75	10.12	0.47	58.46	3	13.60	21.38
5	26	3.59	0.62	0.84	5.05	7.37	9.47	8.67	9.96	0.73	58.75	4	10.31	13.46
6	31	3.19	0.75	0.86	4.80	7.46	9.44	8.67	9.15	0.72	59.76	5	6.88	23.53
7	36	2.80	0.90	0.82	4.52	7.44	9.59	8.52	8.76	0.69	60.49	6	8.10	14.63
8	41	6.54	2.30	1.23	10.07	6.59	9.05	8.11	9.08	0.57	56.53	7	8.25	11.09
9	46	6.95	2.11	1.24	10.30	6.86	8.81	8.19	9.69	0.64	55.51	8	6.21	9.16
10	51	3.40	0.42	0.90	4.72	7.28	9.33	8.79	9.35	0.80	59.73	9	6.05	10.81
11	56	6.61	1.78	1.22	9.61	6.44	8.83	8.42	9.23	0.75	56.73	10	7.21	6.17
12	61	6.81	2.58	1.17	10.56	6.26	8.70	8.04	9.25	0.89	56.29	11	7.17	4.69
13	66	3.87	1.08	0.79	5.74	6.87	9.32	8.36	9.94	0.68	59.10	12	6.16	3.74
14	71	4.75	0.12	0.97	5.84	6.97	9.24	8.35	10.62	0.67	58.31	13	6.49	3.96
15	76	7.59	2.48	1.40	11.47	6.71	8.41	7.79	8.74	0.96	55.91	14	7.36	2.46
16	81	6.45	2.49	1.17	10.11	7.23	8.71	7.95	8.80	0.90	56.29	15	6.32	2.34
17	86	3.73	0.48	0.80	5.01	7.35	9.26	8.58	9.39	0.87	59.55	16	5.78	2.68
18	91	8.86	2.76	1.36	12.98	6.26	8.10	8.04	8.67	0.75	55.20	17	6.94	2.54
19	96	8.06	2.56	1.29	11.91	6.05	8.36	8.02	10.16	0.80	54.69	18	7.20	1.93
20	101	3.63	1.77	0.91	6.31	6.29	9.23	8.35	9.62	0.74	59.46	19	5.79	2.34
21	106	4.66	2.19	1.10	7.95	6.24	8.97	8.15	8.23	0.49	59.98	20	5.22	2.77
22	111	7.35	2.45	1.21	11.01	5.86	8.53	7.92	9.18	0.39	57.12	21	6.62	2.80
23	116	7.22	2.80	1.20	11.22	6.43	8.32	7.50	8.16	0.38	58.00	22	7.01	1.93
24	121	4.38	1.29	0.85	6.52	7.41	8.82	7.80	8.81	0.56	60.08	23	5.75	2.47
25	126	4.10	1.19	0.74	6.03	7.16	8.98	8.23	9.85	0.64	59.10	24	6.08	2.66
26	131	4.16	1.64	0.87	6.67	7.05	8.90	8.25	9.11	0.59	59.44	25	5.78	2.27
27	136	3.95	1.05	0.85	5.85	6.95	9.19	8.47	9.00	0.58	59.95	26	6.09	2.96
28	141	5.74	0.62	1.11	7.47	6.33	9.11	8.46	8.73	0.72	59.20	27	6.09	2.08
29	146	5.57	0.38	1.02	6.97	6.59	9.26	8.33	9.52	0.88	58.45	28	5.82	3.01
30	151	4.13	0.38	0.79	5.30	6.87	9.51	8.55	9.64	0.74	59.39	29	6.41	2.21
31	156	4.67	0.47	0.98	6.12	7.34	9.48	8.78	8.55	0.56	59.17	30	6.07	2.47
32	161	5.32	1.63	1.19	8.14	7.71	9.14	8.58	8.04	0.72	57.66	31	6.12	2.43

j	x /mm	Ta	Hf	Ti	S	Al	Co	Cr	W	Mo	Ni	i	G /K mm^{-1}	R /mm min^{-1}
33	166	8.82	2.58	1.54	12.94	6.25	8.34	8.03	8.97	0.92	54.54	32	5.95	2.48

B4 Partition coefficients along the JMatPro Scheil path

(characteristic values; full table in results/B4_jmatpro_k_vs_T.csv)

Element	k at liquidus (1370.9 ℃)	Mean k over 1370.9–1256 ℃	k (1256 ℃)	Table 4.4 kJ	Table 3.2 kP (ProCAST)	Temperature at which k crosses 1 /℃
W	1.122	1.230	1.391	1.232	1.03	—
Co	1.093	1.111	1.117	1.097	1.14	—
Ta	0.481	0.545	0.632	0.570	0.826	—
Ti	0.729	0.790	0.867	0.756	0.808	—
Al	0.910	0.974	1.042	1.056	0.935	1290
Cr	0.941	1.039	1.155	1.012	0.81	1335
Mo	0.775	0.822	0.887	0.764	0.639	1195
Hf	0.095	0.118	0.142	0.107	0.095	—

Al path: Al in γ is 5.09 wt% at the liquidus; liquid Al reaches a maximum of 6.33 wt% (1290 ℃); γ′ first appears at 1256 ℃, when liquid Al is 6.28 wt%; the last liquid contains 4.79 wt% Al (1125 ℃).

B5 γ-phase composition and phase-stability parameters

(corrected basis; full table in results/B5_gamma_phase_stability.csv)

Element-wise corrections applied to the measured wt% before calculation: Ni (C − 0.092)/0.945; W /1.003; Ta /1.393; Al /1.273; Co /1.008; Cr /1.046; Hf /0.922; Mo /1.348; Ti /1.354; B 0.015, C 0.07 and Zr 0.007 wt% are taken at their nominal values. Empirical partitioning scheme for the γ phase: first M_3B_2, MC and M_6C, then γ′ = 3 × (Al + 0.03Cr + Ti + Ta + Hf − …), with the remainder as γ (Ni, Cr, Co, Mo, W, Zr). Without the correction only 26 of the 34 points can be calculated; the others give negative Ni.

Quantity	Range / mean (34 points)
Ni in γ /at%	Mean 47.0
Cr + Co in γ /at%	Mean 45.8
W in γ /at%	Mean 6.6
$\bar{N}_{v,\gamma}$	1.909–2.358; 4 points > 2.32
$\bar{M}_{d,\gamma}$ /eV	0.867–0.928; 3 points > 0.925
da2 mean / nominal value	Ta 1.61, Ti 1.43, Mo 1.37, Al 1.23, Cr 1.03, Hf 1.01, Co 0.98, W 0.98, Ni 0.95

B6 Correlation coefficients and test statistics

(summary; CSV files for Tables 4.8–4.12 in results/T4_x)

Item	Value
Table 4.12 (x ≥ 46 mm, n = 25), linearly detrended r	Ta +0.66, Ti +0.53, Hf +0.43, S +0.61, Ni −0.62, Co −0.55, Cr −0.39, Al −0.36, W −0.07, Mo +0.14
Bartlett-corrected p (k ≤ n/2)	Ta 0.024, Ti 0.061, Hf 0.14, S 0.040, Ni 0.035, Co 0.052, Cr 0.11, Al 0.13
x ≥ 71 mm (n = 20), r	Ta +0.67, Ni −0.65, S +0.60
Registration scan r(S,G), Δx = +11/+6/+1/−4/−9/−14 mm	−0.56 / +0.08 / +0.61 / −0.12 / −0.43 / +0.15; cosine fit A = 0.55, λ = 19.4 mm, peak position +1.5 mm, rms 0.05
Steady-stage G autocorrelation at 5/10/15/20 mm	−0.03 / −0.66 / +0.06 / +0.48; spectral peak at 20.1 mm (periodogram)
Two-axis coherence r(V1,V2) / r(R1,R2) / r(G1,G2)	0.65 (0.39 without x = 70) / −0.03 / 0.07
Arrival-time residuals (x = 70–175 mm)	sd 11.1 s; autocorrelation 0.09/−0.06/0.01; autocorrelation of the two-step difference at 10 mm −0.67; r(δG, δ(t_{i+2}−t_i)) = 0.90
R–G	r = −0.44 (centres 2.5 mm apart); +0.56 at an offset of 7.5 mm; slope 0.63, 0.92 after correction; predicted CV_R 11.0 % vs measured 11.7 %
Pool frame vs matrix frame δG (axis 2 / axis 1)	+0.34 K/mm, p = 0.047 / +0.12 K/mm, p = 0.42
Axis-1 control r(·, G1)	S +0.16, Ta +0.19, Ni −0.16
Closure dilution, predicted/measured sd	Ni 1.00, Cr 0.77, Co 0.64, W 0.41, Al 0.39
Partial correlation r(W,Ta ; controlling for Ni)	−0.66
Spearman(CV, ·)	abs(k0−1) 0.93 (Table 4.4) / 0.93 (liquidus) / 0.55 (ProCAST kP); Z 0.26; abs(m) −0.38; 1/DL 0.24
Trend Spearman(C, x)	W −0.54 (p 0.001), Ta +0.45 (0.008), Ti +0.35 (0.04), Co −0.33 (0.06), Al −0.25, Cr −0.20, Hf +0.11, Mo +0.09, Ni −0.02
34-point kE [95 %], p	W 1.057 [1.021, 1.093] 0.003; Ta 0.836 [0.658, 1.015] 0.07; Ti 0.881 [0.758, 1.005] 0.06; Hf 0.964 [0.527, 1.402]; Co 1.015 [0.989, 1.041]; Ni 1.003 [0.986, 1.020]
Pc (34 points)	W 1.25, Co 1.77, Ta 1.35, Ti 0.87, Mo 2.47, Hf 5.42; Rδ (W, Co, Ta, Ti) = (2.6 ± 0.7) × 10^{-9} m^2/s, δ = 64 ± 17 μm

Appendix C: Python Program

The package CM247LC_Ch4_supplement.zip contains 1 common module, 7 programs, 10 tests and READMEs. Running python 12_run_all.py takes about 1 minute and generates all 37 result tables and 19 figures.

List of files

File	Corresponding thesis content	Input	Main output
ch4lib.py	Common data layer: reads the thermal-field output (G ÷2), EDS and JMatPro data; registration rules; detrending, autocorrelation, Bartlett n_eff	data/	—
06_thermal_field.py	§4.4.2; Table 4.11; R–G relation and two-axis coherence in §4.4.5; G autocorrelation/spectrum/peak positions; arrival-time residuals	thermal_raw_output.txt	B1; T4_11; 06_*.csv; F01–F04
07_eds_composition.py	§4.2.1, §4.4.4: Tables 4.8–4.10, closure dilution, partial correlation, recomputation of the grey relational grade, Spearman trend	da2.csv, liquid/gamma.dat	B2; T4_8–T4_10; 07_*.csv; F05–F09
08_registration.py	§4.4.5: registration; Table 4.12 under three detrending schemes with p; cosine fit of the registration scan; peak positions; pool-frame test; window sensitivity; axis-1 control; position-mapping sensitivity; apparent kE from closure; R correlation of §4.4.3.3 (R taken at i = j)	da2.csv + thermal field	B3; T4_12; 08_*.csv; F10–F13
09_scheil_fit.py	§4.3.3, §4.4.3.1: Table 4.3 fits for three windows (with CI and p), source check of da4/da7, Table 4.7 Pc inversion and δ	da2/da4/da7.csv	T4_3; T4_7; 09_*.csv; F14–F15

File	Corresponding thesis content	Input	Main output
10_jmatpro.py	§4.3.5: k(T), liquidus k0, Al paths in each phase	liquid/gamma/gammap/Al1.dat	B4; 10_*.csv; F16–F17
11_phase_stability.py	§4.3.6: empirical γ-phase partitioning, N̄v, M̄d (correction factors stated explicitly)	da2.csv	B5; 11_*.csv; F18–F19
12_run_all.py	Master script; thesis values vs recomputed values	All of the above	comparison_thesis_vs_recomputed.csv
tests/test_ch4.py	10 unit tests (G correction, Table 4.11, closure, registration mismatch, Table 4.12 Ta, Spearman 0.93, W kE, da4 source, liquidus k0, n_eff)	—	unittest report

Conventions

Derived point i is at x = 5i mm (0–180); G_i is the average over the interval [x_i, x_i+10], centred at x_i + 5.

1. EDS point j is at x_j = 1 + 5j mm (nominal) and is paired with derived point i = j − 1 (for G), a mismatch of 1 mm; the correlation between R and composition uses i = j (interval centres 1.5 mm apart); the position mapping x_model = (x_j + kerf × segment number)/(1 − shrinkage) is used for the sensitivity analysis.
2. Table 4.12 window: EDS x ≥ 46 mm (n = 25); Table 4.11 window: x ≥ 70 mm; moving-average detrending uses a centred 5-point average within the window, dropping 2 points at each end (n = 21).
3. Bartlett effective sample size: n_eff = n/[1 + 2Σ_{k≤n/2} ρ_C(k)ρ_G(k)].
4. In the Scheil fits, fs is a linear mapping (34 points over 0–0.9), not the simulated solid fraction; OLS is used, and its results agree with the original gradient descent (α = 0.01, 4000 steps).
5. Axis 1 serves only as a reference for axis 2.